BASE24 Release 6.0 Version 7

Implementation Guide

Nov-2006, Release 6.0, version 7 Publication Number BA-AE010-07

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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one version to another are indicated with the symbol "|" in the outside margin.

Release 6.0 Version 7 (D. Barton, S. Eschliman, K. Koeppen, W. Kurzenberger, C. Meyers, C. Rink, R. Schieuer, J. Steffes, L. Watson, J. Wishart; R. Brittain, R. Duvall, R. McCall; M. Ahrens, S. Schmitt,)
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1: Introduction

The ***BASE24 Release 6.0 Version 7 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 release 6.0 version 6 systems to BASE24 release 6.0 version 7.

This section provides an overview of the planning requirements for BASE24 users migrating to release 6.0 version 7. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for release 6.0 version 7, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for migrating their BASE24 systems to release 6.0 version 7. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Version 7 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to release 6.0 version 7.

BASE24 Release 6.0 products include the following:

BASE24-atm
BASE24-pos
BASE24-teller
BASE24-remote banking
BASE24 Interchange Interfaces
BASE24 Value-Added Products:

- BASE24-card
- BASE24-customer service
- BASE24-InfoBase
- BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the migration process and those who implement the migration to release 6.0 version 7. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 3.

To migrate to release 6.0 version 7, users must be on release 6.0 version 6. Refer to the ***BASE24 Release 6.0 Version 6 Implementation Guide*** to assist in planning the tasks necessary to get to release 6.0 version 7.

Using this guide, you can do the following:

- Determine what ACI services are available to assist in the migration to release 6.0 version 7.
- Develop a project plan to migrate to release 6.0 version 7.
- Determine staffing and training needs.
- Gain a detailed understanding of the release 6.0 version 7 enhancements.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

FIX

FIX 2.28 (or newer) is required to apply any new BASE24 fix files produced after 19 October 2006. FIX 2.28, like the previous version of FIX, will allow application of fix files to a specified file name in a concatenated list of fix files. This option can be combined with the FIRSTSCN and LASTSCN options to apply only fixes to the specified file name in a specified SCN range. As of 19 October, FIX 2.28 (or the most current version of FIX) is automatically shipped with each software order. Please replace your existing FIX object with the most current version.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to release 6.0 version 7. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the migration method to be used.

In general, all changes for release 6.0 version 7 can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user’s release 6.0 version 6 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from release 5.3 to the latest version of release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from release 6.0 version 1 to release 6.0 version 2. No changes are made to these programs other than necessary fixes. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from release 6.0 version 2 to release 6.0 version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from release 6.0 version 3 to release 6.0 version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from release 6.0 version 4 to release 6.0 version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from release 6.0 version 5 to release 6.0 version 6. No changes are made to these programs other than necessary fixes.

The *UC0A subvolumes (e.g., AT60UC0A, BA60UC0A, PS60UC0A) contain the conversion program for users migrating from release 6.0 version 6 to release 6.0 version 7. No changes are made to these programs other than necessary fixes.

If a release 5.3 user wants to convert to the latest version of release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 2, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 2 user wants to convert to release 6.0 version 3, the conversion programs on *UC06 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a release 6.0 version 3 user wants to convert to release 6.0 version 4, the conversion programs on *UC07 should be run.

If a release 6.0 version 4 user wants to convert to release 6.0 version 5, the conversion programs on *UC08 should be run.

If a release 6.0 version 5 user wants to convert to release 6.0 version 6, the conversion programs on *UC09 should be run.

If a release 6.0 version 6 user wants to convert to release 6.0 version 7, the conversion programs on *UC0A should be run.

If a release 6.0 version 1 user wants to convert to release 6.0 version 7, the conversion programs on *UC05 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user should run the conversion programs on *UC06, *UC07, *UC08 and on *UC09. Finally, the user should run the conversion programs on *UC0A.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the release 6.0 version 7 software. Such manuals include the formal publications mentioned under the Publications heading in Section 3, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 6.0 Installation Guide (Informal) - BA-AE000-60

A primary reference document for information about the installation of a BASE24 release 6.0 system from scratch. This document is included as a Microsoft Word document and is shipped on the BASE24 6.0 CD-ROM.

The BASE24 6.0 Installation guide is also provided on the software tape with the code. The document is located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to the move the install document to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Switch 6.0 interfaces. It explains how customers can implement Switch 6.0 interfaces in their existing networks as well as in a BASE24 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a 6.0 network. This document is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 release 6.0 network. This document is included as a Microsoft Word document and is shipped with the release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to release 6.0 version 7. This document is included as a HP NSK TGAL document and is shipped with the release 6.0 version 7 software. This document is named AAREAD0A and is located on the BA60UC0A subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc. Internet website (http://www.aciworldwide.com/support).

- A current listing of HELP24 phones worldwide.
- An option to send an e-mail message to HELP24.
- An option to sign up for HELP24 InfoLink.
- An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- Access the HELP24 database to open a new Clarify case.
- Access installation and fix release procedures online.
- Access all BASE24, BASE24-es, and NET24-XPNET documentation online.
- Search the site for information about BASE24.

The **HELP24 eSupport Users Guide** contains procedures for opening a new Clarify case, accessing the Clarify database, and accessing documentation on-line. Once you log on to HELP24, the path to the **HELP24 eSupport Users Guide** is Support Center> InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center> InfoLink > Product Info >BASE24> BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

- Pre-Install Checklists- for BASE24 6.0 and ITS
- Install Guides—Procedures for installing a software release or version for the first time.
- Retail Install Guides—Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center> InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A software routine within existing code that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file (also requires an operator to manually rename the file that has been converted), move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A utility that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program and online conversion) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

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2: Functionality

This section contains an overview of the BASE24 release 6.0 version 7 enhancements.

Functionality has been organized by product, and within each product, according to whether the enhancement is a general enhancement or an optional enhancement.

- General enhancements are those enhancements that are inherent in the release 6.0 version 7 offerings. These enhancements affect all BASE24 users. BASE24 users must review the features that are part of these enhancements to ensure an understanding of the new processing and ensure proper configuration within the BASE24 environment.
- Optional enhancements are those enhancements that can be enabled or disabled in release 6.0 version 7. BASE24 users can implement these enhancements based on their own BASE24 processing and marketing requirements.

Enhancements are described in this section by product as follows:

- BASE24 Base (applicable to all products)
- BASE24-atm
- BASE24-pos
- BASE24-teller
- BASE24-remote banking
- BASE24 Interchange Interfaces
- BASE24 Value Added Products

BASE24 users should acquaint themselves with the BASE24 Base section (Section 4) as well as the applicable product-specific sections in order to obtain an understanding of the overall release 6.0 version 7 offering.

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BASE24 Base Enhancements

The BASE24 Base product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Base product, refer to Section 4.

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General Enhancements

The following enhancements are common to all products or provide the base for product-specific enhancements. As these enhancements are inherent in release 6.0 version 7, BASE24 users must review them to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

Changes for all BASE24 release 6.0 version 7 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

New BASE24 Tokens and Subtypes

BASE24 has been enhanced to support several new tokens and transaction subtypes for new transactions only supported with the BASE24-es IFX Device Handler. These new tokens and subtypes are used in processing these transactions through to a backend Host only.

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Optional Enhancements

The following BASE24 Base release 6.0 version 7 enhancements are optional. BASE24 users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Allow Unsorted Data to be Loaded by Refresh

The BASE24 Refresh Process has been enhanced to allow for input records that are not pre-sorted, a situation which typically exists in a multi-institution processor environment. Full file refresh has been changed to allow unsorted data input. Partial file refresh has been changed to allow data batches to be input in any order. The option to accept unsorted data is set using two new LCONF parameters, REFR-INPUT-DATA-UNSORTED and REFR-MAX-RECS-UNSORTED. This new functionality now gives the customer the option of processing sorted or unsorted data depending on their particular environment.

EMV CCD Issuer Support

EMVCo has produced CCD (Common Core Definition), which provides the basic EMV data definitions, and CPA (Common Payment Application), which define the behavior of the EMV application. With the Visa April 2005 Business enhancement, Visa announced requirements for EMV CCD (Common Core Definition).

Enhancements have been made to BASE24 BASE to support EMV CCD data used by BASE24-atm and BASE24-pos for EMV CCD issuer processing.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required

HISO Full Message Encryption Support

BASE24-atm and BASE24-pos HISO processing has been enhanced to support full message encryption between BASE24 and host systems, and additional detection and correction of key synchronization errors.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required

HISO Miscellaneous Enhancements

BASE24-atm and BASE24-pos HISO processing has been enhanced to support enhanced performance monitoring, Store And Forward (SAF) throttling, and ASCII/EBCDIC conversion between BASE24 and host systems.

Modify Discretionary Data Suppression Enhancement

The utility program that currently truncates discretionary data was enhanced so that it populates the field with zeros instead of truncating. This will enable issuers who depend on field lengths for processing to complete the processing and settlement of the transaction.

BASE24-atm Enhancements

The BASE24-atm product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-atm product, refer to Section 5.

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General Enhancements

The following BASE24-atm enhancements are inherent in the release 6.0 version 7 offerings. All BASE24-atm users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

Add Additional Values to ATD

Additional values have been added to the ATM Terminal Definition (ATD) file to better identify the EMV capabilities of the ATM. This information will be used to better process EMV transactions.

ATDD2 Record Cleanup Enhancement

The BASE24-atm file maintenance system was enhanced to delete the terminal's ATM Terminal Dynamic Data scratch pad (ATDD2) record, in addition to the terminal definition file records currently deleted, when the operator presses the F4 (delete record) key. In order to provide an audit of the deletion, an Online Maintenance File (OMF) record will be logged for the deleted ATDD2 record. The OMF record will contain only the OMF header; it will not contain an image of the ATDD2 record. If the operator presses the F3 (add record) key, no ATDD2 record will be added. If the operator presses the F5 (update record) key, the ATDD2 record will not be updated. The adding and updating of ATDD2 records is handled dynamically by the device handlers during transaction processing.

Previously, when a terminal's definition file records were deleted, the file maintenance system deleted the terminal's ATDD1 and ATDS1 records, but did not delete the terminal's ATDD2 record. This could lead to unnecessary disk space usage and potential problems with sensitive cardholder information being left on the system unnecessarily.

ATM Passbook Update

BASE24-atm has been changed to provide customers with the ability to support a Passbook Update transaction. The Passbook Update transaction enters BASE24 as a Statement Print transaction (transaction code 70) with a subtype of APU0. Currently, only the BASE24-es IFX Device Handler supports the Passbook Update transaction.

Prerequisites: The BASE24-es IFX Device Handler is required

Deposit Hold Notification

BASE24-atm has been enhanced to support the notification to a customer of any deposit holds that will be applied to the deposit transaction the customer is currently engaged in. BASE24-atm accepts deposit hold information from the customer's host system and returns the deposit hold information to the cardholder at the ATM giving them an option of either accepting or canceling the transaction before they insert the deposit into the ATM. Currently, only the BASE24-es IFX Device Handler supports Deposit Hold Notifications.

Prerequisites: The BASE24-es IFX Device Handler is required

Exchange Rate Notification

BASE24-atm has been enhanced to support the notification to the customer at the ATM of the specific exchange rate that is to be used in the transaction the customer is currently engaged in. The cardholder will get the option to accept or decline the exchange rate offered by the Financial Institution. If the cardholder declines, no further processing occurs. If the cardholder accepts, the transaction proceeds normally. Currently, only the BASE24-es IFX Device Handler supports Exchange Rate Notifications.

Prerequisites: The BASE24-es IFX Device Handler is required.

NCR Release 3.00 Upgrade Enhancements

The BASE24-atm NCR 5xxx Device Handler has been enhanced to support version 3.00 of the NCR Aptra Advance NDC+ Message Specification. Some of the enhancements include the new state "m", changes to transaction request state "I", added Operations Code extension table to state "Y", configurable BNA retract functionality, support for 999 states, and the support of 9999 customized screens.

Prerequisites: The NCR 5xxx Device Handler is required

Service Code Validation Enhancement

The purpose of this enhancement is to provide better detection of counterfeit cards. In card-read transactions, the card is authenticated by validating the Card Verification Digits encoded in the track data (CVD1). For the algorithm used to generate the CVD1, the input data includes the Service Code, also encoded in the track data. In manually-entered transactions, the card is authenticated by validating the Card Verification Digits printed or embossed on the card itself (CVD2). For the algorithm used to generate the CVD2, the input data includes a value of “000” in place of the Service Code, which is typically not available in manually-entered transaction data.

A counterfeit card can be created by generating fraudulent track data, which includes the CVD2 printed or embossed on the card and a Service Code value of “000”. The authorizing system may successfully validate the CVD2 (using the Service Code of “000”), believing it to be the CVD1. This means that the card is assumed to be valid when in fact it is fraudulent. To detect these fraudulent cards, the BASE24-pos Router/Authorization module has been modified to reject transactions which contain a Service Code of “000” in track data in card-read transactions.

Tidel General Enhancements

This BASE24-atm enhancement provides additional functionality to the Tidel Device Handler. The Tidel Device Handler is now capable of supporting PIN Change transactions. The storing and retrieving of STM and token data for use in reversal processing has been expanded to allow for more data when necessary. Also, status message processing has been enhanced.

Prerequisites: The Tidel Device Handler is required.

Triton Token Retrieval Option

This BASE24-atm enhancement provides additional functionality to the Triton Device Handler. The storing and retrieving of STM and token data for use in reversal processing has been expanded to allow for more data when necessary.

Prerequisites: The Triton Device Handler is required

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Optional Enhancements

The following BASE24-atm release 6.0 version 7 enhancements are optional. BASE24-atm users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Completion Messages Configuration Enhancement

BASE24-atm has been enhanced to provide for new completion required values which will be used to indicate that completions are to be sent to the host only if BASE24-atm performed the authorization. Customers will now be able to configure the system to either send all BASE24-atm authorized transactions, including declines, or to only send approved BASE24-atm authorized transactions.

Correct Declined Advices

A project was done in BASE24 Release 6.0 version 5 (November, 2004) to send declined advices to the Host when BASE24 or an interchange stood in for the Host and declined the transaction. This functionality was added to both BASE24-atm and BASE24-pos.

A code change was made to this functionality in January, 2005 that has the affect of logging all declined advice/force post transactions to the TLF before an attempt is made to find the CPF. This causes problems for transactions where a CPF record could have been found and routed to the appropriate destination. It also means that records are logged to the TLF with no FIID information.

A declined advice/forced post is a transaction with a STM message type of 0220 and a response code greater than or equal to 050. These messages should be forwarded by Authorization to the host with their original response code as sent in from an interchange. If a declined advice/forced post transaction cannot be processed (for example the CPF, IDF or IPCF record cannot be found), the transaction will be marked by Authorization as an exception and not forwarded to a host, since it will not be possible to determine where to send the transaction. The transaction will be logged to the TLF with the original response code but marked as an exception.

EMV CCD Issuer Support

EMVCo has produced CCD (Common Core Definition), which provides the basic EMV data definitions, and CPA (Common Payment Application), which define the behavior of the EMV application. With the Visa April 2005 Business enhancement, Visa announced requirements for EMV CCD (Common Core Definition).

Enhancements have been made to BASE24-atm to support EMV CCD for issuer processing. The main impacts are in the following areas: identifying CCD cards; processing the CCD cryptogram version number (CCD identifier); passing the required data (to TSS/HSMs) for ARQC verification/ARPC generation; supporting the new format of the CVR; and supporting the CSU data. Any Interface changes will be completed as part of a future project. Support for CPA will be a separate future project.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required

Locked ATD Records Causing Cutover Failure

BASE24-atm has been enhanced to correct a potential issue related to locked ATM Terminal Data (ATD) and Terminal Data File (TDF) records possibly causing system cutover to fail. The ATM Settlement process has been enhanced to optionally skip locked ATD records and proceed with the settlement process. The ATM Settlement process reads a new LCONF parameter which will specify whether or not the ATM Settlement process should skip cutover of a terminal with a locked ATD/TDF record.

Shared 912 AKDS Certificate Support

BASE24-atm has been enhanced to support Automated Key Distribution (AKDS) functionality in those cases where the ATM software vendor utilizes the Shared 912 Remote Key Load (RKL) specification to facilitate remote key downloading to ATMs equipped with certificate-based encrypting PIN pads (EPPs). This specification is managed and used by Wincor Nixdorf, although it is open to all vendors.

AKDS automates the otherwise manual process of initializing and re-keying ATM master keys by eliminating the manual processes related to key generation, dual custodianship and manual key loading at the ATM. AKDS also facilitates compliance with “unique key per ATM” guidelines specified by ANSI X9 TG-3 key management standards mandated by networks and card associations. The AKDS key distribution protocol uses public-key cryptography (RSA encryption) to generate and transport the initial master key (the “A-key”). Use of digital certificates and digital signatures is also used for authentication and to protect against tampering of data.

If AKDS has not been previously purchased, AKDS is a separately billable module.

Prerequisites: The Diebold 10xx/478x Device Handler and the Shared 912 AKDS modules are required. BASE24-es TSS version 05.4 or greater will be required.

Shared 912 Offline PIN Synchronization

The BASE24-atm Shared 912 EMV Extension has been enhanced to provide the functionality of sending an EMV "PIN Update" script back to the EMV chip card when a “Change PIN” transaction is performed at an ATM. EMV cards have the capability of holding the cardholder’s PIN within the Integrated Chip Card (ICC). This allows “offline PIN verification” (i.e. where the PIN entered by the cardholder is verified by the ICC) to be offered as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals will not be enhanced to offer offline PIN verification, and so “online PIN verification” (i.e., where the PIN entered by the cardholder is verified by the issuer’s system) will continue to be widely used. Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion would result if the PIN held on the card’s ICC differed from the PIN held on the issuer’s system.

Also, if a cardholder exceeds the allowable bad PIN tries when entering the PIN, a mechanism is required to allow the bad PIN try counter on the card’s ICC to be reset. Therefore, it is critical that card issuers provide offline PIN management functionality if they support “Change PIN” transactions. The Shared 912 EMV Extension has been modified to send an EMV PIN Update script down to the ATM/card, if a PIN Change transaction is performed by an EMV card, and to support an EMV PIN Unblock script.

In addition, the Diebold EMV Extension has been enhanced to handle error conditions resulting from the above two situations. These error codes are carried in the Issuer Script Results data in either an explicit Solicited Status Message or in the Last Transaction Status buffer of the next request message. The Device Handler then examines this data and if the script results indicate that a script has failed, the Device Handler makes a decision as to whether a reversal is appropriate.

For more information on EMV processing, please refer to “EMV Support” in the BASE24 Base Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 2).

Prerequisites: The Diebold 10xx/478x Device Handler and the Shared 912 EMV Extension modules are required.

Shared NDC+ AKDS Certificate Support

BASE24-atm has been enhanced to support Automated Key Distribution (AKDS) functionality in those cases where the ATM software vendor utilizes the Shared NDC+ Remote Key Load (RKL) specification to facilitate remote key downloading to ATMs equipped with certificate-based encrypting PIN pads (EPPs). This specification is managed and used by Wincor Nixdorf, although it is open to all vendors.

AKDS automates the otherwise manual process of initializing and re-keying ATM master keys by eliminating the manual processes related to key generation, dual custodianship and manual key loading at the ATM. AKDS also facilitates compliance with "unique key per ATM" guidelines specified by ANSI X9 TG-3 key management standards mandated by networks and card associations. The AKDS key distribution protocol uses public-key cryptography (RSA encryption) to generate and transport the initial master key (the "A-key"). Use of digital certificates and digital signatures is also used for authentication and to protect against tampering of data.

If AKDS has not been previously purchased, AKDS is a separately billable module.

Prerequisites: The NCR 5xxx Device Handler and the Shared NDC AKDS modules are required. BASE24-es TSS version 05.4 or greater will be required.

Shared NDC+ EMV Upgrade

BASE24-atm has been enhanced to support EMV-capable devices running in NDC+ emulation according to the Wincor Nixdorf NDC/Diebold 91x Message Format Extension for EMV. The previous version supported only the older protocol defined in the Wincor Nixdorf ProCash/NDC EMV Programmers Reference for NDC+ devices. The new specification offers functionality such as multiple-language support that is unavailable using the previous specification. Because Wincor's EMV specification is open to other vendors, it is anticipated that other vendors will support this new version as well. The BASE24-atm Diebold 10xx Device Handler Shared EMV module already supports the newer Shared EMV specification.

For more information on EMV processing, please refer to "EMV Support" in the BASE24 Base Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 2).

This functionality is provided automatically as part of the Shared NDC EMV Module. If the Shared NDC EMV Module has not previously been purchased, it is a separate billable module.

Prerequisites: The NCR 5XXX Device Handler and the Shared NDC EMV module are required.

Simultaneous Transactions Fraud Enhancement

BASE24-atm has been enhanced to re-check usages when an approved transaction returns from the host system. If usages are exceeded when the transaction usages are checked when returning from the host, the transaction will be denied and an EMS event message is generated. This change is being made to prevent exceeding usages when two transactions for the same account occur at two or more ATMs at the same time.

Triton Mobile Top-Up

This BASE24-atm enhancement provides additional functionality to the Triton Device Handler. In addition to the standard ATM transaction set, the Triton Device Handler is now capable of supporting mobile top-up transactions. This enhancement does not include support for the BCGI mobile top-up scheme (messaged-based model). Only the definitions-based model of mobile top-up will be supported by the Triton Device Handler.

The BASE24-atm Triton Mobile Top-Up Device Handler extension enables cardholders to replenish, or top up, their mobile telephone accounts at an ATM. The replenishment transaction is a non-currency dispense transaction. The institution must purchase both the Non–Currency Dispense add-on product and the Triton Mobile Top-Up Device Handler extension to offer mobile top-up services. A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction.

For more information on Mobile Top-up processing, please refer to “Mobile Top-up Integration” in the BASE24-atm Optional Enhancements of the BASE24 Release 6.0 Implementation Guide (Version 5).

The above enhancement is considered a billable item. Please contact your Customer Manager for further information.

Prerequisites: The Triton Device Handler and the Non-Currency Dispense module are required.

BASE24-pos Enhancements

The BASE24-pos product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24-pos product, refer to Section 6.

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General Enhancements

The following BASE24-pos enhancements are inherent in the release 6.0 version 7 offerings. All BASE24-pos users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

General Changes

Add Additional Values to PTD

New default values have been added to the POS Terminal Definition file (PTD) which will allow retailers to better identify the capabilities of the terminal. New indicators were added related to terminal input capability, terminal EMV capabilities, and terminal authentication capability.

Disallow Leading Zero Approval Code

The BASE24-pos Router/Authorization logic that creates an approval code has been enhanced to manipulate the generated value to ensure that the new approval code does not contain one or more leading zeros. American Express requires that all approval codes start with a non-zero digit.

PTDD2 Record Cleanup Enhancement

The BASE24-pos file maintenance system has been enhanced to delete the terminal's POS Terminal Dynamic Data scratch pad (PTDD2) record, in addition to the terminal definition records currently deleted, when the operator presses the F4 (delete record) key. In order to provide an audit of the deletion, an Online Maintenance File (OMF) record will be logged for the deleted PTDD2 record. The OMF record will contain only the OMF header; it will not contain an image of the PTDD2 record. If the operator presses the F3 (add record) key, no PTDD2 record will be added. If the operator presses the F5 (update record) key, the PTDD2 record will not be updated. The adding and updating of PTDD2 records is handled dynamically by the device handlers during transaction processing.

Previously, when a terminal's definition file records were deleted, the file maintenance system deleted the terminal's PTDD1 and PTDS1 records, but did not delete the terminal's PTDD2 record. This could lead to unnecessary disk space usage and potential problems with sensitive cardholder information being left on the system unnecessarily.

AMEX SE Number Additional Validation

The POS Retailer Definition File (PRDF) AFT processing modules have been enhanced to validate the contents of the Alternate Merchant Id field containing the American Express SE Number (ten numeric digits) and to validate the check digit.

APACS MACing

The BASE24-pos APACS Device Handler has been enhanced to support APACS Message Authentication Code (MACing) processing through BASE24-es TSS.

The APACS Device Handler previously supported MACing by communicating directly with the Thales Hardware Security Module (HSM), but will now utilize TSS to perform its MACing functions. Newly supported TSS MACing messages include:

- Verify MAC on incoming confirmation message from terminal
- Generate MAC and new key register value on response message to the terminal
- Verify MAC on transaction request without PIN and generate MAC residue
- Verify MAC on transaction request with PIN and generate Terminal PIN Key and MAC residue
- Verify MAC on administration request and generate MAC residue
- Key register reset

As part of this enhancement, BASE24-es Transaction Security Services (TSS) has also been modified to support the required APACS MACing commands.

Prerequisites: BASE24-es TSS version 06.2 or greater will be required.

APACS Support for Triple DES

As a side benefit of the APACS MACing enhancement and the ability to use TSS with the BASE24-pos APACS device handler it brings, BASE24 users will now be able to support Triple DES PIN and MAC encryption between the BASE24-pos APACS device handler and an APACS70 compatible terminal. Triple DES support at POS for PIN block encryption is being mandated by Visa and MasterCard.

Prerequisites: BASE24-es TSS version 06.2 or greater will be required. It is an APACS requirement that if Triple DES is used for encryption it must be used for both PINs and MACs.

Institution Level Security for CRT Auth

This enhancement will allow for institution level security, so that an unauthorized CRT Auth operator does not have access to another institution's CRT Authorization files.

The BASE24-pos CRT Authorization product has also been modified to support the configuration of CRT Authorization device handler processes in multiple logical networks when those logical networks share a common Pathway environment.

Service Code Validation Enhancement

The purpose of this enhancement is to provide better detection of counterfeit cards. In card-read transactions, the card is authenticated by validating the Card Verification Digits encoded in the track data (CVD1). For the algorithm used to generate the CVD1, the input data includes the Service Code, also encoded in the track data. In manually-entered transactions, the card is authenticated by validating the Card Verification Digits printed or embossed on the card itself (CVD2). For the algorithm used to generate the CVD2, the input data includes a value of "000" in place of the Service Code, which is typically not available in manually-entered transaction data.

A counterfeit card can be created by generating fraudulent track data, which includes the CVD2 printed or embossed on the card and a Service Code value of "000". The authorizing system may successfully validate the CVD2 (using the Service Code of "000"), believing it to be the CVD1. This means that the card is assumed to be valid when in fact it is fraudulent. To detect these fraudulent cards, the BASE24-pos Router/Authorization module has been modified to reject transactions which contain a Service Code of "000" in track data in card-read transactions.

SPDH & RTAU Enhancements for Visa and MasterCard

BASE24-pos has been enhanced to support a number of business enhancements implemented by Visa and MasterCard. These enhancements include auto-substantiation transactions for healthcare and transit purchases (Visa), healthcare eligibility inquiry transactions (Visa), card level results (Visa), and recurring payment cancellations (Visa and MasterCard).

Auto-Substantiation is the process of verifying that purchase transactions are for expenses permitted under Internal Revenue Service regulations for Flexible Spending Accounts (FSAs) and Healthcare Reimbursement Arrangements (HRAs). Healthcare auto-substantiation transactions enable employers and their third-party healthcare service providers to approve qualified medical expenses at the point of sale for purchases made with FSA and HRA payment cards at participating retailers.

Additionally, participating retailers that sell transit fare media, such as commuter passes, parking passes, and mass transit vouchers and tickets, may also use auto-substantiation transactions to approve purchases made with FSA payment cards.

Partial authorizations are supported for auto-substantiation transactions. Like other prepaid or pre-funded cards, it is difficult for consumers to spend the exact amount available in FSA or HRA accounts. Partial authorizations allow consumers to make purchases that exceed the account balance, with the remainder of the purchase paid by other means (credit card, cash, etc.)

The healthcare/transit auto-substantiation transaction does not require a new transaction type to be defined in BASE24-pos. Rather, the existing financial transaction set is utilized, with healthcare/transit data carried in the Healthcare/Transit Token.

Healthcare eligibility inquiry transactions enable healthcare providers (doctors, dentists, hospitals, etc.) to immediately determine the healthcare insurance coverage of FSA or HRA payment card holders.

The healthcare eligibility inquiry transaction does not require a new transaction type to be defined in BASE24-pos. Rather, the balance inquiry transaction is used, with healthcare eligibility data carried in the Healthcare/Transit token and the Healthcare Eligibility token. Eligibility information for up to five healthcare services may be requested in a single transaction.

Additionally, BASE24-pos has been enhanced to return Visa’s card level results to the acquirer, to return notification of recurring payment cancellations to the acquirer and populate the approval code with Visa’s card level results (if available) when BASE24-pos acts as the card issuer.

SPDH New Fields to Support Network Requirements

MasterCard, American Express (CAPN) and EMV require new data to be passed to and from the merchant. This enhancement adds support for new fields in the ACI Standard POS device handler (SPDH) and new tokens to carry new network data.

The SPDH was enhanced to support additional data in transaction requests bound for the American Express Card Acceptance and Processing Network. This additional data consists of Card Not Present (ITD) data from merchants in mail, telephone, and internet order industries, and Airline Passenger Data (APD) from merchants in the airline industry.

Changes were also made to rename FID 6 SubFID N in SPDH messages from Reason Online Code to Message Reason Code. This includes support for ISO message reason codes 1004 through 1007 for EMV authorization and response code processing. And, to satisfy Interac requirements to return encrypted account balances to the terminal using 32-byte data encryption key, an enhancement was made to expand FID I (Data Encryption Key) and DID j (Data Encryption Key) to support 32 and 48 byte keys.

Visa DUKPT MACing

This enhancement is being provided to customers to address a situation in the initial BASE24-es TSS implementation of DUKPT Message Authentication Code (MAC) processing whereby Atalla and BASE24-pos were unable to support DUKPT MACing in the Scandinavian market. The Atalla MACing commands in use at the time did not provide sufficient flexibility, and as a result, did not return the appropriate MAC values. At ACI's request, Atalla enhanced their software by modifying their new DUKPT MACing commands to also include support for the Visa DUKPT algorithm used by the older Atalla 5C command. With this new Atalla command available, ACI made the corresponding changes in BASE24-pos to use this new functionality.

This enhancement provides the functionality required by Scandinavian customers implementing an older, pre-X9.24 algorithm ("Visa UKPT") where it is mandated. The standard product direction is to support X9.24-1 DUKPT (single DES and triple DES), ISO-9797-1 Algorithm 1 and ISO-9797-1 Algorithm 3. This enhancement and the enhancement made by Atalla only address Scandinavian DUKPT MACing scenarios between the device and BASE24.

It should be noted that as part of this enhancement, BASE24-es TSS has also been enhanced.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required.

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Optional Enhancements

The following BASE24-pos release 6.0 version 7 enhancements are optional. BASE24-pos users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Contactless Chip/Magnetic Stripe Support

This enhancement to BASE24-pos authorization facilitates supporting MasterCard's PayPass and Visa's Visa Wave programs. Chip Cards issued under these programs use RFID technology and can be "waved" over a sensor, eliminating the traditional swipe or insertion of the card.

This enhancement supports dynamic card verification used by the PayPass CVC3 (dynamic) scheme, and the Visa Wave MSD scheme. For MasterCard (or PayPass), the dynamic CVC3 scheme creates a data stream identical to a track 1 or track 2 transaction, except that a dynamic cryptogram (CVC3) generated by the card is placed in the Discretionary Data. Similarly, the Visa Contactless MSD scheme uses a dynamic card verification value (dCVV) rather than the traditional CVV used by magnetic stripe transactions.

Other contactless schemes based on EMV are also available. These include MasterCard's MChip4 Lite, and Visa's VSDC and qVSDC schemes. Support for these schemes is available with a BASE24 EMV license. (See related EMV Product Descriptions)

The dynamic card verification enhancement is considered a billable item. Modifications to the EMV enhancement to support PayPass and Visa Wave are available as a maintenance change to this enhancement. Please contact your Customer Manager for further information.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required.

Correct Declined Advices

A project was done in BASE24 Release 6.0 version 5 (November, 2004) to send declined advices to the Host when BASE24 or an interchange stood in for the Host and declined the transaction. This functionality was added to both BASE24-atm and BASE24-pos.

A code change was made to this functionality in January, 2005 that has the affect of logging all declined advice/force post transactions to the PTLF before an attempt is made to find the CPF. This causes problems for transactions where a CPF record could have been found and routed to the appropriate destination. It also means that records are logged to the PTLF with no FIID information.

A declined advice/forced post is a transaction with a PSTM message type of 0220 and a response code greater than or equal to 050. These messages should be forwarded by Router/Authorization to the host with their original response code as sent in from an interchange. If a declined advice/forced post transaction cannot be processed (for example the CPF, IDF or IPCF record cannot be found), the transaction will be marked by Router/Authorization as an exception and not forwarded to a host, since it will not be possible to determine where to send the transaction. The transaction will be logged to the PTLF with the original response code but marked as an exception.

CVD/CVD2 Checking in a Single Request

BASE24-pos has been enhanced to support the validation of CVD1 and CVD2 in the same transaction, for card types in addition to AMEX. Logic has been added to ensure that the card verification results are recorded in the correct field for card-read and manually entered transactions. This enhancement allows issuers to decline manually entered transactions where CVD2 is not present, and to configure the validation of CVD2 separately to CVD1.

Today BASE24-pos supports the validation of card verification digits (CVD) that are held in the track data (CVD1) and on the signature panel (CVD2). MasterCard requires that both of these values be validated if present in a single authorization request. BASE24-pos has already been enhanced to store the results in separate tokens, but it was believed that CVD1 would be present in card-present transactions, and that CVD2 would be available for card-not-present transactions. Due to increasing fraud, card-present merchants are prompting for the CVD2 value to be entered at the POS terminal.

EMV CCD Issuer Support

EMVCo has produced CCD (Common Core Definition), which provides the basic EMV data definitions, and CPA (Common Payment Application), which define the behavior of the EMV application. With the Visa April 2005 Business enhancement, Visa announced requirements for EMV CCD (Common Core Definition).

Enhancements have been made to BASE24-pos to support EMV CCD for issuer processing. The main impacts are in the following areas: identifying CCD cards; processing the CCD cryptogram version number (CCD identifier); passing the required data (to TSS/HSMs) for ARQC verification/ARPC generation; supporting the new format of the CVR; and supporting the CSU data. Any Interface changes will be completed as part of a future project. Support for CPA will be a separate future project.

Prerequisites: BASE24-es TSS version 06.4 or greater will be required

Hypercom Software MACing Support

The POS Terminal Data Files Maintenance Server (SVPTDTS) has been enhanced to allow configuration of MACing in software and PIN management in hardware for the same terminal.

Hypercom terminals use a proprietary MACing algorithm that is supported by the BASE24-pos Hypercom device handler (HPDH) in software but not hardware (i.e., a host security module). Some customers who perform PIN encryption in hardware using Transaction Security Services (TSS) requested the capability to enable software MACing in their Hypercom terminals.

For Hypercom terminal types, this enhancement allows SVPTDTS to support MAC encryption in software while at the same time allowing PIN encryption through a security device and TSS. This combination of MAC type and PIN encryption type remains prohibited for all other terminal types.

Improve EMV Transaction Certificates in SPDH

The ACI Standard POS device handler (SPDH) has been enhanced to allow the terminal to send an EMV log-only cancellation message to reverse an initial EMV log-only request.

Log Stored Value Dormancy/Escheatment Transaction to PTLF

This enhancement enables BASE24-pos to be configured to log dormancy and escheatment stored value transactions to the POS Transaction Log File (PTLF). Dormancy and escheatment stored value transactions are currently only logged to the Stored Value History File (SVHF). This enhancement will provide a key component for an integrated Stored Value solution by enabling settlement and reconciliation data to more easily be provided to ACI Payments Manager.

BASE24 Interchange Interface Enhancements

All BASE24 Interchange Interface product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Interchange Interfaces, refer to Section 9.

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General Enhancements

The following BASE24 Interchange Interfaces enhancements are inherent in the release 6.0 version 7 offerings. All BASE24 Interchange Interfaces users must review these enhancements in order to ensure an understanding of the new processing and proper configuration within the BASE24 environment.

SPDH & RTAU Enhancements for Visa and MasterCard

The following BASE24 Interchange Interface enhancements have been made in conjunction with SPDH and RTAU enhancements to support Healthcare/Transit Auto-substantiation transactions and Healthcare Eligibility transactions.

Healthcare/Transit Auto-substantiation and Healthcare Eligibility transactions were added to the VisaNet Interface initially, and now other networks have added similar processing. ACI has taken this opportunity to normalize processing by altering the initial token support used for these transactions. The Healthcare Token (CP) will no longer be used and has been replaced by the Healthcare/Transit token (CV) and the Healthcare Service token (CW).

The new tokens will be able to provide eligibility information for up to 5 healthcare services in a single transaction and can be used by multiple interchange interfaces. As of version 7, the VisaNet, Banknet, Visa DPS, and Plus ISO/Interlink Interfaces will be using the new Healthcare/Transit and Healthcare Service tokens.

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Optional Enhancements

The following BASE24 Interchange Interface release 6.0 version 7 enhancements are optional. BASE24 Interchange Interface users should review these changes to ensure an understanding of the new processing and proper configuration within the BASE24 environment before deciding to implement any of the changes.

Contactless Chip/Magnetic Stripe Support

Multiple interfaces have been enhanced to support the acquiring and authorizing of Contactless Chip/Magnetic Stripe transactions by BASE24 or network members. These changes allow for better recognition and reporting of contactless transactions. Several interfaces have been modified to set the POS Entry Mode in the POS-DATA1-TKN from the inbound message and to read the same value when constructing the outbound message.

The following interfaces have been modified:

- Banknet
- EuroPay
- MAC MASM
- MDS Maestro
- NYCE ISO
- STAR ISO
- VisaNet

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BASE24 Value Added Products

All BASE24 value added product enhancements are divided into the following groups:

- General Enhancements
- Optional Enhancements

For more information about the enhancements in the BASE24 Value Added Products, refer to Section 10.

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General Enhancements

There are no general enhancements affecting the following BASE24 value-added product users:

BASE24-card
BASE24-customer service
BASE24-InfoBase
BASE24-inventory Voucher Manager

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Optional Enhancements

There are no optional enhancements affecting the following BASE24 value-added product users:

BASE24-card
BASE24-customer service
BASE24-InfoBase
BASE24-inventory Voucher Manager

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3: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- Product consulting and project management
- Publications
- Education
- HELP24
- ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned; please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 release 6.0 version 7. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading, after reviewing all materials.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to release 6.0 version 7. Information contained in this guide provides insight into required planning and control of the entire project.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user.

1. IDENTIFY PROJECT TEAM

   One of the initial tasks for the BASE24 upgrade is for the user to assign a project team who will be responsible for the success of the project. A project manager and team members will need to be assigned and specific responsibilities established.

2. DETERMINE HARDWARE REQUIREMENTS

   An ACI sales representative can assist in determining whether additional hardware will be required for the BASE24 upgrade. It is the BASE24 user's responsibility to order the hardware and establish the delivery date with HP NSK. HP NSK will perform diagnostic testing on the hardware when it is installed at the site. Should hardware encryption be required, it will be necessary to determine the hardware requirements. Additional hardware should be ordered for installation before testing can occur.

   If applicable, the BASE24 user should order any new device firmware/software required to implement the enhancements they have chosen to support. The delivery date for the device firmware/software should be agreed upon with the vendor, as this time frame will affect the entire project.

3. IDENTIFY COMMUNICATION REQUIREMENTS

   If the BASE24 user has identified additional protocol requirements, the communications lines and modems should be ordered. The customer must coordinate this effort with the local telephone company to ensure that the lines are configured in the least costly manner. The lines and modems must be installed and tested prior to the scheduled software installation date.

4. UPDATE HP NSK SOFTWARE REQUIREMENTS

   ACI recommends that the BASE24 user be on a current HP NSK supported operating system to obtain maximum performance and support. BASE24 release 6.0 requires a minimum of HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46. In addition, BASE24 release 6.0 requires TMF for several files. Customers choosing to implement BASE24-es Transaction Security Services (known as TSS) with BASE24 release 6.0 must review the BASE24-es Hardware and Software Requirements for the BASE24-es release they will deploy.

5. UPGRADE TO XPNET 3.0 or 3.1

   Customers should note that XPNET 3.0 is the minimum requirement for BASE24 release 6.0.

6. PROJECT SCOPE MEETING

   ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 7.

7. MIGRATION PLANNING

   Users will need to use this document, and any references within it, for detailed analysis when creating their migration plan.

8. DEVELOP PROJECT SCOPE DOCUMENT

   ACI does not believe this step is necessary for customers upgrading to BASE24 release 6.0 version 7.

9. DEFINE REQUIREMENTS DEFINITION

   The customer will be required to review existing CSMs to determine which ones will require retrofitting to the new version. With ACI's Fix Release methodology, BASE24 users should evaluate their CSMs to determine if they are handled as standard product in release 6.0 version 7 or if they are specific to the customer's environment and will need to be upgraded to the new version.

10. DEVELOP PROJECT PLAN

    The customer's project manager is responsible for developing an internal project plan. A review of the ACI project guidelines will help this person identify job responsibilities and establish time frames. A tracking mechanism should be included in the plan to ensure that the project is proceeding on schedule.

11. IDENTIFY CUSTOMER SPECIFIC MODIFICATION DEVELOPMENT

If the customer elects to have ACI develop modifications identified as required for the new version, a request will be submitted to ACI Software Development for a cost estimate. The functional descriptions and estimates will be sent to the customer for review and approval. Resources will be assigned and the final delivery dates established upon receipt of the signed contract.

Should the customer decide to develop any modifications in-house, resources will need to be assigned and scheduled. Once delivery dates are established, impacts to system testing can be determined.

12. DEFINE CERTIFICATION REQUIREMENTS

Customers currently on Fortify.3 will need to verify with each of the interchanges they connect to, to determine if they need to re-certify with the Interchanges with release 6.0 version 7 code.

If re-certification is required, preliminary test times should be determined and a copy of the interchange test script secured.

The customer needs to set up test time with the interchange well in advance in order to insure time frames are met.

13. ESTABLISH TEST FACILITY

It is recommended that an area be designated specifically for testing the BASE24 release 6.0 version 7 software. The following items are recommended for the test area:

- 3 PCs
- Test devices
- Hard copy printer
- Test cards

14. INSTALL PRODUCTION HARDWARE

Installation of any additional computer hardware or peripherals is generally performed by the local hardware service organization or customer staff.

15. ATTEND BASE24 EDUCATION CLASSES

ACI offers a full suite of instructor led courses for BASE24. An enhancement course is also available upon request. The course descriptions and outlines can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

16. DEVELOP INTERNAL PROCEDURES

There are no new internal procedures that ACI is aware of for BASE24 release 6.0 version 7 enhancements. User/Operational impacts are noted in the detailed sections for each enhancement described within this document.

17. DEVELOP SYSTEM TEST PLAN

The customer is responsible for developing a system test plan. Included in the test plan should be a description of each step specifying the test card to be used, the account to be used, the transaction to be executed, and the expected results. A procedure for retaining the results of the transaction processing should also be specified. Controlled testing and result verification procedures are key elements for successful system testing.

The execution of the system test plan will occur over a period of weeks to fully test all aspects of the system. BASE24 product functionality should be the first application tested. Any system modifications should then be thoroughly tested. The interface between the host and BASE24 should also be tested. This includes testing on-line authorization and the refresh and extract programs.

18. DEVELOP HOST SOFTWARE

During the course of a business day, BASE24 logs all completed and declined transactions to a transaction log file. After network settlement, the transaction log file can be extracted and used as input into the customer's nightly processing cycle on the host. If the customer would like to take advantage of the new features in release 6.0 version 7, a host program will need to be developed to interpret the new token data that can be extracted with release 6.0 version 7.

After the nightly processing cycle on the host or periodically during the business day the host may need to update card and account files used by BASE24 for pre-authorization or stand-in processing. If the customer would like to take advantage of the new features in release 6.0 version 7 in these files, a host program will need to be developed to initialize the new data fields in BASE24 files.

19. INSTALL BASE24

Release 6.0 version 7 software will need to be installed at the customer site. Licensed release 6.0 version 7 software modules will be provided to the customer upon request for installation on their system.

20. CONVERT DATA FILES

ACI provides database conversion programs for selected BASE24 files. These files are discussed in the detail sections for each enhancement described within this document.

21. CERTIFY INTERCHANGE

This task involves implementing the interchange test script and receiving notice of certification from the interchange. Current BASE24 users should not have to re-certify their Interchange Interfaces, unless they are implementing new functionality not previously tested or as directed by the interchange.

22. PERFORM SYSTEM TESTING

Full business day cycle testing of the system should be performed as determined by the customer testing plan and procedures.

23. DEVELOP CONVERSION PLAN

The customer should develop an implementation plan for converting existing devices and interchange links to release 6.0 version 7. The installation of new devices and interchanges should also be included in this plan. A method of informing users of any changes to existing procedures should also be developed. The conversion schedule should be routed to all departments.

24. GO-LIVE

At the customer's request, an operational support analyst will visit the site one day prior to the "go-live" date to review the current software status and the BASE24 facilities. This individual will remain onsite to assist the customer with production cutover. The migration plan developed by the customer will be utilized in production cutover. On-site go-live support is a billable service.

25. COMPLETE POST IMPLEMENTATION REVIEW

In general, the purpose of this meeting is to ensure overall satisfaction of the customer’s project.

26. TURN OVER TO CUSTOMER MANAGEMENT

If an ACI Project Manager has handled the BASE24 upgrade, a formal turnover meeting is held between the ACI Project Manager and the ACI Customer Manager to ensure a smooth transition in turnover of responsibilities for on-going customer support.

Publications

The BASE24 documentation set has been updated to include the release 6.0 version 7 enhancements. The updated BASE24 documentation is available in three formats.

- **HELP24 eSupport InfoLink.** All files for the manuals in the BASE24 6.0 documentation set are located on the HELP24 eSupport InfoLink database. You have the option of copying the files for individual manuals from the HELP24 eSupport InfoLink database to your PC.

  In addition, the HELP24 eSupport InfoLink database contains documentation that is added or revised between ZIP files or CD-ROMs.

- **Software Management Electronic Distribution.** ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI electronic distribution system.
- **CD-ROM.** The CD-ROM contains all PDF files for the manuals in the documentation set.

The BASE24 6.0 documentation set contains the most current version of each document. If a BASE24 manual has not been updated for release 6.0, the manual for the prior release appears on the release 6.0 menu with a note identifying the prior release.

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one- to four-day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 6.0 version 7.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

Please use ACI Worldwide's eSupport site to report problems or ask questions about BASE24 release 6.0 version 7. eSupport is an Internet Web based application that interfaces with the HELP24 Clarify database. Using eSupport, you can search our KnowledgeBase Solutions for information and resolutions to problems you are experiencing. You can also directly open Clarify cases. An eSupport option, InfoLink, is your link to a wealth of information that can help you in your day to day business. It allows you on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. Contact your local Help24 office for an eSupport logon.

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows us to troubleshoot issues in a secure, online environment. Compared to the dial-up environment we have used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows you to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from your PC to ours, or vice versa. This feature can be used in the event the file(s) are too large to email.

For more information, please refer to the WebEx User's Guide located under Product Info on Infolink. Contact your local HELP24 office to schedule your WebEx demonstration.

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are example descriptions of the various services ASTech could undertake to assist you during the BASE24 upgrade. The number of staff weeks identified to accomplish these tasks is estimated and will vary depending on the amount of customer specific code to be developed and delivered by ACI.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Effort: Two to four staff weeks

ASTech personnel will perform a complete review of the customer's BASE24 software system and determine what current modifications (CSM code) exist at the customer location. ASTech personnel will perform a detailed comparison of existing customer specific software modifications to release 6.0 version 7 functions. Upon completion of the review, the customer will be provided with a document that identifies all modules that contain code unique to their BASE24 system (file names, custom functions, custom function applicability, etc.).

Conversion Planning

Effort: One and one-half to three staff weeks depending on the size and current system configuration.

This service option provides the customer with comprehensive support during the system migration to the new version. ASTech assigns the customer a Technical Project Lead, or Tech Lead from ACI. The Tech Lead will be responsible for ensuring the success of the BASE24 upgrade project.

The Tech Lead will:

Review the migration options and develop a detailed migration plan for the conversions project
Review the Project Scope Document on behalf of the customer (if applicable)
Review the customer's internal procedures to accommodate release 6.0 version 7 functions.

Upgrade Custom Code

Effort: The amount of time necessary for this task varies depending on the number of modules identified as having modifications unique to the customer. Systems with a great number of customizations may take up to sixteen staff weeks. Actual estimates would be provided upon completion of the Customer Software Analysis.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

With this service, ASTech will upgrade all custom code to release 6.0 version 7. This includes unit testing of the code. Customizations that are no longer necessary will be removed. Customizations that must be carried forward can either be upgraded in their current form or be reengineered to take advantage of new functionality in release 6.0 version 7. Reengineering customizations will add time to the estimate.

System Test Planning

Effort: One to four staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products to be tested. Some of this time will be on-site to perform the System Test Plan.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

ASTech personnel will develop and execute a BASE24 System test plan designed for the customer's specific needs. This service also includes the performance and documentation of the system acceptance testing.

Switch and Host Certification Support

Effort: Two to five staff weeks depending on the results of the Customer Software Analysis and the number of BASE24 products and interfaces to be certified. All of this time will be on-site time.

Prerequisite: The Customer Software Analysis to identify all code unique to the customer.

This service option allows ASTech personnel to provide complete support of the various switch and/or host communication certification testing. This includes reviewing and refining the test scripts, performance of certification testing functions, evaluation of the testing results, and documentation and presentation of the results obtained during the certification testing.

Go-Live Support

Effort: One week (on-site).

ASTech personnel will be at the customer site assisting in all conversion tasks and being present for go-live support. This includes remaining on-site for a couple of days to ensure system stability.

Post Go-Live Support

Effort: Varies.

After a successful go-live, a Tech Lead will provide on-going support which may be required by the customer. Tech Lead support can be provided at the customer location and/or remotely through ACI.

4: BASE24 Base

Section 4 discusses the changes and enhancements made to the BASE24 Base modules in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24 users must consider when upgrading from release 6.0 version 6.

Changes and enhancements to the BASE24 Base modules impact all users. Consequently, all users should review the material in this section to determine how it applies to them.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the Base modules changes that affect all BASE24 users, along with the associated conversion and implementation requirements that all BASE24 users must address.
- Optional enhancements describe the Base modules changes that affect only those BASE24 users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 users must address to implement the enhancements.

Section 4 must be used in conjunction with Sections 5 through 10 to identify the changes that BASE24 users need to consider when upgrading to release 6.0 version 7.

Standard Information Format

A standard format is used in Sections 4 through 10 to highlight and describe the various changes, conversion requirements, and implementation requirements that BASE24 users need to consider when upgrading to release 6.0 version 7. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Software Impacts

Describes the software modules impacted for a given change or enhancement and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, only those implementing the enhancement).

CUSTMACS Changes

Describes the CUSTMACS changes required to implement a given change or enhancement. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes the LCONF assign changes required to implement a given change or enhancement. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes the LCONF param changes required to implement a given change or enhancement. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes the data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given change or enhancement.

Template Changes

Identifies file templates that must be created or modified in order to implement a given change or enhancement. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given change or enhancement.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given change or enhancement.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given change or enhancement.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given change or enhancement.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given change or enhancement.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given change or enhancement.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given change or enhancement. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given change or enhancement. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given change or enhancement. This includes NEF changes.

File Conversions

Identifies the files that need to be converted in order to implement a given change or enhancement. This implies that a given file could be impacted by multiple changes and/or enhancements. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given change or enhancement. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract tape formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given change or enhancement. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given change or enhancement.

Reference Documents

Lists any outside sources used in the design of a given change or enhancement. BASE24 users can refer to these documents for additional information.

General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 users must address in order to upgrade to release 6.0 version 7.

General Changes

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described in section 4.

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General Changes

Changes for all BASE24 release 6.0 version 7 enhancements affecting the DDL files and the BASE24 AFT Subsystem files (i.e., screens, requesters, servers) have been merged and cataloged under one SCN for each affected file. Applying changes for individual enhancements to the screen files does not allow BASE24 users to identify which changes go with a particular enhancement, as screen files do not have sequence numbers at the end of each line. Since the screen file changes usually involve new fields being added, all associated requester, server and DDL changes are required. Even though requester, server and DDL files do contain sequence numbers, all SCNs would need to be applied in order to accept the changes to the screen files. Therefore, all requester, server and DDL file changes, for all enhancements, were done under one SCN per file as well.

The Transaction Based Pricing (TBP) reports have been modified to support the Deposit Hold Notification enhancement in BASE24-pos. These changes impact all BASE24 TBP customers. All BASE24 TBP customers must implement the updated version of the TBP report objects and TBP documentation (subvolume BA60TBPD). Changes to the Transaction Based Pricing (TBP) reports requiring an MTAF conversion were introduced with the BASE24 release 6.0 version 5 BASE24-atm Mobile Top-Up Integration enhancement. BASE24 TBP customers who performed this MTAF conversion previously will not convert the MTAF again.

BASE24-atm has been enhanced to support several new tokens and transaction subtypes for the new BASE24-es IFX Device Handler functionality. Currently there are no processes in standard BASE24 product that utilize these tokens.

The following new transaction subtypes have also been added:

ABL0 – Electronic Bill Payment Payee List
ABP0 – Electronic Bill Payment
AIS0 – IFX Interim Statement
AMA0 – Multiple Account Inquiry

The Host Configuration File (HCF) has been modified to include additional fields on screen 1 for the HISO Miscellaneous Changes enhancement. The Host Configuration File screen and requester have been modified to process these new fields. For further information, see **HISO Miscellaneous Changes** later in Section 4.

The Key File (KEYF) has been modified to include additional fields on screen 4 for the HISO Full Message Encryption enhancement. The Key File screen, requester and server have been modified to process these new fields. For further information, see **HISO Full Message Encryption** later in Section 4.

The Card Prefix File (CPF) has been modified to include additional fields on screen 3 for the Contactless Chip/Magnetic Stripe support enhancement, on screen 2 for the

CVD/CVD2 Checking in a Single Request enhancement and on screen 11 for the EMV CCD Issuer Support enhancement. The Card Prefix File screen, requester and server have been modified to process these new fields. For further information, see **Contactless Chip/Magnetic Stripe support** and **CVD/CVD2 Checking in a Single Request** in Section 6 where the impacts of these enhancements are described and **EMV CCD Issuer Support** later in Section 4.

The Cardholder Authorization File (CAF) has been modified to include additional fields on screen 2 and screen 7 for the Contactless Chip/Magnetic Stripe support enhancement. The Cardholder Authorization File screen and requester have been modified to process these new fields. For further information, see **Contactless Chip/Magnetic Stripe support** in Section 6 where the impacts of this enhancement are described.

As part of the Completion Messages Configuration enhancement, the IPCF has been modified to provide for a default value of “0” for the Completion Required field. The ICPF requester and server have been modified to process this change. For further information, see **Completion Messages Configuration** later in Section 4.

Software Impacts

All BASE24 customers:

- Inline changes to BA60DDL.DDLATTKN, DDLFCPF, DDLFHCF, DDLFIPCF, DDLFKEYF, and DDLGISO
- Inline changes to BA60AFT.COBTKN, RQCPFS, RQHCFS, RQIPCFSS, RQIPCFXS , RQKEYFS, SCRNCPFS, SCRNHCF, SCRNKEYF, SECTBL, SVCPFS, SVIPCFTG, SVIPCFTS, and SVKEYFS
- Inline changes to BA60SRC.ATTKNCVS, ATTKNID and SUBTPTBL

All BASE24 Transaction Based Pricing (TBP) customers:

- New executable objects for Transaction Based Pricing (TBP) reports and updates to TBP documentation (subvolume BA60TBPD)

Note: The list of files impacted by the merged changes is included in the **Software Impacts** section of the applicable enhancement(s).

A complete list of the merged files can be found in the SCN file BA60UD0A.SCNMERGE.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support the Completion Messages Configuration enhancement.

<u>Issuer Processing Code File (IPCF)</u>

Refer to the section regarding the **Completion Messages Configuration** enhancement for details.

<u>Token file (TKN)</u>

The following BASE24-atm tokens have been added for this general change:

- The Bill Payment Payee List Token (token ID “AE”) carries bill payment payee information.
- The Bill Payment Confirmation Token (token ID “AF”) is a confirmation token sent by the host containing the payment date, confirmation number, the payee and from which account the bill was paid.
- The Multiple Account Inquiry Token (token ID “AH”) carries account data for multiple accounts instead of a standard BASE24 Account Inquiry for multiple accounts where the account data is stored in the STM.
- The Interim Statement Processing Token (token ID “AJ”) carries statement processing data instead of a standard BASE24 Statement Print transaction where the data is stored in the STM.
- The Custom Response Code Token (token ID “AR”) carries a custom response code. When this token is sent from the host, it will contain an IFX response code. When it is sent from the IFX device handler, it will contain the IFX reversal reason code.

If the tokens for this general change should not be logged to the BASE24-atm Transaction Log File (TLF), the appropriate token records must be configured in the Token file.

If the tokens for this general change are to be extracted with the TLF or sent to the host, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Cardholder Authorization File (CAF)

For customers converting from release 5.3 to release 6.0 version 7, the CAF conversion program must be run to convert the CAF. Running the conversion program is required for any BASE24-pos customer. Users must create the new CAF file using the BADDLFUP file.
No separate conversion program was created to move the CAF from version 6 to version 7. Customers must do a full file refresh using the new CAF layout. This will initialize the new ATC fields to zero.

The CAF conversion program on the BA60UC04 subvolume can be used to convert the CAF records from the release 5.3 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVCAFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#cafcnv.t,name $cafm,nowait/ cnvcafm –force –list –listreplace

Release 5.3 users should update the CNVCAFR obey file on the BA60UC04 subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Running the conversion program is required for any BASE24-pos customer. Users must create the new CPF file using the BADDLFUP file.

The CPF conversion program on the BA60UC04 subvolume can be used to convert the CPF records from the release 5.3 format to the release 6.0 version 7 format. The CPF conversion program on the BA60UC0A subvolume can be used to convert the CPF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVCPFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#cpfcnv.t,name $cpfm,nowait/ cnvcpfm –force –list –listreplace

Release 5.3 users should update the CNVCPFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVCPFR obey file on the BA60UC0A subvolume with the appropriate information. The CNVCPFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0A on the BA60UC0A subvolume for release 6.0 version 6 customers.

Host Configuration File (HCF)

The HCF conversion program must be run to convert the HCF. Running the conversion program is required for any BASE24 ISO Host Interface (HISO) user. Users must create the new HCF file using the BADDLFUP file.

The HCF conversion program on the BA60UC04 subvolume can be used to convert the HCF records from the release 5.3 format to the release 6.0 version 7 format. The HCF conversion program on the BA60UC0A subvolume can be used to convert the HCF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVHCFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#hcfcnv.t,name $hcfm,nowait/ cnvhcfm –force –list –listreplace

Release 5.3 users should update the CNVHCFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVHCFR obey file on the BA60UC0A subvolume with the appropriate information. The CNVHCFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0A on the BA60UC0A subvolume for release 6.0 version 6 customers.

Key File (KEYF)

The KEYF conversion program must be run to convert the KEYF. Running the conversion program is required for any BASE24 ISO Host Interface (HISO) user. Users must create the new KEYF file using the BADDLFUP file.

The KEYF conversion program on the BA60UC04 subvolume can be used to convert the KEYF records from the release 5.3 format to the release 6.0 version 7 format. The KEYF conversion program on the BA60UC0A subvolume can be used to convert the KEYF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVKEYFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#keycnv.t,name $keym,nowait/ cnvkeyfm –force –list –listreplace

Release 5.3 users should update the CNVKEYFR obey file on the BA60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVKEYFR obey file on the BA60UC0A subvolume with the appropriate information. The CNVKEYFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0A on the BA60UC0A subvolume for release 6.0 version 6 customers.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3, release 6.0 version 4 and release 6.0 version 5 customers should refer to the ***Conversion Program Guidelines*** within Section 1 of this document.

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing the use of the new BASE24-atm tokens. Modifications to existing files and processing impact the External Interfaces and Super Extract Process.

External Interfaces

Host code changes are required if a customers chooses to support the use of the new BASE24-atm tokens.

Super Extract Process

If the new tokens for this general change are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

System Limitations

BASE24 users implementing the use of the new tokens for this general enhancement must be aware of their internal system limitations for messaging to the STM and TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the Base modules enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements. These enhancements are the following:

Allow Unsorted Data to be Loaded by Refresh

EMV CCD Issuer Support

HISO Full Message Encryption

HISO Miscellaneous Changes

Modify Discretionary Data Suppression Enhancement

BASE24 users can choose whether or not to implement the above enhancements. Users who plan to implement these particular enhancements must address the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements. Users who do not plan to implement these particular enhancements can review the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements for future reference.

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Allow Unsorted Data to be Loaded by Refresh

This section describes the implementation requirements associated with the Allow Unsorted Data to be Loaded by Refresh enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Allow Unsorted Data to be Loaded by Refresh enhancement.

The Refresh process has been modified to allow refresh input files to contain unsorted data when performing full or partial refreshes of the Card Authorization File (CAF) and Positive Balance File (PBF) files. The refresh input file is considered to be unsorted if the primary key for each record within the file, consisting of the Primary Account Number (PAN) and member number, is not in ascending order. Previously, EMS event messages would be raised and the Refresh process would stop processing when it encountered unsorted data.

Software Impacts

All BASE24 customers:

- Inline changes BA60REFR.REFRG and BA60REFR.REFRS
- Updated event (log) message manual BA60LOGM.BAREFR
- Updated documentation files BA60MISC.LCONFBA and BA60MISC.BAMISCTD

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support these changes. Details of the new parameters are provided in the LCONFBA file located on the BA60MISC subvolume.

REFR-INPUT-DATA-UNSORTED

This param specifies whether or not the Refresh process will allow refresh input files to contain unsorted data when performing full or partial refreshes of the Card Authorization File (CAF) and Positive Balance File (PBF) files. The refresh input file is considered to be unsorted if the primary key for each record within the file, consisting of the Primary Account Number (PAN) and member number, is not in ascending order. This is an optional param.

Valid values are:

- "Y" (yes, allow unsorted data)
- "N" (no, do not allow unsorted data)

If the param is not found or is invalid, the default value of "N" is used.

REFR-MAX-RECS-UNSORTED

This param specifies the number of records in the refresh input file. The File Utility Program (FUP) uses this value to determine the size of the scratch file used by the SORT process. This value does not need to be exact, but it should be equal to or greater than the actual number of records in the refresh input file. This is an optional param, used only when the REFR-INPUT-DATA-UNSORTED param is set to "Y" (yes, allow unsorted data).

Valid values are 1 through 2147483647. If the param is not found, the default value of 10000 is used. If the param is invalid, the Refresh process will abend.

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

If the value of REFR-INPUT-DATA-UNSORTED is not "Y", the host application must supply BASE24 with CAF and PBF data that is sorted. Otherwise, the host has the option of supplying BASE24 with CAF and PBF data that is unsorted.

User/Operational Impacts

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

There will be a negative performance impact on the Refresh process if the REFR-INPUT-DATA-UNSORTED param is set “Y” (yes, allow unsorted data). This is due to a scratch file being created and the sorting of the unsorted data in the refresh input file.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

EMV CCD Issuer Support

This section describes the implementation requirements associated with the EMV CCD Issuer Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV CCD Issuer Support enhancement.

Section 5 contains BASE24-atm specific requirements associated with the EMV CCD Issuer Support enhancement.

Section 6 contains BASE24-pos specific requirements associated with the EMV CCD Issuer Support enhancement.

This enhancement is provided to support the CCD implementation as defined in version 4.1 of the EMV specifications. BASE24 has been enhanced to support the functions described in this section.

CCD Issuer Application Data Format

In order to include CCD as a valid Issuer Application Data Format, the EMV IAD Format field on the CPF has been enhanced to include CCD as an option.

New cards will be issued containing a CCD-compliant application. Therefore, there is likely to be a period of time when an issuer has some CCD-complaint cards and some cards containing another application, e.g. VIS or M/Chip, in circulation. This means that cards with the same prefix could contain a different EMV application. In order to ease migration to the new application, an “alternate” EMV IAD Format field has been added to the CPF. The actual IAD format applicable to a specific transaction will be derived from the transaction data.

The following table shows how the Issuer Application Data (IAD) format on cards compliant with specific applications can be identified from the transaction data.

Application (IAD Format)	IAD Length	IAD Byte 1	IAD Byte 2
VIS	>= 7	06	N/A
M/Chip 2.1	>= 8	N/A	0x (see note 1)
M/Chip 4.0	>= 18	N/A	1y (see note 1)
CCD	32	0F	Az (see note 2)

Note 1: x, y = any hexadecimal digit.
Note 2: z = a hexadecimal digit in the range 5 – F.

The following table shows how the Issuer Application Data (IAD) format is derived for a specific transaction, using the two IAD Format parameters in the appropriate CPF record and the format identified from the transaction data (as described in the table above).

CPF IAD Format	CPF “Alternate” IAD Format	IAD Format Derived From Transaction Data	IAD Format Used For Transaction
Any Value	Matches CPF IAD Format	N/A	CPF IAD Format
Non-Zero	Zero	Matches CPF IAD Format	CPF IAD Format
Non-Zero	Zero	Does Not Match CPF IAD Format	Zero
Zero	Non-Zero	Matches CPF “Alternate” IAD Format	CPF “Alternate” IAD Format
Zero	Non-Zero	Does Not Match CPF “Alternate” IAD Format	Zero
Non-Zero	Non-Zero	Matches CPF IAD Format	CPF IAD Format
Non-Zero	Non-Zero	Matches “Alternate” CPF IAD Format	CPF “Alternate” IAD Format
Non-Zero	Non-Zero	Does Not Match Either CPF IAD Format	Zero

The first check must be on the two IAD Format fields in the CPF record (the first row in the table above). If these values match, then the specified format will be assumed, regardless of the format of the Issuer Application Data in the transaction message.

It is possible for the Issuer Application Data in a transaction to match more than one defined format. For example, a 10-byte IAD beginning %h06 %h00 in the first two bytes could indicate either a VIS or an M/Chip 2.1 format. If both IAD formats are configured on the CPF, then BASE24 uses the value in the EMV IAD FORMAT field, rather than the value in the ALTERNATE IAD FORMAT field.

5-Byte Card Verification Results (CVR) Field

For CCD, the Card Verification Results (CVR) field is 5 bytes in length. The Card Verification Results is a transaction dependent data element reflecting the current state of the application and the results of several internal checks performed on the current transaction parameters.

BASE24 can be configured to check the bit settings in the CVR as part of EMV authorization processing by using a CVR table. The table contains an authorization action to be performed whenever a bit within the CVR in the transaction message is set to '1'. BASE24-atm and BASE24-pos has been enhanced to support a table that reflects the CVR length and bit definitions for CCD.

BASE24 has also been enhanced to perform some special additional processing of the CVR, in order to identify whether offline PIN verification has been performed on a transaction.

Additional EMV Scheme

A new EMV scheme has been defined for CCD, to allow the application's security profile to be configured. In conjunction with the Cryptogram Version Number (included in the Issuer Application Data), the EMV scheme specifies the security options implemented for the application.

In BASE24, the EMV scheme is configured on the EMV Key File (KEYI). However, the KEYI is not used by systems that utilize BASE24-es Transaction Security Services (TSS) for security processing. Since CCD security processing is being supported via TSS only, there is no requirement to allow the CCD EMV scheme to be configured on the KEYI. Instead, TSS message 47 is used to retrieve the EMV scheme from the TSS database, and BASE24 must support the new value to identify the required security processing.

Additional Cryptogram Version Numbers

The Cryptogram Version Number is transmitted in the Issuer Application Data. In conjunction with the application's EMV security scheme, it specifies the security options implemented for the application, in particular:

- The card key derivation method
- The session key derivation method for the Application Cryptogram (AC) generation
- The data elements used in the AC generation
- The method used for the ARPC generation
- The session key derivation method for the secure messaging
- The format used for the secure messaging

The new Cryptogram Version Numbers that must be supported are shown in the table below.

EMV Scheme	Cryptogram Version Numbers
M/Chip	%h14 and %h15
CCD	%hA5

Where BASE24 currently performs processing based on the Cryptogram Version Number, the logic has been enhanced to cater for the above values.

Card Key Derivation Using "Option B"

For CCD, the ICC Master Key (i.e. the card key) must be derived using the "Option B" algorithm defined in EMV version 4.1.

BASE24 previously supported "Option A" only, so EMV authorization processing has been enhanced to support "Option B".

The main complexity is that the HSM requires the full PAN and Application PAN Sequence Number (APSN) when deriving the card key using "Option B". For "Option A", the HSM can accept the PAN and APSN in a pre-formatted data block. The ATM and POS EMV authorization modules currently format this data block, so this logic has to be inhibited when processing a transaction from a card compliant with CCD.

The PAN data block is currently formatted very early in transaction processing. Ideally, this logic would be moved further down the transaction path, where it would be more appropriate to perform processing based on the EMV scheme. However, this would affect all existing EMV implementations, so the current logic remains where either or both of the following conditions are satisfied:

- TSS is not being used (“Option B” card key derivation will be supported via TSS only);
- The PAN is less than or equal to 14 characters in length (the “Option B” and “Option A” card key derivation methods are identical for this length of PAN).

By following this approach, the transaction flow for existing customers is impacted only if they use TSS and issue cards with a PAN greater than 14 digits in length.

With “Option A” card key derivation, a 16-byte PAN buffer (comprising 14 bytes of PAN data, padded or truncated if necessary, and the 2-byte APSN) is passed to the CAM Utilities by the ATM Authorization or POS Router/Authorization EMV extension module. For “Option B” card key derivation, the whole PAN (together with the APSN) must be passed to the CAM Utilities, so the PAN buffer has been increased to 21 bytes in length. The PAN is stored in the first 19 bytes of the extended PAN buffer (left-justified and space-filled), and the APSN is stored in the last two bytes of the extended PAN buffer (defaulted to ‘00’ if the data element is not present). This allows the code to identify the PAN length: a value other than spaces in the last two bytes of the extended PAN buffer indicates that the full PAN is present.

Session Key Derivation Using the “Common” Method

For CCD, the Session Key must be derived using the “Common” method defined in the “EMV Specification Update Bulletin No. 46”.

BASE24 previously supported the MasterCard-proprietary and EMV 2000 methods only, so the EMV authorization processing has been enhanced to support the “Common” method.

The “Common” method is very similar to the MasterCard-proprietary method, the only difference being that zeros replace the Unpredictable Number (UN) used in the MasterCard-proprietary method.

MasterCard has also adopted the “Common” method of session key derivation into its latest specification (M/Chip 4.0). Therefore, BASE24 has also been enhanced to support the “Common” session key derivation method for M/Chip cards.

Support ARPC Generation Using “Method 2” and the CSU

For CCD, the ARPC generation logic must use “Method 2”, as described in EMV version 4.1.

BASE24 previously supported “Method 1” only, so the EMV authorization processing has been enhanced to support “Method 2”.

For CCD, the ARPC is 4 bytes in length, as opposed to the 8 bytes used for other EMV schemes supported by BASE24, e.g. VIS and M/Chip. Also, data used in generating the ARPC is 4 bytes in length, as opposed to the 2 bytes used for other EMV schemes supported by BASE24, e.g. VIS and M/Chip. BASE24 has been enhanced to use the 4-byte Card Status Update (CSU) field in generating the ARPC, and then to populate the Issuer Authentication Data with the 4-byte ARPC concatenated with the 4-byte CSU.

In the enhanced TSS command message for ARQC verification/ARPC generation (22), the CSU must be included in the “ARPC Data Block” field (renamed from “Converted Response Code”), replacing the ARPC Response Code.

All bits in the CSU, except the “Issuer Approves Online Transaction” bit (the highest order bit in the second byte of the CSU), are set to zero by BASE24. The “Issuer Approves Online Transaction” is determined from the response code for the transaction.

When a transaction is received, the ARQC is verified and the ARPC is generated with the assumption that the transaction will be successful, i.e. the “Issuer Approves Online Transaction” bit in the CSU is set to ‘1’. If the transaction is subsequently declined, for whatever reason, the ARPC needs to be re-generated. Here, the “Issuer Approves Online Transaction” bit in the CSU is set to ‘0’.

When generating an ARPC, BASE24 uses only the CSU as input to the generation algorithm, i.e. the Issuer Authentication Data does not include either an Authorization Response Code (ARC) or Proprietary Authentication Data (PAD).

Secure Messaging Using “Format 1”

For CCD, the issuer script processing logic must use Secure Messaging Format 1, as described in EMV version 4.1.

BASE24 previously supported “Format 2” only, so the EMV authorization processing has been enhanced to support “Format 1”.

The structure of the script data sent differs between “Format 1” and “Format 2”, and the required secure messaging format is specified in the script command itself. As the script data is configured on the LCONF (i.e. system-wide) in BASE24, the existing LCONF script parameters must be manipulated based on the format required. The current LCONF parameters are used as “templates”, with values “adjusted” where appropriate (this already occurs for offline PIN management, where the PIN CHANGE script format uses the PIN UNBLOCK script configured in the LCONF as a template).

The structure of the script data used in generating the MAC for “Format 1” is very different to that for “Format 2”. This is true both for data enciphered for confidentiality and for data not enciphered for confidentiality. This means that the buffer passed to TSS has to be formatted differently by BASE24. In addition, the length of the MAC generated must be 4 bytes for CCD, whereas BASE24 previously generated MACs of 8 bytes only.

Software Impacts

All BASE24 customers:

- Inline changes to Base DDL changes (DDLFCPF)
- Inline changes to Base Token changes (DDLBATKN)
- Inline changes to the Card Prefix File (CPF) User Interface module screen, requester and server.

All BASE24-pos customers:

- Inline changes to the Router/Authorization Globals file (RTAUG)

All BASE24 EMV customers:

- Inline changes to the Base Card Authentication Method Utilities (BA60IEMV.CAMUTILS, CAMUTILG, and EMVG)

All BASE24-atm EMV customers:

- Inline changes to the Authorization EMV Extension module (AT60IATH.AUTHEMVS, and CVRTBLS)

All BASE24-pos EMV customers:

- Inline changes to the Router/Authorization EMV Extension module (PS60IRTA.RTAUEMVS, RTAUEMVG, and CVRTBLS)

A complete list of the files affected by the EMV CCD Issuer Support enhancement can be found in the SCN file BA60UD0A.SCNECCD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Card Prefix File (CPF)

CPF screen 11 has been modified to allow new values in the following existing fields:

Field Name: EMV IAD FORMAT
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF. EMV-ISS-APPL-DATA-FRMT

This field is used to identify the format of the Issuer Application Data for this prefix. A of “4” indicates that CCD format is supported.

Field Name: STATUS CHECK ACTION INDEX
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF. ACTION-TABLE-INDEX

This field is used to identify which of the preset rules for status field processing are to be used for this prefix. Values of “5” and “6” allow more sets of preset rules to be configured.

CPF screen 11 has been modified to include the following new fields:

Field Name: ALTERNATE IAD FORMAT
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF. EMV-ISS-APPL-DATA-FRMT-ALT

This field is used to identify the alternative format of the Issuer Application Data for this prefix. Valid values are the same as those supported for the EMV IAD FORMAT field.

Field Name: ALTERNATE ACTION INDEX
Screen: CPF Screen 11
DDL Name: CPF.EMVCPF. ACTION-TABLE-INDEX-ALT

This field is used to identify which of the preset rules for status field processing are to be used for this prefix when the transaction contains the alternative format of the Issuer Application Data. Valid values are the same as those supported for the STATUS CHECK ACTION INDEX field.

The full list of values for these fields can be found in the ***BASE24-atm EMV Support Manual*** and the ***BASE24-pos EMV Support Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

<u>Card Prefix File (CPF)</u>

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

<u>TSS Considerations</u>

Issuers requiring support for CCD and/or “Common” Session Key Derivation must implement version 06.4 of TSS.

Issuers not requiring support for CCD and “Common” Session Key Derivation may implement the enhanced BASE24 code, but do not have to implement a new version of TSS.

Issuers not requiring support for CCD or “Common” Session Key Derivation or EMV 2000 Session Key Derivation or offline PIN management (changing the offline PIN on an EMV card) do not have to implement TSS to support EMV processing.

User/Operational Impacts

Not applicable

Vendor Impacts

Issuers requiring support for CCD and/or “Common” session key derivation must implement one of the following HSMs, with the specified minimum software release level:

- Atalla Axx100-VAR version 2.80 or Axx100-AKB version 2.80
- Thales RG8000 version 2.1, 2.2, 2.3

Reference Documents

- EMVCo - EMV 2004 version 4.1, May 2004
- EMVCo - EMV Application Note Bulletin No. 29, August 2005
- EMVCo - EMV Specification Update Bulletin No. 46, October 2005
- MasterCard - Addendum to M/Chip 4 Card Application Specifications, December 2005

HISO Full Message Encryption

This section describes the implementation requirements associated with the HISO Full Message Encryption enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the HISO Full Message Encryption enhancement.

BASE24-atm and BASE24-pos HISO processing has been enhanced to support full message encryption between BASE24 and host systems.

Prerequisite: The Transaction Security Services (TSS) enhancement must be installed prior to implementing the HISO Full Message Encryption enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 1 and 2)***.

Software Impacts

All BASE24 customers:
- Inline changes to the Security Utilities (BA60SRC.SECUTUILS)
- Inline changes to BA60DDL.DDLFKEYF and DDLGISO
- Inline changes to BA60AFT.RQKEYFS, SCRNKEYF, SECTBL, and SVKEYFS
- Updated event (log) message manual BA60LOGM.BAHISO, ATHISO, and PSHISO

All BASE24 ISO Host Interface customers:
- Inline changes to BA60HISO.BAHISOG, BAHISOM, and BAHISOS

All BASE24-atm ISO Host Interface customers:
- Inline changes to AT60HISO.ATHISOS

All BASE24-pos ISO Host Interface customers:
- Inline changes to PS60HISO.PSHISOS

A complete list of the files affected by the HISO Full Message Encryption enhancement can be found in the SCN file BA60UD0A.SCNHISO.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that are required to support this enhancement.

Interchange Security file (ISECD)

The security keys for the MSG will reside in the Interchange Security file (ISECD). Instead of passing the security key data in the calls to the HSM, the BASE24 ISO Host Interface will (via SECUTILS) pass an index value that will be used as the primary key to read the Interchange Security file. The index value will be stored in the KEYF. The key locator value is set to the primary key value for the record (e.g., an index value created by the conversion program for the file (i.e., for the Key File)). The TSS process uses the key locator value as the primary key value to retrieve the actual hardware-encrypted key information from its database.

For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 1 and 2)***.

Key File (KEYF)

The index value for locating the ISECD record is stored in the KEYF in the MAC key field. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 1 and 2)***.

The following fields contain the values used for MSG management in transactions sent from the BASE24 Interface process to DPCs or interchanges and received from the DPC or interchange.

Screen Name:	FULL ENCRYPTION
Screen:	KEYF Screen 4
DDL Name:	KEYF.INTERFACE.MSG.FULL-ENCRYPT

A flag indicating what part of the message will be encrypted. Valid values are:

0 = No MSG encryption (default value)

1 = Full MSG encryption

2 = Encryption on specified fields. This option is not currently supported.

Screen Name: ENCRYPT TYPE MSG
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.ENCRYPT-TYP-MSG

A flag indicating the type of message encryption used or expected by the DPC or interchange. Valid values are:

0 = Clear data

1 = Secure device

Screen Name: CHECK DIGITS
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.KEY-CHK-VALUE

The check digits corresponding to the value in the MSG KEY field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters. The check digits can be obtained from the utility used to encrypt the key. They are also available from the BASE24 ASMCOM and RSMCOM utilities.

Screen Name: OUTBOUND KEY COUNTER
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.OUTBND-KEY-CNTR

The number of times the outbound message key has been changed by dynamic key management processing. This counter is increased whenever the outbound message key is changed by the BASE24 transaction processing system. This value should require adjustment using files maintenance only if the DPC expects a value other than zero when dynamic key management is established; otherwise, the BASE24 ISO Host Interface increments it automatically and attempts to resynchronize it with the DPC. Valid values for each position in this field are 0 through 9 and A through F.

Screen Name: INBOUND KEY COUNTER
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.INBND-KEY-CNTR

The number of times the inbound message key has been changed by dynamic key management processing. This counter is increased whenever the inbound message key is changed by the BASE24 transaction processing system. This value should require adjustment using files maintenance only if the DPC expects a value other than zero when dynamic key management is established; otherwise, the BASE24 ISO Host Interface increments it automatically and attempts to resynchronize it with the DPC. Valid values for each position in this field are 0 through 9 and A through F.

Screen Name: MSG KEY TIMER VALUE
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.KEY-TIMER-LMT

The maximum length of time a message working key should be used. The time period can be expressed in minutes, hours, or days, depending on the value in the MESSAGE KEY TIMER INTERVAL field. The shortest length of time is five minutes. The timer resets each time a successful key exchange response is received. Valid time periods are:

5 through 1500 if the MESSAGE KEY TIMER INTERVAL field is set to the value M (minutes)

1 through 1000 if the MESSAGE KEY TIMER INTERVAL field is set to the value H (hours)

1 through 180 if the MESSAGE KEY TIMER INTERVAL field is set to the value D (days)

The interface does not use this parameter if the value in this field is 0. This parameter must be set to the value 0 unless the MESSAGE ENCRYPT TYPE field on this KEYF screen is set to the value 1 (Secure device).

Screen Name: MSG KEY TIMER VALUE
Screen: KEYF Screen 4
DDL Name: Not applicable

A code specifying the unit of time used with the value in the MESSAGE KEY TIMER VALUE set to the maximum length of time that a MSG key should be used. Valid values are:

D = Days

H = Hours

M = Minutes (default value)

BASE24 products check the values in the MSG KEY TIMER VALUE and MSG KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 products change the value in the MSG KEY TIMER INTERVAL field to the longer interval and redisplay the value in the MSG KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

Screen Name: MSG KEY TRAN
Screen: KEYF Screen 4
DDL Name: KEYF.INTERFACE.MSG.KEY-TRAN-LMT

The maximum number of usages with the current message key before changing the key. Valid values are:

0 = This parameter is not used (default value)

50 – 100000 = The maximum number of transactions

This parameter must be set to the value 0 unless the MESSAGE ENCRYPT TYPE field is set to the value 1 (secure device).

Screen Name:	MSG KEY ERROR
Screen:	KEYF Screen 4
DDL Name:	KEYF.INTERFACE.MSG.KEY-ERR-LMT

The maximum number of message encryption/decryption errors that occur with the present message key before changing the key. Valid values are as follows:

0 = This parameter is not used (default value)

50 – 999 = The maximum number of errors

This parameter must be set to the value 0 unless the MESSAGE ENCRYPT TYPE field is set to the value 1 (secure device).

Screen Name:	CONSECUTIVE MSG KEY ERROR
Screen:	KEYF Screen 4
DDL Name:	KEYF.INTERFACE.MSG.CONS-ERR-LMT

The maximum number of consecutive encryption/decryption errors that can occur with the present MSG key before changing the key. A successful message encryption/decryption, as well as a change in the MSG key resets the counter associated with this limit. Valid values are:

0 = This parameter is not used (default value)

50 – 999 = The maximum number of errors

This parameter must be set to the value 0 unless the MESSAGE ENCRYPT TYPE field is set to the value 1 (secure device).

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

A SECTBL entry for the new KEYF screen 4 has been added..

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Access to the new KEYF screen must be added through the SEC requester for the appropriate users.

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) must be installed in order to use this enhancement.

File Conversions

Key File (KEYF)

The KEYF conversion program must be run to convert the KEYF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Host code changes are required for all BASE24 users that wish to utilize full message encryption. The check digits for the encrypted message must be placed prior to the header definition because they must be placed outside of the encrypted portion of the message. Customers who choose not to perform full message encryption are not required to place additional characters in front of the header.

The BASE24 ISO Host Interface will not perform message encryption on the start-of BASE24-header, the BASE24 header or the message type identifier in messages to the host. Nor will it expect the above fields to be encrypted in messages from the host. All 0800 Network Management messages will NOT be encrypted. This includes Logons, Logoffs, Echo Tests, and Key Exchanges.

This enhancement also adds support for better key synchronization error detection and correction. The check digits of the PIN and MAC keys will be placed in bit 53 of any financial messages exchanged between BASE24 and the Host. These check digits will be compared against the current check digits in use. If they are different, a Key Synchronization error will be returned by setting an appropriate status code in the BASE24 SEM Header and returning a 9xxx message. BASE24 will generate and send a new key to the Host whenever a key synchronization error occurs. Note that a 9800 can still be sent but if it is a MAC sync error, the reject will be followed by a MAC key exchange.

The Host must use the new reject values that are received and sent in reject messages.

> The reject value of 197 (invalid MAC error) is being replaced with specific values for each type of invalid key error. The value of 211 is used for invalid MSG error. The value of 221 is used for invalid MAC error. The value of 231 is used for invalid PIN error. However, 221 will also be used generically for instances in Telebanking ISO Host Interface and the Teller ISO Host Interface processing where 197 was previously used.
>
> The value for reject 196 (key sync error) will be differentiated with new reject values of 210 (MSG key sync error), 220 (MAC key sync error), and 230 (PIN key sync error). However, the value of 196 will also be used generically for instances in Telebanking ISO Host Interface and the Teller ISO Host Interface processing.

User/Operational Impacts

<u>Performance Impacts</u>

The full message authentication and encryption will have an impact on system performance because of the extra I/O to the Hardware Security Module (HSM). An Inter-Process Communication (IPC) message is added for each call to an HSM.

Operational Impacts

For SNA, the ASCII/EBCDIC conversion can happen at the controller. However customer's implementing the full message encryption within BASE24, should not use the ASCII/EBCDIC conversion at the controller. Instead use the ASCII/EBCDIC conversion enhancement within the BASE24 ISO Host Interface instead. This is because the conversion should occur before the message is encrypted.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

To implement this enhancement, BASE24 must have access to a hardware security module. The enhancement supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of software that is capable of supporting the Transaction Security Services enhancement.

Reference Documents

Not applicable

HISO Miscellaneous Changes

This section describes the implementation requirements associated with the HISO Miscellaneous Changes enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the HISO Miscellaneous Changes enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.RQHCFS, and SCRNHCF
- Inline changes to BA60DDL.DDLFHCF
- Updated event (log) message manual BA60LOGM.BAHISO, ATHISO, and PSHISO

All BASE24 HISO customers:

- Inline changes to BA60HISO.BAHISOG and BAHISOS

All BASE24-atm HISO customers:

- Inline changes to AT60HISO.ATHISOS

All BASE24-pos HISO customers:

- Inline changes to PS60HISO.PSHISOS

A complete list of the files affected by HISO Miscellaneous Changes enhancement can be found in the SCN file BA60UD0A.SCNHISM.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new fields in the files that are required to support this enhancement.

Host Configuration File (HCF)

Screen Name: CHARACTER FORMAT
Screen: HCF Screen 1
DDL Name: HCF.SEG0.CHAR-FRMT

A flag indicating whether the BASE24 ISO Host Interface must perform an ASCII/EBCDIC conversion for outbound messages and an EBCDIC/ASCII conversion for inbound messages. For ASCII character format, no conversion is necessary. Valid values are:

A = ASCII character format (default value)
E = EBCDIC character format

Screen Name: CHARACTER FORMAT
Screen: HCF Screen 1
DDL Name: HCF.SEG0.ENHNC-STAT

A flag indicating whether the combined status and performance command is to be displayed for the current performance period. Valid values are:

N = No, do not display the enhanced status message (default value)
Y = Yes, display the enhanced status message

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Host Configuration File (HCF)

The HCF conversion program must be run to convert the HCF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

The host must communicate in the same character format that is configured on the BASE24 HCF.

User/Operational Impacts

<u>Operational Impacts</u>

For SNA, the ASCII/EBCDIC conversion can happen at the controller. However customer's implementing the full message encryption within BASE24, should not use the ASCII/EBCDIC conversion at the controller. Instead use the ASCII/EBCDIC conversion enhancement within the BASE24 ISO Host Interface instead. This is because the conversion should occur before the message is encrypted.

New operational commands were created as a result of this enhancement. Operators must be trained on the new SAFPACING, PACINGSTAT, and PERFSTAT commands and to recognize when it is appropriate to utilize them.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Modify Discretionary Data Suppression Enhancement

This section describes the implementation requirements associated with the Modify Discretionary Data Suppression Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Modify Discretionary Data Suppression Enhancement.

The Discretionary Data Suppression Enhancement has been modified to overwrite the discretionary data area of track 1 and track 2 with zeroes rather than truncating it. Discretionary data is replaced with zeroes up to, but not including, the end sentinel, thus preserving the original track length. Previously, discretionary data was truncated by moving the end sentinel to the beginning of the discretionary data area and padding the remainder of the track 1 or track 2 fields with blanks, changing the original track length.

Software Impacts

All BASE24 customers:

- Inline changes to BA60SRC.BAUTILS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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5: BASE24-atm

Section 5 discusses the changes and enhancements made to BASE24-atm in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24-atm users must consider when upgrading from release 6.0 version 6.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-atm users, along with the associated conversion and implementation requirements that all BASE24-atm users must address.
- Optional enhancements describe the changes that affect only those BASE24-atm users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-atm users must address to implement the enhancements.

Section 5 must be used in conjunction with Section 4 to identify the changes that BASE24-atm users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-atm users must address in order to upgrade to release 6.0 version 7.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- ATM Passbook Update Enhancement
- Deposit Hold Notification Enhancement
- Exchange Rate Notification Enhancement
- NCR Release 3.00 Upgrade Enhancements
- Service Code Validation Enhancement
- Tidel General Enhancements
- Triton Token Retrieval Option Enhancement

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

Add Additional Values to PTD/ATD. ATD Screen 21 has been enhanced to support the configuration of the EMV Terminal Capabilities as defined in EMV v4.1 (2004).

The BASE24-atm Terminal Data files (ATD) server has been modified to delete the BASE24-atm Terminal Data Dynamic Scratch Pad File (ATDD2) record when a terminal is deleted using the ATD file maintenance screen. An Online Maintenance File (OMF) record containing only the OMF header is logged for the deleted ATDD2 record. Previously, only the terminal's BASE24-atm Terminal Data Dynamic General File (ATDD1) record and BASE24-atm Terminal Data Static General File (ATDS1) record were deleted.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.SVTDFLTG and SVTDFLTS
- Inline changes to BA60DDL.DDLFOMF

All BASE24-atm customers:

- Inline changes to AT60AFT.RQATDS, SCRNATD, SVATDTBL, SVATDTG and SVATDTS

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not Applicable

Database Configuration

Terminal Data File (ATD)

The following additional EMV Terminal Capabilities may be configured:

Screen Name:	NO CVM REQUIRED
Screen:	ATD Screen 21
DDL Name:	ATDX .EMV_TERM_CAP.CVM (i.e., TERM-CAP byte 2, bit 4)
Valid Values are:	Y and N (default is N)

Screen Name:	COMBINED DATA AUTHENTICATION
Screen:	ATD Screen 21
DDL Name:	ATDX.EMV_TERM_CAP.SEC (i.e., TERM-CAP byte 3, bit 4)
Valid Values are:	Y and N (default is N)

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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ATM Passbook Update Enhancement

This section describes the implementation requirements associated with the ATM Passbook Update Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the ATM Passbook Update Enhancement.

The purpose of this enhancement is to provide BASE24 customers with the ability to support a Passbook Update transaction. The Passbook Update transaction will enter BASE24 as a Statement Print transaction (transaction code 70) with a transaction subtype of APU0. The BASE24-atm Authorization module and the BASE24-atm ISO Host Interface (HISO) module have been modified to process the new transaction subtype for online only authorization (i.e., pass through to Host).

Prerequisites: This enhancement will not be supported by any BASE24-atm device handlers. Only the BASE24-es IFX device handler will support this enhancement.

Software Impacts

All BASE24 customers:
• Inline changes to BA60DDL.DDLATTKN
• Inline changes to BA60AFT.COBTKN
• Inline changes to BA60SRC.ATTKNCVS, ATTKNID and SUBTPTBL
• Updated event log message manual BA60LOGM.ATHISO

All BASE24-atm customers:
• Inline code changes to AT60AUTH.AUTHG and AUTHS

All BASE24-atm HISO customers:
• Inline changes to AT60HISO.ATHISOG and ATHISOS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNPSBK.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24-atm tokens have been added for this enhancement:

• The Passbook Processing Token (token ID “AK”) carries processing information pertaining to the type of passbook, the location of the next available line, etc. It is added by the BASE24-es IFX device handler.

• The Interim Statement/Passbook Data Token (token ID “AI”) carries the information to be printed in the cardholder’s passbook. It is added by the host and used by the BASE24-es IFX device handler.

The following existing BASE24 token is utilized for this enhancement:

• Transaction Subtype Token (token ID “BM”)

The Passbook Processing Token and Interim Statement/Passbook Data Token must be configured to be carried in the ISO message sent by the ISO Host Interface.

If the tokens for this enhancement should not be logged to the BASE24-atm Transaction Log File (TLF), the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the TLF or sent to the host, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract Process.

<u>External Interfaces</u>

Host code changes are required to support the ATM Passbook Update transaction. The Passbook Processing token is added and sent by the BASE24-es IFX Device Handler on the request and is updated by the host system on the response. The Interim Statement/Passbook Data token is added by the host system on the response.

The ATM Passbook Update transaction is only supported online, i.e., BASE24-atm does not stand-in for this transaction. BASE24-atm accepts and sends ATM Passbook Update completions (from the ATM) and reversals to the host. BASE24-atm generates ATM Passbook Update reversals as it does for financial messages such as deposits, transfers and withdrawals. Reversals and ATM completions are required for the ATM Passbook Update transactions as the host needs to know the last item printed in the passbook.

Super Extract Process

If the new tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

System Limitations

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Operational Impacts

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on BA60LOGM.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Deposit Hold Notification Enhancement

This section describes the implementation requirements associated with the Deposit Hold Notification Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Deposit Hold Notification Enhancement.

The BASE24-atm Authorization module and the BASE24-atm ISO Host Interface (HISO) module have been modified to accept a new token from a Host authorizer indicating what part of the deposit amount in an 'Envelope Deposit' transaction request will be put on hold and the date the hold will last until. The cardholder can then decline the transaction, so no deposit happens, or accept the hold and then complete the deposit transaction.

Prerequisites: This enhancement will not be supported by any BASE24-atm device handlers. Only the BASE24-es IFX device handler will support this enhancement.

Software Impacts

All BASE24 customers:
- Inline changes to BA60DDL.DDLATTKN
- Inline changes to BA60AFT.COBTKN
- Inline changes to BA60SRC.ATTKNCVS and ATTKNID

All BASE24 Transaction Based Pricing (TBP) customers:
- New executable objects for Transaction Based Pricing (TBP) reports and update to TBP documentation (subvolume BA60TBPD)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNDEPH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24-atm tokens have been added for this enhancement:

• The Hold Token (token ID "AL") carries processing information pertaining to the hold amount. It is added by the host system.

The Hold Token must be configured to be carried in the ISO message sent by the ISO Host Interface.

If the tokens for this enhancement should not be logged to the BASE24-atm Transaction Log File (TLF), the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the TLF or sent to the host, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the External Interfaces and Super Extract processes.

External Interfaces

The host or co-network may require modification to detect that the transaction is a deposit hold notification and to add and process the new token. The new data will be passed in the ISO message to/from the BASE24-atm ISO Host Interface (HISO).

Super Extract Process

If the new tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts.

There will be multiple records on the TLF representing a single transaction

Disk Impacts

The introduction of an additional record logged to the TLF will result in the use of additional disk space. The additional disk space required for the TLF is dependent upon the volume of deposit hold notification transactions processed. This impact is customer-specific and must be evaluated by each BASE24-atm customer.

Existing BASE24-atm users that implement this enhancement will experience a change in messaging or logging to the TLF as follows:

- Each successful envelope deposit transaction will be recorded in two TLF records: one for the deposit hold notification and one for the deposit; However, if the host is not applying a hold to the deposit, the transaction is processed as a standard deposit and only one TLF record will be logged.
- Declined transactions may have as many as two TLF records: one for the deposit hold notification and one for the deposit.
- Reversed transactions will have an additional TLF record if a problem occurs during the envelope deposit or if the cardholder cancels the transaction.

The TLF requires additional disk space as the new token for this enhancement can be included in the transaction records logged.

System Limitations

Existing BASE24-atm users that implement this enhancement will experience a change in messaging or logging to the TLF. Each successful envelope deposit transaction will be recorded in one or two TLF records: one for the initial envelope deposit request which will contain the deposit hold amount and one for the actual envelope deposit. If the host will not apply a hold to the deposit, then the transaction will be processed as a standard deposit. (Refer to the Disk Impact section above for more information).

BASE24-atm users installing this enhancement must be aware of their internal system limitations for messaging to the TLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-atm users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Cardholder Impacts

The Deposit Hold Notification enhancement will provide the BASE24-atm customers the ability to provide a notification service to the cardholder at the ATM. Whenever the Hold Token is present, the BASE24-es IFX Device Handler will present the cardholder with the option to accept or decline the transaction based on the amount being held by their institution.

Performance Impacts

Performance will be impacted in systems running Deposit Hold notification transactions due to the increase in the transaction path length. The transaction path may be longer than other transactions because deposit hold notifications are only supported online and will require an additional interaction with the cardholder at the ATM. BASE24 does not stand-in for this transaction.

Transaction Based Pricing Reports

The Transaction Based Pricing (TBP) reports have been modified to support the Deposit Hold Notification enhancement in BASE24-atm. These changes necessitate that all BASE24 TBP customers implement the updated version of the TBP report objects into their system. When the transaction is an envelope Deposit transaction, the utility will determine whether the Hold token is present. If a cardholder has neither accepted nor rejected the hold, or the cardholder has rejected the hold, then the transaction is not billable. The notification portion of the transaction will be counted as a non-transaction.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Exchange Rate Notification Enhancement

This section describes the implementation requirements associated with the Exchange Rate Notification Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Exchange Rate Notification Enhancement.

The Exchange Rate Notification enhancement will provide the BASE24-atm customer the ability to provide a notification service to the cardholder at the ATM of the exchange rate that will be used on the transaction in progress.

Whenever BASE24-atm Authorization receives a Message to FI (Transaction Code 60) with the Exchange Rate Notification transaction subtype, BASE24-atm Authorization returns the exchange rate from the Exchange Rate File (ERF) to the device handler. The cardholder will get the option to accept or decline the exchange rate offered by the Financial Institution. If the cardholder declines, no further processing occurs.

No Transaction Log File (TLF) record is created for the Exchange Rate Notification message.

Prerequisites: This enhancement will not be supported by any BASE24-atm device handlers. Only the BASE24-es IFX device handler will support this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60SRC.SUBTPTBL

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHG and AUTHS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNEXCH.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts.

Cardholder Impacts

The Exchange Rate Notification enhancement will provide the BASE24-atm customer the ability to provide a notification service to the cardholder at the ATM of the exchange rate that will be used on the transaction in progress.

Whenever BASE24-atm Authorization receives a Message to FI (Transaction Code 60) with the Exchange Rate Notification transaction subtype AER0, BASE24-atm Authorization will return the exchange rate from the Exchange Rate File (ERF) to the device handler. The cardholder will get the option to accept or decline the exchange rate offered by the Financial Institution. If the cardholder declines, no further processing occurs.

Performance Impacts

Performance will be impacted in systems running exchange rate notification transactions due to the increase in the transaction path length. The transaction path may be longer than other transactions because the exchange rate must be presented to the cardholder for acceptance before continuing with the transaction.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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NCR Release 3.00 Upgrade Enhancements

This section describes the implementation requirements associated with the NCR Release 3.00 Upgrade Enhancements.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the NCR Release 3.0x Upgrade Enhancements.

With this enhancement, support was added to the BASE24-atm NCR 5XXX device handler to allow customers to take advantage of various enhancements made to Aptra Advance NDC, up to and including version 03.00.03.

Prerequisites: The NCR 5XXX Device Handler is required.

Software Impacts

All BASE24-atm NCR 5XXX device handler customers:

Inline changes to N50DDLS, RECIMGCS, RQ50ECPS, RQ50STS, RQ50SUPS, RQN50LS, SCRNECP, SCRNST, SVDN50S, SVLIBS, N50G, N50NAMS, N50S, AT60DN50.N50SSDS
Updated event (log) message manual BA60LOGM.AT5070
New load group in the configuration file (i.e., AT60N50.CNFG5070)
Changes to the documentation file (i.e., AT60N50.CNN50DOC)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNAA30.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

NCR 5XXX Enhanced Configuration Parameter Screen (SCRNECP)

Support was added for the following (refer to the Device Configuration for further details):

- Option 45 - Bunch Note Acceptor Message Settings,
- Timer 72 - Card Removal from DASH Card Reader Timeout,
- Timer 92 - Fault Display Timeout.

NCR 5XXX State Screens (SCRNST)

Support was added for a new state type, the PIN and Language Select (m). Customers wanting to take advantage of this state will need to make their own configuration changes.

The maximum Next State Number has also been increased from 750 to 999. The range is now 000-254 or 256-999.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

With this enhancement, support was added to allow customers to take advantage of various enhancements made to Aptra Advance NDC, up to and including version 03.00.03. Some of these enhancements can be configured by accessing the NCR 5XXX device specific screen of the Device Control Terminal (DCT), screen 45, making changes to the NCR 5XXX Configuration File (CNFG5070), and downloading these changes to the terminal. An overview of the changes is noted below:

Support was added for a new state type, the PIN and Language Select (m). Customers wanting to take advantage of this state will need to make their own configuration changes.

The maximum Next State Number has been increased from 750 to 999. The range is now 000-254 or 256-999.

Aptra Advance NDC 2.06 includes support for up to 9,999 multi-language screens. While NCR provides two solutions for implementing this support, the US NDC solution and the 'u' Screen Group solution, the NCR 5XXX device handler already supports 9,999 multi-language screens through the US NDC solution and will continue to do so.

Additional codes have been added to Option 45 (Bunch Note Acceptor (BNA) Message Settings) supporting note retract functionality. This is in addition to the previously supported inclusion of transaction counts. However, acceptance of more than 90 notes is not currently supported.

Support was added for two new timers, Timer 72 (DASH Card Removal) and Timer 92 (Fault Display).

Support was added for Message Authentication (MACing) of the Dispenser Currency Cassette Mapping Table (DCCMT).

Support was added for three new product classes, 6674, 6676, and Personas 87.

The following file was modified to include a new load group:

<u>NCR 5XXX Configuration File (CNFG5070)</u>

- Load Group 603

  Load group 603 provides support for the Dip and Smart Hardware (DASH) card reader. It is intended to be loaded on top of the standard EMV load group 600 and produces a slightly modified state flow that allows the transaction to be initiated through a dip reader, rather than a motorized reader.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

A new field, BNA-MSG-SETTINGS, was defined in the ATDD1-CPM-BNA-DATA-NCR-5XXX structure. This structure is used to define the BNA-CPM additional offset in the data area of the ATDD1. An inline conversion of the ATDD1 will be performed to shift any data following the BNA-CPM definition back to allow space for the new field, initialize this field to blanks indicating acceptance of more than 90 notes is not configured, and increase the length of the BNA-CPM definition to include the new field. Note that acceptance of more than 90 notes is not currently supported.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Operational Impacts:</u>

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

The appropriate level of Aptra Advance NDC must be installed on a device in order to take advantage of any of the previously mentioned version 2.06 – 03.00.03 enhancements. Customers running prior versions of Aptra Advance NDC will still be supported but will not be able to take advantage of any of the new enhancements.

Reference Documents

Aptra Advance NDC Reference Manual (B006-6180-F000), August 2003

Aptra Advance NDC Reference Manual (B006-6180-H000), June 2005

Aptra Advance NDC Reference Manual (B006-6180-H000), Issue 3, December 2005

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Service Code Validation Enhancement

This section describes the implementation requirements associated with the Service Code Validation Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Service Code Validation Enhancement.

The purpose of this enhancement is to provide better detection of counterfeit cards. In card-read transactions, the card is authenticated by validating the Card Verification Digits encoded in the track data (CVD1). For the algorithm used to generate the CVD1, the input data includes the Service Code, also encoded in the track data. In manually-entered transactions, the card is authenticated by validating the Card Verification Digits printed or embossed on the card itself (CVD2). For the algorithm used to generate the CVD2, the input data includes a value of “000” in place of the Service Code, which is typically not available in manually-entered transaction data.

A counterfeit card can be created by generating fraudulent track data, which includes the CVD2 printed or embossed on the card and a Service Code value of “000”. The authorizing system may successfully validate the CVD2 (using the Service Code of “000”), believing it to be the CVD1. This means that the card is assumed to be valid when in fact it is fraudulent. To detect these fraudulent cards, the BASE24-atm Authorization module has been modified to reject transactions which contain a Service Code of “000” in card-read transactions.

Software Impacts

All BASE24-atm customers:
• Inline code changes to AT60AUTH.AUTHLIBS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Tidel General Enhancements

This section describes the implementation requirements associated with the Tidel General Enhancements.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Tidel General Enhancements.

The BASE24-atm Tidel Device Handler has been modified to support the BASE24-atm standard PIN Change transaction. This will enable users to provide the capability to cardholders to change their own PINs at a Tidel ATM connected to BASE24-atm.

The BASE24-atm Tidel Device Handler has been modified to allow a new option for storing/retrieving token data for use in reversal processing. Previously, the Tidel device handler could only store token data for a transaction in the space allowed in the TDF for such data. This could limit customers abilities to add new features/functions to their Tidel ATMs, as all data required to process a reversal may not be available, having been truncated to fit in the TDF.

With this enhancement, the BASE24-atm Tidel Device Handler can utilize the Transaction Log File (TLF) record of a completed transaction to retrieve all data required to process a reversal of the transaction. If the new option is set to use the TLF and the length of the tokens to store are greater than the LAST-MSG area of the TDF, the BASE24-atm Tidel Device Handler saves off the TLF record address of the last transaction in the TDF.LAST-MSG area and reads the TLF with the record address saved to get the last transaction data when required. Although there is a space limit in TLF records as well, there is much more room than in the current space in the TDF, which should allow customers to implement new transactions/features for some time to come.

If the tokens to store/retrieve for a transaction will fit in the TDF.LAST-MSG field the Tidel device handler will continue to store and retrieve the data from the TDF, even if the new option is set to use the TLF. This will avoid an extra disk hit to read the TLF record when it is not necessary.

The BASE24-atm Tidel Device Handler has also been enhanced to support the processing of ATM Error Codes sent by the terminal in a Status Change Comm (SCC) Request message. This is in addition to the status indicators in the SCC message that are already supported. This enhancement will allow for up to 20 error codes to be received in a single request, with an event message being raised for each describing its meaning. The sending of the SCC message by the ATM can be enabled/disabled via setup at the ATM, providing the ATM is running AnyCard release 5.30 or later.

Software Impacts

All BASE24-atm customers:

Inline changes to AT60DDL.DDLFTDF, AT60AFT.RQATDS, SCRNATD1, SVATDTG, SVATDTS, SVATDTBL, and AT60SRC.ATDXG

All BASE24-atm Tidel device handler customers:

Inline changes to RQTTIDLS, TIDELS, TIDELG, TIDELM, TIDLNAMS
Updated event (log) message manual BA60LOGM.ATTIDL
Updated informal device handler manual AT60TIDL.TIDLMAN

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNTIDL.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ATM Terminal Definition Files (ATD)

The Token Retrieval Option, on screen 3, is used to configure the method used by the Tidel Device Handler to retrieve token information for reversal messages.

Screen Name:	TOKEN RETRIEVAL OPTION
Screen:	TDF Screen 3
DDL Name:	TDF.ADDL-DATA.TKN-RETRV-OPT
Valid Values are:	1 – TDF (Default)
	2 – TLF

Receipt file (RCPT)

Add information for a Pin Change receipt (PC). Information on how to configure this receipt is contained in the ***BASE24-atm Files Maintenance Manual***.

Token file (TKN)

The following existing BASE24 tokens are utilized for this enhancement:

- The PIN Change Token (token ID “06”)
- The TLF Token (token ID “B7”)

The PIN Change Token will contain the new PIN offset. This should be taken into account when determining if the token should be logged to the TLF.

Token records must be configured to ensure the TLF token is logged to the TLF for token retrieval from the TLF to work.

If the tokens for this enhancement are to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

To support Status Change Communication (SCC) messages, this option must be enabled at the ATM.

To support PIN change transactions, this option must be enabled at the ATM. This will allow the transaction type to appear on the transactions menu screen.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Data File (TDF)

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF records from the release 5.3 format to the release 6.0 format. The TDF conversion program on the AT60UC0A subvolume can be used to convert the TDF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVATDS file on the AT60UC04 subvolume or the CNVTDFS file on the AT60UC0A subvolume.

To compile the conversion program on the AT60UC04 subvolume, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvatd.t,name $atdm,nowait/
cnvatdm –force –list –listreplace
```

To compile the conversion program on the AT60UC0A subvolume, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#cnvtdf.t,name $tdfm,nowait/ cnvtdfm
–force –list –listreplace
```

Release 5.3 users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVTDFR obey file on the AT60UC0A subvolume with the appropriate information. The appropriate obey file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume for release 5.3 customers and the BA60UC0A subvolume for release 6.0 version 6 customers.

Host/Co-Network Impacts

External Interfaces

The host or co-network may require modification to process the tokens for this enhancement. The tokens are passed in the ISO message from the BASE24-atm ISO Host Interface (HISO) if configured in the system to be sent.

Super Extract Process

If the tokens for this enhancement are to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Operational Impacts:

New EMS messages will be added to report status information contained in SCC messages from the terminal.

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

Performance may be impacted for those terminals configured to retrieve token data from the Transaction Log File (TLF). If the amount of token data on a single transaction exceeds the 528 storage limit in the TDF.LAST-MSG area, only the TLF token will be stored in memory. If a reversal is required, the TLF token will be used as a locater to the appropriate record in the TLF. The TLF record will contain the token data to be used in the reversal. The additional I/O on the TLF could cause a slight decrease in performance. In those cases where token data is to be retrieved from the TLF, but the token data does not exceed the 528 storage limit or reversal processing is not required, the I/O on the TLF is not necessary and there should be no impact on performance.

Vendor Impacts

This enhancement may require device firmware changes to be able to capture and send additional data. To implement the functionality from these enhancements, the Tidel terminal must be capable of performing PIN change transactions and sending SCC messages.

Reference Documents

- ***ATM Communication Specification, Tidel, Inc.*** (Part Number 201-3653-013, version M, November 2, 2004)

Triton Token Retrieval Option Enhancement

This section describes the implementation requirements associated with the Triton Token Retrieval Option Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton Token Retrieval Option Enhancement.

The BASE24-atm Triton Device Handler has been modified to allow a new option for storing/retrieving token data for use in reversal processing. Previously, the Triton device handler could only store token data for a transaction in the space allowed in the TDF for such data. This could limit customers abilities to add new features/functions to their Triton ATMs, as all data required to process a reversal may not be available, having been truncated to fit in the TDF.

With this enhancement, the BASE24-atm Triton Device Handler can utilize the Transaction Log File (TLF) record of a completed transaction to retrieve all data required to process a reversal of the transaction. If the new option is set to use the TLF and the length of the tokens to store is greater than the LAST-MSG area of the TDF, the BASE24-atm Triton Device Handler saves off the TLF record address of the last transaction in the TDF.LAST-MSG area and reads the TLF with the record address saved to get the last transaction data when required. Although there is a space limit in TLF records as well, there is much more room than in the current space in the TDF, which should allow customers to implement new transactions/features for some time to come.

If the tokens to store/retrieve for a transaction will fit in the TDF.LAST-MSG field the Triton device handler will continue to store and retrieve the data from the TDF, even if the new option is set to use the TLF. This will avoid an extra disk hit to read the TLF record when it is not necessary.

Software Impacts

All BASE24-atm customers:

Inline changes to AT60DDL.DDLFTDF, AT60AFT.RQATDS, SCRNATD1, SVATDTG, SVATDTS, SVATDTBL, and AT60SRC.ATDXG

All BASE24-atm Triton device handler customers:

Inline changes to TRITONG, TRITONM, TRITONS
Updated event (log) message manual BA60LOGM.ATTRIT
Updated informal device handler manual AT60TRIT.TRITMAN

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNTTRO.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ATM Terminal Definition Files (ATD)

The Token Retrieval Option, on screen 3, is used to configure the method used by the Triton Device Handler to retrieve token information for reversal messages.

Screen Name:	TOKEN RETRIEVAL OPTION
Screen:	TDF Screen 3
DDL Name:	TDF.ADDL-DATA.TKN-RETRV-OPT
Valid Values are:	1 – TDF (default)
	2 - TLF

Token file (TKN)

The following existing BASE24 token is utilized for this enhancement:

- The TLF Token (token ID “B7”)

Token records must be configured to ensure the TLF token is logged to the TLF for token retrieval from the TLF to work.

If the tokens for this enhancement are to be extracted with the TLF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

BASE24-atm Terminal Data File (TDF)

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF records from the release 5.3 format to the release 6.0 format. The TDF conversion program on the AT60UC0A subvolume can be used to convert the TDF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVATDS file on the AT60UC04 subvolume or the CNVTDFS file on the AT60UC0A subvolume.

To compile the conversion program on the AT60UC04 subvolume, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#cnvatd.t,name $atdm,nowait/ cnvatdm –force –list –listreplace

To compile the conversion program on the AT60UC0A subvolume, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#cnvtdf.t,name $tdfm,nowait/ cnvtdfm –force –list –listreplace

Release 5.3 users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVTDFR obey file on the AT60UC0A subvolume with the appropriate information. The appropriate obey file should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREADME file on the BA60UC04 subvolume for release 5.3 customers and the BA60UC0A subvolume for release 6.0 version 6 customers.

Host/Co-Network Impacts

External Interfaces

The host or co-network may require modification to process the token for this enhancement. The token is passed in the ISO message from the BASE24-atm ISO Host Interface (HISO) if configured in the system to be sent.

Super Extract Process

If the token for this enhancement is to be included in the TLF Extracts, BASE24-atm users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Operational Impacts:

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

Performance may be impacted for those terminals configured to retrieve token data from the Transaction Log File (TLF). If the amount of token data on a single transaction exceeds the 528 storage limit in the TDF.LAST-MSG area, only the TLF token will be stored in memory. If a reversal is required, the TLF token will be used as a locater to the appropriate record in the TLF. The TLF record will contain the token data to be used in the reversal. The additional I/O on the TLF could cause a slight decrease in performance. In those cases where token data is to be retrieved from the TLF, but the token data does not exceed the 528 storage limit or reversal processing is not required, the I/O on the TLF is not necessary and there should be no impact on performance.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Optional Enhancements

This section identifies the BASE24-atm enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Completion Messages Configuration Enhancement
- Correct Declined Advices to Host
- EMV CCD Issuer Support
- Locked ATD Records Causing Cutover Failure Enhancement
- Shared 912 AKDS Certificates
- Shared 912 Offline PIN Synchronization
- Shared NDC+ AKDS Certificates
- Shared NDC+ EMV Upgrade
- Simultaneous Transactions Fraud Enhancement
- Triton Mobile Top Up Enhancement

BASE24-atm users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements for future reference.

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Completion Messages Configuration

This section describes the implementation requirements associated with the Completion Messages Configuration enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Completion Messages Configuration enhancement.

BASE24-atm has been enhanced to provide for new COMPL REQ values which will be used to indicate that completions are to be sent to the host only if BASE24 performed the authorization. Customers will now be able to configure the system to either send all BASE24 authorized transactions, including declines or to only send approved BASE24 authorized transactions.

Software Impacts

All BASE24 customers:
- Changes to comments in BA60DDL.DDLFIPCF, DDLGPSTM and DDLGSTM
- Inline changes to BA60AFT.RQIPCFSS, RQIPCFXS, SVIPCFTG, and SVIPCFTS

All BASE24-atm customers:
- Inline changes to AT60AUTH.AUTHG, AUTHS and AUTHLIBS

All BASE24-atm ISO Host Interface customers:
- Inline changes to AT60HISO.ATHISOS and ATHISOG

A complete list of the files affected by the Completion Messages Configuration enhancement can be found in the SCN file BA60UD0A.SCNCOMP.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the new and enhanced fields in the files that are required to support this enhancement.

Issuer Processing Code File (IPCF)

Field Name: COMPLETION REQUIRED TO HOST
Screen: IPCF Screen 2
DDL Name: IPCF.COMPL-REQ

A code used to control the production of 0220 financial transaction advice messages. Valid values are:

0 = Do not send 0220 messages
1 = Send 0220 messages for approved transactions
4 = Send 0220 messages for all transactions
5 = Send 0220 message for approved BASE24 authorized transactions only
6 = Send 0220 message for all BASE24 authorized transactions (approved/denied)

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

If the BASE24 user wishes to enable the new value of “5” or “6”, then the host system must be capable of accepting and processing the appropriate approved or non-approved advices. The host system must also be prepared to handle the increase in message traffic if applicable.

User/Operational Impacts

If the BASE24 user wishes to enable the new value of “5” or “6”, then the COMPLETION REQUIRED TO HOST field on IPCF screen 2 must be set appropriately.

Performance Impacts

If the new value of “5” or “6” is implemented, there will be an increase or decrease in the number of messages sent to the Host depending upon the previous value. The Host must be prepared to handle the increase in message traffic if applicable.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Correct Declined Advices to Host

This section describes the implementation requirements associated with the Correct Declined Advices to Host enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Forward Declined Advices to Host enhancement.

Previously, it was not possible for advices of transactions declined before they reached the BASE24-atm Authorization process to be forwarded to a Host system. Logic has been added to the BASE24-atm Authorization process to allow declined advices (e.g. transactions declined by an interchange or interchange interface) to be sent to a HISO process, if required.

This enhancement is an extension to the “Forward Declined Advices to Host” enhancement, implemented in Release 6.0 Version 5. Please refer to the “BASE24 Release 6.0 Version 5 Implementation Guide” for full details of this enhancement.

The following considerations apply to this enhancement:

- The ability to forward and log declined advices will be available for all authorization levels.
- Declined advices for which a CPF record cannot be successfully retrieved and processed will not be forwarded as declined advices to the Host.
- All advices rejected before the IDF record is successfully retrieved and processed cannot be forwarded as declined advices to the Host, since no routing information is available without an IDF record.
- Declined advices for which an IPCF record cannot be successfully retrieved and processed will not be forwarded as declined advices to the Host.

Software Impacts

All BASE24-atm customers:

- Inline changes to AUTHS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0A.SCNDADV.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

If the BASE24 user wishes to forward declined advices, then the Host system must be capable of accepting and processing non-approved advices.

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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EMV CCD Issuer Support

This section describes the implementation requirements associated with the EMV CCD Issuer Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV CCD Issuer Support enhancement.

For additional detailed information on this enhancement see Section 4 (Base).

Software Impacts

For full details on the software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNECCD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

For full details on the software impacts for this enhancement see Database Configuration in Section 4 (Base) of this document.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

For full details on the software impacts for this enhancement see File Conversions in Section 4 (Base) of this document.

Host/Co-Network Impacts

For full details on the software impacts for this enhancement see Host/Co-Network Impacts in Section 4 (Base) of this document.

User/Operational Impacts

<u>Operational Impacts:</u>

The Card Verification Results field is a different format on a CCD transaction. It is a 5 byte field (all bits in Byte 5 are reserved for future use). The decisional information contained in these bytes is as follows:

Action Index	Byte	Bit	Auth Action	Meaning
“n”	1	4	auth_act_cont_d	CDA performed
“n”	1	3	auth_act_cont_d	Offline DDA performed
“n”	1	2	auth_act_cont_d	Issuer authentication not performed
“n”	1	1	auth_act_cont_d	Issuer authentication failed
“n”	2	4	auth_act_cont_d	Offline PIN verification performed
“n”	2	3	auth_act_cont_d	Offline PIN verification performed and PIN not successfully verified
“n”	2	2	auth_act_cont_d	PIN try limit exceeded
“n”	2	1	auth_act_cont_d	Last online transaction not completed
“n”	3	8	auth_act_cont_d	Lower offline transaction count limit exceeded
“n”	3	7	auth_act_cont_d	Upper offline transaction count limit exceeded
“n”	3	6	auth_act_cont_d	Lower cumulative offline amount limit exceeded
“n”	3	5	auth_act_cont_d	Upper cumulative offline amount limit exceeded
“n”	4	4	auth_act_cont_d	Issuer script processing failed
“n”	4	3	auth_act_cont_d	Offline data authentication failed on previous transaction
“n”	4	2	auth_act_cont_d	Go online on next transaction was set
“n”	4	1	auth_act_cont_d	Unable to go online

Users may set the “Auth Action” for any bit setting, according to business requirements.

The default base set for CCD cards is:

"4", 1, 1, auth_act_dcln_return_d

This is interpreted as Issuer Authentication failed on the last transaction, and will result in the transaction being declined and the card returned.

Vendor Impacts

For full details on the software impacts for this enhancement see Vendor Impacts in Section 4 (Base) of this document.

Reference Documents

- EMVCo – EMV 2004 version 4.1, May 2004

Locked ATD Records Causing Cutover Failure Enhancement

This section describes the implementation requirements associated with the Locked ATD Records Causing Cutover Failure Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Locked ATD Records Causing Cutover Failure Enhancement.

The BASE24-atm Settlement module has been amended to enhance ATM Settlement (SETL) so that Cutover will not fail because one ATD/TDF record is locked. ATM Settlement can now be configured to skip records in this condition.

ATM Settlement will read a new optional LCONF parameter ATM-SETL-ATD-LOCKED-RECS-SKIP. If the new LCONF param is set to "N" then locked ATD/TDF records will not be skipped during cutover, thus ATM Settlement could fail because of a locked ATD/TDF record during cutover.

If the new LCONF param is set to "Y" then locked ATD/TDF records will get skipped, thus ATM Settlement will not fail because of a locked ATD/TDF record.

An EMS event is raised when a locked ATD/TDF record is skipped.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60SETL.SETLS and SETLG.
- Updated event (log) message manual BA60LOGM.ATSETL

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNLOCK.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFAT file located on BA60MISC subvolume.

<u>ATM-SETL-ATD-LOCKED-RECS-SKIP</u>

Identifies whether the BASE24-atm Settlement module will skip locked ATD/TDF records during cutover.

A value of “Y” (yes) indicates that locked ATD/TDF records will get skipped. Any other value indicates that locked ATD/TDF records will not get skipped. The default is “N” (no).

Y = Skip locked ATD/TDF records during cutover and raise an EMS event.

N = Do not skip locked ATD/TDF records during cutover. The default value is “N”.

Database Configuration

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not Applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not Applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not Applicable

Reference Documents

Not Applicable

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Shared 912 AKDS Certificate Support

This section describes the implementation requirements associated with the Shared 912 AKDS Certificate Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared 912 AKDS Certificate Support enhancement.

The BASE24-atm Diebold 10XX/478X device handler has been enhanced to support the Shared 912 Automated Key Distribution System (AKDS) Certificate Support enhancement. Changes to an existing bindable extension, AT60AW10.C10WAKDS, have been developed to support the Shared 912 AKDS Certificate Support enhancement.

This enhancement enables customers to perform AKDS downloads from the Diebold 10XX/478X device handler to Diebold ATMs loaded with firmware that conforms to the Shared 912 Certificate specification created by Wincor Nixdorf.

As chosen by Diebold, the Certificate Authority (CA) to be used will be Digital Signature Trust Co. (DST).

Prerequisite: The Transaction Security Services (TSS) version 05.4 or greater must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to the ATM Constant DDL file (BA60DDL.DDLACNST)

All BASE24-atm Diebold 10XX/478X device handler customers:

Minor inline changes to the main device handler module AT601000.C1000S, to amend a hook to a Shared 912 AKDS procedure.
Minor inline changes to AT601000.C1000G
Updated event (log) message manual BA60LOGM.AT1000

All BASE24-atm Diebold 10XX/478X device handler customers using the existing Shared 912 Signature functionality:

Minor inline changes to the Shared 912 AKDS globals file AT60AW10.C10WAKDG
Inline changes to the Shared 912 AKDS Source file AT60AW10.C10WAKDS to add the functionality required for the Shared 912 Certificate Support enhancement.

All BASE24-atm Diebold 10XX/478X device handler customers requiring the new

Shared 912 Certificate Support functionality:

An extension subvolume, AT60AW10, with all of the files required to support the Shared 912 AKDS Certificate Support functionality.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNWDCT.

CUSTMACS Changes

For customers who do not currently support the Shared 912 Signature enhancement, the following existing Make flag should be set to true to bind the Shared 912 AKDS module to the Diebold 10XX/478X device handler.

- atm_dh_c1000_shrd_akd_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement. Note that these params already exist for the Diebold 10XX/478X AKDS module and are now being used by the Shared 912 AKDS module as well.

<u>ATM-DH-AKDS-TSS-TIMER</u>

This param specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

<u>ATM-DH-1000-AKDS-SERIAL-NUM-VRFY</u>

This param specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the fields in the files that may need to be modified to support this enhancement. Note that these fields already exist, and were not added for this enhancement. They are simply being used by the Shared 912 AKDS Certificate Support enhancement, too. The settings of each field will depend on the configuration desired by the institution.

ATM Terminal Definition Files (ATD)

The Native Message Format field, on screen 1, is used by the Shared 912 AKDS Certificate Support enhancement.

For each terminal that runs firmware that conforms to the Shared 912 Certificate specification created by Wincor Nixdorf, ensure that the Native Message Format field is set to “1”. This will ensure that Shared 912 AKDS or Shared NDC+ is performed instead of standard Diebold AKDS or standard NCR AKDS processing.

Screen Name:	NATIVE MESSAGE FORMAT
Screen:	ATD Screen 1
DDL Name:	ATDS1-CORE.NATV-MSG-FRMT

The Hardware field, on screen 1, is also used by the Shared 912 AKDS Certificate Support enhancement.

This field will be updated by the Diebold 10XX/478X device handler during Shared 912 AKDS Certificate processing, and set to the value that is returned by the ATM in response to an “EPP Capabilities” request message.

Screen Name:	HARDWARE
Screen:	ATD Screen 1
DDL Name:	ATDD1-CORE.HDWR
Valid Values are:	0 – Unknown 1 – Wincor 2 – NCR 3 – Diebold 4 – Fujitsu 5 – De La Rue 6 – Triton 7 – Olivetti
Default Value:	3 if the device type is 30 (Diebold) 2 if the device type is 22 (NCR) 0 for all other device types.

The Key Signer field on screen 22 is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Primary Datasource (HSPKPD). If multiple Host Public Keys are required for an individual vendor's ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

Screen Name:	KEY SIGNER
Screen:	ATD Screen 22
DDL Name:	ATDS1-CORE.KEY-SIGNER

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed. The default value is "HOST".

The EPP Format field, also on screen 22, is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. This field will be updated by the Diebold 10XX/478X device handler during Shared 912 AKDS Certificate processing, and set to the value that is returned by the ATM in response to an "EPP Capabilities" request message.

Screen Name:	EPP FORMAT
Screen:	ATD Screen 22
DDL Name:	ATDD1-CORE.EPP-FORMAT
Valid Values are:	0 – No RKL Supported 1 – Signature 2 – Certificate 3 – Both
Default Value:	2 if the device type is 30 (Diebold) 1 if the device type is 22 (NCR). 0 for all other device types.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM implementing AKDS.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS), version 05.4 or greater, must be installed in order to use this enhancement.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Security Device Configuration in the Transaction Security Services database</u>

For each security device to be used for public key cryptography (PKI) functions, the 'Enable PK'I field must be enabled on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not enabled for a security device, it is not used for PKI functions. This enables the consolidation of all PKI functions in just one or a few devices.

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain certificates signed by the CA’s primary private key for both the EPP’s public signature verification key, and its public encipherment key, certificates signed by the CA’s secondary private key for both the EPP’s public signature verification key and its public encipherment key, along with a certificate signed by the CA’s primary private key containing the CA’s primary public signature verification key, and a certificate signed by the CA’s secondary private key containing the CA’s secondary public signature verification key. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Primary Datasource (HSPKPD). Use the “Key Generate (KEYGEN) Command” to generate and store this RSA key pair.

The public key must be sent to the appropriate Certificate Authority to be verified and signed. The Certificate Authority must return primary and secondary certificates, containing the BASE24's public key, as well as the primary and secondary certificates containing the Certificate Authority's own public keys. These values must be entered into the Transaction Security Services database.

As chosen by Diebold, the Certificate Authority (CA) to be used will be Digital Signature Trust Co. (DST).

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the Diebold 10XX/478X device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the Shared 912 AKDS certificate enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the Diebold 10XX/478X device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Shared 912 AKDS Certificate Support enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared 912 AKDS Certificate Support enhancement.

- Diebold ATMs running firmware compliant with Wincor Nixdorf's Shared Certificate Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for RKL release 1.5.
- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to “off” in the security policy delivered from the factory due to security exposure. BASE24

requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in the system security policy using the 108 and 109 commands:

- Generate RSA key pair (Command 120)
- Generate AKB for root public key (Command 122)
- Verify public key and generate AKB of public key (Command 123)
- Generate digital signature (Command 124)
- Verify digital signature (Command 125)
- Generate message digest (Command 126)
- Generate ATM master key and encrypt with public key (Command 12F)

- Thales e-Security version 6.02 or greater firmware and the optional DSP plug-in (for RSA functions) provide public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:
  - Generate a key (Command A0)
  - Generate RSA key set (Command EI)
  - Load a secret key (Command EK)
  - Generate a MAC on a public key (Command EO)
  - Validate a certificate and generate MAC on its public key (Command ES)
  - Generate a signature (Command EW)
  - Validate a signature (Command EY)
  - Export a DES key (Command GK)
  - Hash a Block of Data (GM)

Reference Documents

- Atalla: ***Atalla Key Block Banking Command Reference Manual*** (524130-002, July 2002)
- Thales: ***Host Security Module RG7000 Programmer's Manual*** (1270A514 Issue 4, November 2001)
- ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL*** (Release 1.5. 15th December 2004)

Shared 912 Offline PIN Synchronization

This section describes the changes associated with the Shared 912 Offline PIN Synchronization enhancement for BASE24-atm. The standard BASE24-atm Shared 912 EMV Extension device handler module has been modified to support this enhancement.

Section 2 contains a functional overview of the Shared 912 Offline PIN Synchronization enhancement.

EMV cards have the capability to hold the cardholder’s PIN within the Integrated Circuit Card (ICC). This allows “offline PIN verification” (i.e. where the PIN entered by the cardholder is verified by the ICC) to be offered as a Cardholder Verification Method (CVM) for electronic transactions. Many payment terminals will not be enhanced to offer offline PIN verification, and so “online PIN verification” (i.e., where the PIN entered by the cardholder is verified by the issuer’s system) will continue to be widely used.

Where both online and offline PIN verification can be used as the CVM for an EMV card, cardholder confusion would result if the PIN held on the card’s ICC differed from the PIN held on the issuer’s system. Also, if a cardholder exceeds the allowable bad PIN tries when entering the PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide offline PIN management functionality.

The Offline PIN Synchronization enhancement introduces two new administrative functions for card issuers.

- Support for PIN change on the ICC, where the cardholder wants to change their PIN to a new value.
- Support for PIN unblocking on the ICC, where the PIN Try Counter on the card has reached zero and needs to be reset to the normal value (the PIN Try Limit).

This additional functionality is achieved by delivering a script(s) to the card during an authorization response message.

BASE24-atm Offline PIN Synchronization support includes both PIN change and PIN unblock (offline PIN try counter reset). BASE24-atm also ensures that the issuer system is notified if the card cannot be updated with the PIN management information in the script message.

PIN Unblock

The PIN on the card may be unblocked either implicitly or explicitly.

- An EMV transaction request initiated at an ATM contains data retrieved from the EMV Integrated Circuit card. This data includes the Card Verification Results (CVR), which contains an indicator (byte 3, bit 7) of whether the cardholder has exceeded the maximum number of PIN tries allowed for offline transactions. The checking of this indicator on all EMV transactions allows BASE24 to provide an "implicit" mechanism to unblock the PIN on the card.
- A PIN Unblock transaction request may be initiated by a customer, using an EMV card at an EMV-capable terminal. This provides for an "explicit" mechanism to unblock the PIN on the ICC.

A Host system may reset the bad PIN try counter on an EMV card by including a 'PIN Unblock' script command in any EMV transaction response message. In BASE24, issuers are able to specify whether offline PIN unblocking is permitted.

- A parameter, configurable by card prefix, allows an issuer to specify whether "implicit" offline PIN unblocking is permitted. One setting of this parameter requires the cardholder perform an "explicit" PIN Unblock transaction or PIN Change transaction to reset the bad PIN try counter on the card.
- Another parameter, configurable by card prefix, allows an issuer to specify whether "explicit" offline PIN unblocking is permitted.

BASE24 is capable of generating a 'PIN Unblock' script command, either "implicitly" (if byte 3, bit 7 of the CVR is set to '1') or "explicitly" (in response to a 'PIN Unblock' transaction request). A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a 'PIN Unblock' script should be generated. A 'PIN Unblock' script command may be returned to the card only if the cardholder's online PIN is verified successfully and the transaction's Authorization Request Cryptogram (ARQC) is verified successfully.

BASE24 is also capable of generating a 'PIN Change' script command in response to a 'PIN Unblock' transaction request. For example, having forgotten the PIN, a cardholder may be issued a new PIN (but not a new card). This processing will then allow a cardholder to resynchronize the ICC held PIN with the PIN held by the issuer. (Note that a 'PIN Change' script command will unblock the PIN on the ICC in addition to changing the PIN.) A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a 'PIN Change' script should be generated. A 'PIN Change' script command may be returned to the card only if the cardholder's PIN, and the transaction's Authorization Request Cryptogram (ARQC) are successfully verified.

PIN Change

The PIN on the card may be changed using an "explicit" mechanism only. A PIN Change transaction request (and optionally a PIN Unblock transaction request), which may be initiated by an EMV-card at an EMV-capable ATM, provides the required functionality.

A Host system may change the PIN on an EMV card by including a 'PIN Change' script command in a transaction response message. A parameter in BASE24, configurable by card prefix, allows an issuer to specify whether an offline PIN change is permitted.

The parameter also indicates whether the new PIN data should be logged to the BASE24 database, as this information may be required if the card subsequently has to be re-issued with the new PIN. Current PIN Change transaction logic allows the new PIN verification digits to be written to the BASE24-atm Transaction Log File (TLF). This information can then be extracted and provided to a Card Management System, which may require the new PIN verification digits if the card subsequently has to be re-issued. For Visa PVV PIN verification, however, the new PIN cannot be derived from the new PVKI/PVV; therefore, additional information (e.g. the new PIN data itself) would be required.

BASE24 is capable of generating a 'PIN Change' script command, if the PIN Change request is processed successfully (i.e. the cardholder's PIN verification digits are updated successfully) in the BASE24 database. A parameter in BASE24, configurable by card, allows an issuer to specify the conditions under which a 'PIN Change' script should be generated. A 'PIN Change' script command may be returned to the card only if the cardholder's online PIN is verified successfully and the transaction's Authorization Request Cryptogram (ARQC) is verified successfully.

For an EMV card that does not contain an active offline PIN, it would be inappropriate to return a 'PIN Change' script command, as the card may not be able to process this type of script command. Conversely, for an EMV card that does contain an active offline PIN, it would be inappropriate to authorize non-EMV PIN Change transactions, as a 'PIN Change' script command could not be returned in the transaction response (resulting in non-synchronization of the online PIN and the offline PIN). Another parameter in BASE24, configurable by card, allows an issuer to specify whether synchronization of the online and offline PIN is required for the card. The parameter specifies the action to take when a PIN Change request (either EMV or non-EMV) is performed on the EMV card.

Script Failure Notification

Once authorized by BASE24 or the issuer Host system, it is possible for script processing to fail at the ATM.

If BASE24 is notified that the card failed to unblock the offline PIN, following “implicit” PIN Unblock processing, BASE24 will raise an event message so that appropriate action can be taken.

If BASE24 is notified that the card failed to change or unblock the offline PIN (following a PIN Change transaction or an “explicit” PIN Unblock transaction), then the PIN Change/Unblock transaction is reversed. This has the effect of either restoring the cardholder’s original PIN verification digits to the BASE24 database, or notifying the issuer that the PIN on the card has not been unblocked. Similarly, the reversal is forwarded to an issuer Host system (if external to BASE24), if the BASE24 system did not authorize the transaction.

Prerequisite: The ATM Terminal Definition Files (ATD) Infrastructure enhancement must be installed prior to implementing the Shared 912 Offline PIN Synchronization enhancement in the Shared 912 device handler. In the ATD Infrastructure enhancement, the ATD records were expanded and new offsets were added.

Prerequisite: The Transaction Security Services (TSS) enhancement must be installed prior to implementing the Shared 912 Offline PIN Synchronization enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 2)***.

Prerequisites: The NCR 5XXX Device Handler and the Shared NDC EMV module are required.

Software Impacts

All BASE24-atm Shared 912 device handler customers:

Hooks to EMV procedures from the main device handler module (AT601000.C1000S)
Inline changes to AT601000.C1000D
Added a new load group to AT601000.CN100DOC
Added new states and screens to AT601000.CNFGIX

All BASE24-atm Shared 912 device handler EMV customers:

Inline changes to AT60IW10.C10WEMVG and C10WEMVS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNDPIN.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token File (TKN)

The following BASE24 Base token is used for this enhancement:

- The EMV Issuer Script Results Token (Token ID “BJ”) is used by the Diebold 10XX ATM Device Handler. The EMV Issuer Scripts Results Token contains information about the processing of EMV Script data. This token will be created by the acquirer process (e.g. device handler or interchange). It will contain information about the results of EMV script processing.

If the token for this enhancement is not to be logged to the TLF or BASE24 Interchange Log File (ILF), the appropriate token records must be configured.

If the token for this enhancement is to be extracted with the TLF or ILF, sent to the host (via HISO), or passed through BIC ISO Interchange Interface, the appropriate token records must be configured.

Receipt File (RCPT)

A new PIN Unblock receipt template is required for this enhancement.

Please refer to AT60MISC.RCPT for an example of the new template required.

Configuration Requirements

The following configuration requirements apply when the PIN Change transaction is authorized on BASE24-atm:

- The PIN verification method must be IBM 3624-format or Visa PVV.
- The BASE24-atm authorization method must be Positive or Positive Balance, so the cardholder has a Cardholder Authorization File (CAF) record.
- PIN verification digits must be carried in the CAF.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

ATMs

Where they are directly connected to BASE24, ATMs supporting offline PIN changes must be compliant with Diebold's EMV Smart Card application:

- Diebold's EMV Smart Card Application requires a Diebold ix, Opteva Series, or foreign vendor ATM, operating in 912 mode, equipped with the following:
    - HTP or newer processor board
    - Any EMV Level 1 certified Smart Card reader
    - Microsoft Windows NT (with Service Pack 6) or Microsoft Window 2000 or Microsoft Windows XP Embedded (XPe) operating system
    - Communication Subsystem (CSS) required for TCS Plus
    - Agilis Base Communications (ABC) 4.0
    - Terminal Control Software (TCS Plus 1.4.0 or greater) and/or Agilis 91x Software, version 1.2 or greater for Opteva terminals or version 1.1 or greater for Diebold ix series terminals
    - EMV Kernel, latest version

Security Devices

In order to generate a PIN change script command, Transaction Security Services (TSS) is required to support communication with the HSMs. The following Security devices and software are required:

- Thales HSMs with version 6 software or higher (supporting the enhanced ‘KU’ command)
- Atalla HSMs (supporting the new ‘351’ command)

Diebold 10XX Configuration File (CNFGIX)

The Diebold 10XX Configuration File (CNFGIX) requires new states and screens to support the new EMV PIN Unblock transaction type. For more detailed information, see the CN100DOC file on the AT601000 subvolume. A default set of screens is provided, but these may require specific modifications depending upon the institution’s requirements.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Host/Co-Network Impacts are incurred by those BASE24 users implementing this enhancement. Modifications to existing files and processing impact the BIC ISO Interface, ISO Host Interface and Super Extract processes.

<u>BIC ISO Interface Process</u>

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a ‘PIN Change’ or ‘PIN Unblock’ script) may be included in the BASE24 external message. When the BIC ISO module sends a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token. When the BIC ISO module receives a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token.

If the EMV Issuer Script Results Token is configured to be included in the ISO message to the co-network, this change to the message needs to be handled by the co-network programs.

<u>ISO Host Interface Process</u>

In approved PIN management transaction responses, the EMV Response Token and EMV Script token (containing a ‘PIN Change’ or ‘PIN Unblock’ script) may be included in the BASE24 external message. When the HISO module sends a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token. When the HISO module receives a PIN Change or PIN Unblock response message, the SEND-CRD-BLK field in the EMV Response token in field S-126 will contain either ‘C’ or ‘U’ to indicate the type of script present in the EMV Script token.

If the EMV Issuer Script Results Token is configured to be included in the ISO message to the host, this change to the message needs to be handled by the host programs.

Super Extract Process

The new EMV PIN Unblock transactions are supported with an internal transaction code of "820000". Hosts should be analyzed for any impacts that may result from receiving TLF Extract records containing the new transaction code.

If the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the TLF and ILF Extracts, BASE24 users must consider this additional data in host programs that use the data from the log files.

User/Operational Impacts

Disk Impacts

The TLF and ILF require additional disk space if the PIN Change Token (containing new PIN data) and/or new EMV Issuer Script Results Token is configured to be included in the transaction records logged to these files. The total impact varies for each BASE24 system depending on the transaction volume in the BASE24 system.

Operational Impacts

New event (log) messages were created as result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Performance Impacts

If BASE24 is configured to generate the ‘PIN Change’ and/or ‘PIN Unblock’ script command, then the additional HSM command will have an impact on system performance.

Vendor Impacts

Not applicable

Reference Documents

- ***Agilis 91X/TCS EMV Programmers' Guide***, Version 1.0.0, Preliminary Draft Copy, November 2003
- ***Agilis 91X/TCS EMV User's Guide***, Version 1.0.0, Preliminary Draft Copy, November 2003
- ***Diebold 91X Terminal Control Software (TCS), TCS Plus, and 91X TCS CSP Terminal Programming Manual***, July 2002, edition N
- Europay, MasterCard, Visa (EMV) Integrated Circuit Card Specification for Payment Systems (Books 1–4), version 4.0 December 2000

Shared NDC+ AKDS Certificate Support

This section describes the implementation requirements associated with the Shared NDC+AKDS Certificate Support enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared NDC+ AKDS Certificate Support enhancement.

The BASE24-atm NCR 5XXX device handler has been enhanced to support the Shared NDC+ Automated Key Distribution System (AKDS) Certificate Support enhancement. Changes to an existing bindable extension, AT60AWN5.N50WAKDS, have been developed to support the Shared NDC+ AKDS Certificate Support enhancement.

This enhancement enables customers to perform AKDS downloads from the NCR 5XXX device handler to Diebold ATMs loaded with firmware that conforms to the Shared NDC+ Certificate specification created by Wincor Nixdorf.

As chosen by Diebold, the Certificate Authority (CA) to be used will be Digital Signature Trust Co. (DST).

Prerequisite: The Transaction Security Services (TSS) version 05.4 or greater must be installed prior to implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to the ATM Constant DDL file (BA60DDL.DDLACNST)

All BASE24-atm NCR 5XXX device handler customers:

Minor inline changes to the main device handler module AT60N50.N50S, to amend a hook to a Shared NDC+ AKDS procedure.
Minor inline changes to AT60N50.N50G
Updated event (log) message manual BA60LOGM.AT5070

All BASE24-atm NCR 5XXX device handler customers using the existing Shared NDC+ Signature functionality:

Minor inline changes to the Shared NDC+ AKDS globals file AT60AWN5.N50WAKDG
Inline changes to the Shared NDC+ AKDS Source file AT60AWN5.N50WAKDS to add the functionality required for the Shared NDC+ Certificate Support enhancement.

All BASE24-atm NCR 5XXX device handler customers requiring the new Shared NDC+ Certificate Support functionality:

An extension subvolume, AT60AWN5, with all of the files required to support the Shared NDC+ AKDS Certificate Support functionality.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNWNCT.

CUSTMACS Changes

For customers who do not currently support the Shared NDC+ Signature enhancement, the following existing Make flag should be set to true to bind the Shared NDC+AKDS module to the NCR 5XXX device handler.

- atm_dh_n50_shrd_akd_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param records that may be added to the LCONF to support this enhancement. Note that these params already exist for the NCR 5XXX

ATM-DH-AKDS-TSS-TIMER

This param specifies the number of seconds the device handler waits for a reply from Transaction Security Services before aborting the load. This is an optional param. Valid values are 10 through 600. If the param is not found or is invalid, the default value of 600 is used.

ATM-DH-N50-AKDS-SERIAL-NUM-VRFY

This param specifies whether or not Transaction Security Services will verify the Encrypting PIN Pad's (EPP's) serial number during the exchange of public keys between the EPP and BASE24. This is an optional param. Valid values are "Y" (yes, verify the serial number) and "N" (no, do not verify the serial number). If the param is not found or is invalid, the default value of "N" is used.

Database Configuration

This section describes the fields in the files that may need to be modified to support this enhancement. Note that these fields already exist, and were not added for this enhancement. They are simply being used by the Shared NDC+AKDS Certificate Support enhancement, too. The settings of each field will depend on the configuration desired by the institution.

<u>ATM Terminal Definition Files (ATD)</u>

The Native Message Format field, on screen 1, is used by the Shared NDC+AKDS Certificate Support enhancement.

For each terminal that runs firmware that conforms to the Shared NDC+ Certificate specification created by Wincor Nixdorf, ensure that the Native Message Format field is set to “1”. This will ensure that Shared 912 AKDS or Shared NDC+ is performed instead of standard Diebold AKDS or standard NCR AKDS processing.

Screen Name:	NATIVE MESSAGE FORMAT
Screen:	ATD Screen 1
DDL Name:	ATDS1-CORE.NATV-MSG-FRMT

The Hardware field, on screen 1, is also used by the Shared NDC+AKDS Certificate Support enhancement.

This field will be updated by the NCR 5XXX device handler during Shared NDC+ AKDS Certificate processing, and set to the value that is returned by the ATM in response to an “EPP Capabilities” request message.

Screen Name:	HARDWARE
Screen:	ATD Screen 1
DDL Name:	ATDD1-CORE.HDWR
Valid Values are:	0 – Unknown 1 – Wincor 2 - NCR 3 - Diebold 4 - Fujitsu 5 – De La Rue 6 – Triton 7 – Olivetti
Default Value:	3 if the device type is 30 (Diebold) 2 if the device type is 22 (NCR) 0 for all other device types.

The Key Signer field on screen 22 is used to indicate the name to be used in the HOST field of the TSS Host Public Key Security Primary Datasource (HSPKPD). If multiple Host Public Keys are required for an individual vendor's ATMs, the Key Signer field can be used to assign a separate HOST name for each of those Public Keys.

Screen Name:	KEY SIGNER
Screen:	ATD Screen 22
DDL Name:	ATDS1-CORE.KEY-SIGNER

This field allows up to 16 alphanumeric characters of free-format text to be entered. Embedded spaces are allowed. The default value is "Host".

The EPP Format field, also on screen 22, is used to indicate whether the EPP, and therefore the ATM, supports Signature- or Certificate-based AKDS processing. This field will be updated by the NCR 5XXX device handler during Shared NDC+ AKDS Certificate processing, and set to the value that is returned by the ATM in response to an "EPP Capabilities" request message.

Screen Name:	EPP FORMAT
Screen:	ATD Screen 22
DDL Name:	ATDD1-CORE.EPP-FORMAT
Valid Values are:	0 – No RKL Supported 1 – Signature 2 - Certificate 3 – Both
Default Value:	2 if the device type is 30 (Diebold) 1 if the device type is 22 (NCR). 0 for all other device types.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM implementing AKDS.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The Transaction Security Services (TSS) version 05.4 or greater must be installed in order to use this enhancement.

File Conversions

Not applicable Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, the ‘Enable PK’I field must be enabled on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not enabled for a security device, it is not used for PKI functions. This enables the consolidation of all PKI functions in just one or a few devices.

Setup for Transaction Security Services

Initial setup procedures must be carried out for TSS before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain certificates signed by the CA’s primary private key for both the EPP’s public signature verification key, and its public encipherment key, certificates signed by the CA’s secondary private key for both the EPP’s public signature verification key and its public encipherment key, along with a certificate signed by the CA’s primary private key containing the CA’s primary public signature verification key, and a certificate signed by the CA’s secondary private key containing the CA’s secondary public signature verification key. All of these are created when the EPP is manufactured.

BASE24's public/private key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Primary Datasource (HSPKPD). Use the “Key Generate (KEYGEN) Command” to generate and store this RSA key pair.

The public key must be sent to the appropriate Certificate Authority to be verified and signed. The Certificate Authority must return primary and secondary certificates, containing the BASE24's public key, as well as the primary and secondary certificates containing the Certificate Authority's own public keys. These values must be entered into the Transaction Security Services database.

As chosen by Diebold, the Certificate Authority (CA) to be used will be Digital Signature Trust Co. (DST).

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Power-Up Considerations

Prior to the Automated Key Distribution System enhancement, the NCR 5XXX device handler automatically performed a full load (or a load of just the PIN encryption and MAC keys) after receiving a power-up message from an ATM. With this enhancement the device handler will continue to do so, but there is an exception: The load will not be automatically performed if the ATM has an AKDS-enabled EPP and a master key has not yet been downloaded to the EPP. Instead, the device handler will generate an event message informing the operator that the master key must be loaded. The master key must be present before a PIN encryption key or MAC key can be loaded.

Financial Institution Table (FIT) Considerations

The local PIN verification key (PEKEY), if used, is encrypted under the master key and stored in the FIT. If a new master key is downloaded to an ATM, every PEKEY in the FIT must be re-encrypted under the new master key and the FIT must be downloaded to the ATM.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the Shared NDC+ AKDS certificate enhancement. AKDS does not support the downloading of the power-up COMM key (B-key) because this key is not currently supported by the Diebold 10XX/478X device handler. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this enhancement.

Performance Considerations

If the Shared NDC+ AKDS Certificate Support enhancement is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's Shared NDC+ AKDS Certificate Support enhancement.

- Diebold ATMs running firmware compliant with Wincor Nixdorf's Shared Certificate Specification – Wincor Nixdorf NCD/Diebold D91x Message Format Extension for RKL release 1.5.
- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. AKB products require the use of a Secure Configuration Terminal (SCT), release 4.0 or later. SCT 4.0 devices support the Atalla AKB products while remaining fully backward compatible with older Atalla releases. All of the public key cryptography commands are defaulted to “off” in the security policy delivered from the factory due to security exposure. BASE24

requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application); nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in the system security policy using the 108 and 109 commands:

  - Generate RSA key pair (Command 120)
  - Generate AKB for root public key (Command 122)
  - Verify public key and generate AKB of public key (Command 123)
  - Generate digital signature (Command 124)
  - Verify digital signature (Command 125)
  - Generate message digest (Command 126)
  - Generate ATM master key and encrypt with public key (Command 12F)

- Thales e-Security version 6.02 or greater firmware and the optional DSP plug-in (for RSA functions) provide public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The commands are as follows:
  - Generate a key (Command A0)
  - Generate RSA key set (Command EI)
  - Load a secret key (Command EK)
  - Generate a MAC on a public key (Command EO)
  - Validate a certificate and generate MAC on its public key (Command ES)
  - Generate a signature (Command EW)
  - Validate a signature (Command EY)
  - Export a DES key (Command GK)
  - Hash a Block of Data (GM)

Reference Documents

- Atalla: ***Atalla Key Block Banking Command Reference Manual*** (524130-002, July 2002)
- Thales: ***Host Security Module RG7000 Programmer's Manual*** (1270A514 Issue 4, November 2001)
- Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL (Release 1.5. 15th December 2004)

Shared NDC+ EMV Upgrade

This section describes the implementation requirements associated with the Shared NDC+ EMV Upgrade enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Shared NDC+ EMV Upgrade enhancement.

The BASE24-atm NCR 5XXX device handler has been modified to support the Shared NDC+ EMV Upgrade enhancement. Changes to an existing bindable extension, AT60IWN5.N50WEMVS, have been developed to support the Shared NDC+ EMV Upgrade enhancement.

This enhancement enables customers to support ATMs that run NDC+ emulation using the EMV format defined in the Wincor Nixdorf NDC/Diebold 91x Message Format Extension for EMV specification. ATMs that follow this format may be used exclusively or in combination with ATMs configured to follow the older format defined in the Wincor Nixdorf ProCash/NDC EMV Programmers Reference. Since Wincor terminals are compatible with both standards, the customer may change an ATM from one mode to the other simply by downloading the appropriate configuration data. Customers who have also licensed the standard BASE24 NCR 5XXX EMV extension may support NCR devices along with devices following either or both Wincor standards.

Software Impacts

All BASE24-atm NCR 5XXX device handler customers:

Inline changes to Make files ATN50M, N50DDLM
Inline changes to N50DDLS
Inline changes to N50G, N50S, and N50D.
Inline changes to RECIMGCS, SV50PRIS, RQN50STS, RQ50ACSS, SCRNACS, RQ50STS, SCRNST, SVLIBS, and SVN50S to support the new Wincor Chip Card State (e) and the Wincor EMV Application Identifier Table.
Updated event (log) message manual BA60LOGM.AT5070.
Updates to the configuration file CNFG5070 to support new load groups for Shared EMV support. These are described below under Database Configuration.

All BASE24-atm NCR 5XXX device handler customers using the existing Shared NDC+ EMV functionality:

Inline changes to ATIWN5MM
New files RQWEMAIM, RQEMWAIS, and SCNWEMAI to support the Wincor EMV Application Identifier Table.
Inline changes to N50WEMVG and N50WEMVS.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNIWN5.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

The NCR 5XXX configuration file CNFG5070 on the AT60N50 subvolume has been updated to include new terminal groups to support Shared EMV using the new format. These include state tables and other data as required for the following groups:

- 630 - Base EMV
- 631 - Encrypted PIN Change support
- 634 - Surcharging
- 635 - Non-Currency Dispense

Group 630 can be loaded directly on top of group 0. Groups 631-635 should be loaded on top of group 0 and group 630. The EMV groups that support standard NCR terminals and Wincor terminals using the older format (600-624) are not required for or compatible with the newer format.

Customers may have to modify or add to the provided data as required for their own processing. Configuring an ATM to use the new format requires use of the new Wincor EMV Chip Card state (e) table and the Wincor Application Identifier table. See the BASE24-atm EMV Support Manual and the appropriate Wincor documentation for information on these tables.

The Shared EMV module will dynamically monitor the configuration data being downloaded to each Shared EMV-compatible ATM in order to determine whether the ATM follows the older or the newer format. No configuration changes to BASE24 are required other than loading the correct groups.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

All Wincor hardware that runs in NDC+ emulation and supports EMV is compatible with both the older and newer formats, depending on the state tables that are downloaded. No new software or configuration changes at the ATM site are required to support this enhancement.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Load procedures may have to be modified to load the new terminal groups described above under Database Configuration.

Vendor Impacts

Wincor hardware is already compatible with the newly supported format. Since Wincor has made its standards available to other vendors, it is anticipated that other vendors will support it as well.

Reference Documents

- ***Wincor Nixdorf NDC/Diebold 91x Message Format Extension for EMV*** (Release 1.1. 20th December 2004)

Simultaneous Transactions Fraud Enhancement

This section describes the implementation requirements associated with the Simultaneous Transactions Fraud Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Simultaneous Transactions Fraud Enhancement.

The BASE24-atm Authorization module has been amended to help prevent a fraud situation known as the Simultaneous Transaction Fraud. This fraud situation occurs when two identical fraudulent cards are used at ATMs adjacent to each other, at exactly the same time.

Without these changes, it is likely that the full withdrawal daily limit can be dispensed at each of these ATMs, causing the Issuer Institution to debit the cardholder's account for double the expected amount of money.

These changes allow the Issuer Institution to configure a new LCONF param, ATM-RESP-LMTS-CHK, which if set, allows the BASE24-atm Authorization module to recheck the daily limits on a response from the Host, and so deny the second of the transactions to be processed by the Authorization module. An EMS event is raised, which allows the Issuer Institution to investigate the transaction, and manually post a reversal to the cardholder's account if necessary. This fraud scenario only affects customers using an Authorization level of online/offline, and an authorization method of CAF or CAF/PBF.

Software Impacts

All BASE24-atm customers:

- Inline changes to AT60AUTH.AUTHS, AUTHLIBS and AUTHG.
- Updated event (log) message manual BA60LOGM.ATAUTH

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFAT file located on BA60MISC subvolume.

ATM-RESP-LMTS-CHK

This param, used by the BASE24-atm Authorization module, specifies whether the BASE24-atm Authorization module should recheck the daily limits on the processing of any response from a Host, if the Authorization level is online/offline, and the Authorization method is CAF or CAF/PBF.

Y = Recheck the daily limits on the receipt of a response from the Host. If the limits have now been exceeded, deny the transaction back to the ATM.

N = Do not recheck the daily limits on the receipt of a response from the Host. Processing continues as it did before this enhancement was introduced. The default value is “N”.

Database Configuration

Not Applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not Applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not Applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

Vendor Impacts

Not Applicable

Reference Documents

Not Applicable

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Triton Mobile Top-UP Enhancement

This section describes the implementation requirements associated with the Triton Mobile Top-Up enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Triton Mobile Top-Up enhancement.

The BASE24-atm Triton Mobile Top-Up Device Handler extension enables cardholders to replenish, or top up, their mobile telephone accounts at an ATM. The replenishment transaction is a non-currency dispense transaction. A telecommunications services provider or an individual mobile operator, rather than the institution that owns the ATM or issues cards, is responsible for authorizing the minutes purchase, maintaining the consumer mobile telephone account, and calculating any taxes applied to the transaction. The BASE24-atm product authorizes the funds portion of the transaction.

NOTE: The BASE changes in the BASE24-atm Mobile Top-Up Integration optional enhancement must be installed prior to implementation of the Triton Mobile Top-Up enhancement. For more information refer to the ***BASE24 Release 6.0 Implementation Guide (Version 5)***. A complete list of the files affected by the release 6.0 version 5 mobile top-up support can be found in the SCN file BA60UD08.SCNMTOP.

Software Impacts

All BASE24 customers:

Changes to BA60MAKE.CUSTMACS and FLGSDOC

All BASE24-atm Triton customers:

Hooks in the main device handler module to support Mobile Top-Up (AT60TRIT.TRITONS)
Inline changes to AT60TRIT.TRITONG
Additional procedures in AT60TRIT.TRITOND
Inline changes to AT60TRIT.TRITONM
Updated event (log) message manuals BA60LOGM.ATTRIT

All BASE24-atm Triton customers taking the Mobile Top-Up enhancement:

New subvolume, AT60TTRT, with files for the Triton device handler Mobile Top-Up extension

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNTTRT.

Prerequisite: The Triton Device Handler, Non-Currency Dispense module and the Mobile Top-up extensions for BASE24-atm are required.

Customers who take this enhancement should also install the enhancement Triton Token Retrieval enhancement. The large tokens used for Mobile Top-Up will need more space than is available in the Last Message Area of the TDF record. See the section in this Implementation Guide for more information on this enhancement.

CUSTMACS Changes

Set the following flag to true to bind the Mobile Top-Up module to the Triton device handler.

atm_dh_triton_mtu_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to or already exist in the LCONF to support this enhancement. Samples of the parameters are provided in the LCONFBA and LCONFAT files located on BA60MISC subvolume.

Different param records can be specified for each device handler process if desired.

<u>MOBILE-PARAM-VERSION-NUM</u>

A value used by the device handler to determine whether the mobile top-up information stored at the device must be replaced during a download request/response. This param value must be manually incremented when one or more of the following parameters are changed, otherwise the device handler will not send the updated information to the device:

> CARRIER-VT*identifier*
>
> CARRIER-AMTS-VT*identifier*
>
> MOBILE-OPERATOR-ID-VT*identifier*

This param is required by the BASE24-atm Triton Mobile Top-Up add-on product. Valid values are 0001-9999. There is no default value.

CARRIER-AMTS-VT*identifier*

This param is used by the Triton device handler and should only be added for the definitions-based model of mobile top-up. It identifies the six amounts supported for this mobile operator.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with eight amounts and each amount must be six digits separated by semicolons.

Example: 000030;000050;000075;000000;000000;000000

Represents $30, $50, and $75 as valid amounts for this carrier.

CARRIER-VT*identifier*

This param is used by the Triton device handler. It identifies the four-digit mobile phone operator ID.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

The value of the param links a mobile operator with an entry in the Mobile Operator File (MOF) and enables the mobile operator to be identified for the transaction.

MOBILE-NUM-MAX-LGTH

This param, used by the Triton device handler, specifies the maximum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 11.

MOBILE-NUM-MIN-LGTH

This param, used by the Triton device handler, specifies the minimum length allowed for the mobile phone number entered by the cardholder. Valid values are 0 - 15. The default value is 10.

MOBILE-OPERATOR-ID-VT*identifier*

This param is used by the Triton device handler. It specifies the mobile phone operator IIN (7 digits) plus the mobile operator name (up to 20 characters) associated with the mobile phone operator.

The *identifier* is a two-digit numeric value, 00 – 08, representing the voucher type. It is also used to associate the Operator IIN, name, mobile phone operator ID, and valid amounts for each carrier.

Valid values must be formatted with a seven-digit operator IIN followed immediately by a mobile operator name.

Example: 1200991CARRIERNAME

Database Configuration

Mobile Operator File (MOF)

The Mobile Operator File (MOF) is a required file for the Mobile Top-Up for Triton Device Handler Enhancement and contains one record for each telecommunications provider supplying mobile top-up services for its customers. It defines details of the mobile operators and the services provided. Information on use of the MOF file may be found in the ***Base Files Maintenance Manual*****.**

Split Transaction Routing File (STRF)

The Split Transaction Routing File (STRF) is a required file for the Triton Mobile Top-Up Enhancement. It contains one record for each Transaction Subtype in the BASE24 network that requires unique routing. The STRF enables the routing of transaction requests to two or more destinations based upon information obtained from a device for a single transaction. Information on use of the STRF file may be found in the ***Base Files Maintenance Manual*****.**

Token file (TKN)

The following existing BASE24 tokens are utilized for this enhancement:

- The Non-Currency Dispense Token (token ID “A5”)
- The Pre-Pay Receipt Token (token ID “BF”)
- The Pre-Pay Top-Up Token (token ID “BI”)
- The Transaction Subtype Token (token ID “BM”)
- The Split Transaction Routing Token (token ID “BR”)
- The Pre-Pay Original Data Token (token ID “BX”)

If the tokens for this enhancement should not be logged to the TLF or the ILF, the appropriate token records must be configured in the Token file.

If the tokens for this enhancement are to be extracted with the TLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Configure the AT60TTRT.TTRTNAMS table. This table controls mapping a Service Provider ID to a BASE24 Transaction Subtype. ALL IDs supported at the Triton Terminal for Mobile Top-Up Transactions must be entered in this table.

Device Configuration

To utilize this enhancement, terminals must run a version of firmware that supports Mobile Top-Up transactions.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

In a standard BASE24-atm system, the device handlers always route transactions directly to the BASE24-atm Authorization module. In a mobile top-up environment, the device handler must accommodate the possibility of multiple authorization destinations. The BASE24-atm Triton Device Handler will route mobile top-up transactions to the destination specified in the Split Transaction Routing File (STRF). This destination should be the process name of the BASE24-atm Transaction Context Manager (TCM).

The TCM will utilize the new Split Transaction Routing File (STRF) for purposes of routing. When the transaction is not a mobile top-up transaction, normal transaction processing and routing occurs.

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

<u>Operational Impacts:</u>

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate response for a given message. Refer to the event message documentation files on the BA60LOGM subvolume.

If the contents of any of the following parameters change for a Triton device handler process, an additional step must be performed.

CARRIER-VT*identifier*

CARRIER-AMTS-VT*identifier*

MOBILE-OPERATOR-ID-VT*identifier*

The contents of the LCONF param MOBILE-PARAM-VERSION-NUM for the same process(es) must also be incremented. This param is a value used by the device handler to determine whether the mobile top-up information stored at the device must be replaced during a download request/response. If the param is not changed, the device handler will not send the updated information to the device.

<u>System Limitations</u>

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the TLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the BASE24 Tokens Manual provides information on assessing the impacts of tokens on log file disk space requirements.

<u>Performance Impacts</u>

Performance will be impacted in systems running mobile top-up transactions due to the increase in the transaction path length. The transaction path is longer because mobile top-up transactions are routed to two destinations (i.e., a funds authorizer and a mobile I/F). Additionally, there may be multiple interactions between the BASE24 device handler process, the BASE24-atm Transaction Context Manager process, and the Mobile Operator Interface process prior to the funds authorization and prior to the final top-up authorization. While each interaction with the device handler process resets the timer, BASE24-atm customers may need to examine the value of the LCONF parameter ATM-DH-AUTH-TIMER and determine if the increased length of the transaction path necessitates an increase in the value assigned to ATM-DH-AUTH-TIMER.

Vendor Impacts

Not applicable

Reference Documents

- ***Triton Terminal Communications Protocol And Message Format Specification,*** *version TSDC 5.29, July 14, 2006*

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6: BASE24-pos

Section 6 discusses the changes and enhancements made to BASE24-pos in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24-pos users must consider when upgrading from release 6.0 version 6.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-pos users, along with the associated conversion and implementation requirements that all BASE24-pos users must address.
- Optional enhancements describe the changes that affect only those BASE24-pos users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-pos users must address to implement the enhancements.

Section 6 must be used in conjunction with Section 4 to identify the changes that BASE24-pos users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-pos users must address in order to upgrade to release 6.0 version 7.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

- General Changes
- AMEX SE Number Additional Validation
- APACS MACing
- APACS Triple DES
- Institution Level Security for CRT Auth Enhancement
- Service Code Validation Enhancement
- SPDH & RTAU Enhancements for Visa and MasterCard
- SPDH New Fields to Support Network Requirements Enhancement
- Visa DUKPT MACing Enhancement

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

- The BASE24-pos Terminal Data files (PTD) server has been modified to delete the BASE24-pos Terminal Data Dynamic Scratch Pad File (PTDD2) record when a terminal record is deleted using the PTD file maintenance screen. An Online Maintenance File (OMF) record containing only the OMF header is logged for the deleted PTDD2 record. Previously, only the terminal's BASE24-pos Terminal Data Dynamic General File (PTDD1) record and BASE24-pos Terminal Data Static General File (PTDS1) record were deleted.
- Add Additional Values to PTD/ATD:

  PTD Screen 17 has been enhanced to support the configuration of the EMV Terminal Capabilities as defined in EMV v4.1 (2004).

  PTD Screen 19 has been enhanced to support configuration options for the ***cardholder authentication capability*** and the ***terminal output capability***, defined in the POS Data Code (data element 022) in the ISO 8583 (1993) Standard. The cardholder authentication capability indicates the method(s) supported by the terminal for verifying the identity of the cardholder, and the terminal output capability indicates whether the terminal can display messages, print messages or both.

  New token fields have been implemented to carry the new data on PTD Screen 19 through the BASE24-pos system.

  New values have been added to the existing Cardholder ID Method in the SPDH message (subFID 6X) to allow the terminal to specify the cardholder authentication method used, from which it is possible to derive the cardholder authentication entity.
- Disallow Leading Zero Approval Code. The BASE24-pos Router/Authorization logic that creates an approval code has been enhanced to manipulate the generated value to ensure that the new approval code does not contain one or more leading zeros.
- The POS Retailer Definition File (PRDF) has been modified to include additional fields on screen 4 for the AMEX SE Number Additional Validation enhancement. The POS Retailer Definition File screen, requester and server have been modified to process these new fields. For further information, see **AMEX SE Number Additional Validation** later in Section 6.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.SVTDFLTG and SVTDFLTS
- Inline changes to BA60DDL.DDLFOMF
- Inline changes to BA60DDL.DDLPSTKN and BA60DEFS.DDLGDEFS

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPTDTG and SVPTDTS
- Inline changes to PS60AFT.SCRNPTD, RQPTDS, SVPTDTS, SVPTDTBL
- Inline changes to PS60DDL.DDLFPTD and DDLGPTD
- Inline changes to PS60RTAU.AUTHLIBS

All BASE24-pos SPDH customers:

- Inline changes to PS60SPDH.ASPDHS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

This section describes the files that have been affected by the **General Changes**.

<u>POS Terminal Data File (PTD)</u>

The following additional EMV Terminal Capabilities may be configured:

Screen Name:	NO CVM REQUIRED
Screen:	PTD Screen 17
DDL Name:	TERM-CAP (byte 2, bit 4)
Valid Values are:	Y and N (Default value is N)

Screen Name:	COMBINED DATA AUTHENTICATION
Screen:	PTD Screen 17
DDL Name:	TERM-CAP (byte 3, bit 4)
Valid Values are:	Y and N (Default value is N)

The following additional Terminal Capabilities may be configured:

Screen Name:	TERMINAL OUTPUT CAPABILITIES
Screen:	PTD Screen 19
DDL Name:	TERM-OUTPUT-CAP-IND
Valid Values are:	Space, 0, 1, 2, 3, 4 (default is Space)

Screen Name:	CARDHOLDER AUTHENTICATION
Screen:	PTD Screen 19
DDL Name:	CRDHLDR-AUTHN-CAP-IND
Valid Values are:	Space, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (default is Space)

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

SPDH devices may be enhanced to support the new values in subFID 6X in the SPDH message.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

POS Retailer Definition File (PRDF)

The PRDF conversion program must be run to convert the PRDF. Running the conversion program is required for any BASE24-pos customer utilizing the PRDF in transaction processing. Users must create the new PRDF file using the BADDLFUP file.

The PRDF conversion program on the PS60UC04 subvolume can be used to convert the PRDF records from the release 5.3 format to the release 6.0 version 7 format. The PRDF conversion program on the PS60UC0A subvolume can be used to convert the PRDF records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPRDFS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#prdcnv.t,name $prdm,nowait/ cnvprdfm –force –list –listreplace

Release 5.3 users should update the CNVPRDFR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVPRDFR obey file on the PS60UC0A subvolume with the appropriate information. The CNVPRDFR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0A on the BA60UC0A subvolume for release 6.0 version 6 customers.

POS Terminal Data File (PTD)

The PTD conversion program must be run to convert the PTD. Running the conversion program is required for any BASE24-pos customer utilizing the PTD in transaction processing. Users must create the new PTD file using the PSDDLFUP file.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTD records from the release 5.3 format to the release 6.0 version 7 format. The PTD conversion program on the PS60UC0A subvolume can be used to convert the PTD records from the release 6.0 version 6 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPTDS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#ptdcnv.t,name $ptdm,nowait/ cnvptdm –force –list –listreplace

Release 5.3 users should update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. Release 6.0 version 6 users should update the CNVPTDR obey file on the PS60UC0A subvolume with the appropriate information. The CNVPTDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume for release 5.3 customers and to the AAREAD0A on the BA60UC0A subvolume for release 6.0 version 6 customers.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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AMEX SE Number Additional Validation

This section describes the implementation requirements associated with the AMEX SE Number Additional Validation enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the AMEX SE Number Additional Validation enhancement.

The POS Retailer Definition File (PRDF) AFT processing modules have been enhanced to validate the contents of the Alternate Merchant Id field containing the American Express Service Establishment (SE) Number. The enhanced logic ensures that the field contains ten numeric digits, and validates the check digit.

The new processing is performed on the American Express SE Number before it is stored in a PRDF record in the ALTERNATE MERCHANT ID field for card type ‘AX’. No additional validation will be performed on either the RETAILER ID (the AMEX SE Number may be configured as the actual BASE24-pos Retailer Id) or the SERVICE ESTABLISHMENT # on PRDF Screen 1.

It is not possible to highlight the field in error on the PRDF screen when the check digit on the American Express SE Number is found to be incorrect. However, a message is output to the screen to identify the field in error.

Software Impacts

All BASE24 customers:

- Inline changes to BA60SRC.BAUTILS

All BASE24-pos customers:

- Inline changes to PS60DDL.DDLPRDF
- Inline changes to PS60AFT.SCRNPRDF, RQPRDFS, SVPRDFS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNAXSE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

POS Retailer Definition File (PRDF)

The following new field makes it possible to configure the enhanced validation (including check digit verification) on the American Express SE Number:

Screen Name:	AMEX SE NUMBER CHECK
Screen:	PRDF Screen 4
DDL Name:	SRV-EST-NUM-CHK
Valid Values are:	Space, N, Y (default is Space)

A full description of this field can be found in the ***BASE24-pos Files Maintenance Manual.***

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

POS Retailer Definition File (PRDF)

The PRDF conversion program must be run to initialize the new AMEX SE Number Check field. Details of this conversion program can be found in the **General Changes** earlier in Section 6.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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APACS MACing

This section describes the implementation requirements associated with the APACS MACing enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the APACS MACing enhancement.

The BASE24-pos APACS Device Handler has been enhanced to support APACS MACing via TSS. Previously, the BASE24-pos APACS Device Handler only supported MACing via direct calls to a hardware security module (HSM) via its own set of libraries.

Prerequisite: The minimum version of TSS required for the implementation of this enhancement is version 06.2. The Thales/Racal 8000 family HSM v2.3 is the only HSM that supports APACS MACing.

Software Impacts

All BASE24-pos customers:

- Inline changes to the TSS Command Message DDL file (BA60DDL.DDLGTSS)

All BASE24-pos APACS device handler customers:

- Inline changes to the APACS device handler module (PS60APDH.APACSDHS)
- Inline changes to the MAC Utilities Make file (PS60APDH.MACUTILM)
- Inline changes to the MAC Utlities (PS60APDH.MACUTILS)
- Inline changes to the EMV extension for the APACS device handler module (PS60IAPD.APDHEMVS)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNAPAC.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

BASE24-pos Terminal Data File (PTDF)

All key fields in the PTDF records must be updated to contain key locators. The key locators are used by TSS to read the Channel Security Data Source (CSEC) records to retrieve the actual key registers used by the APACS device handler's terminals.

TSS Channel Security Data Source (CSEC)

A CSEC record must exist for every set of key registers used by the APACS device handler's terminals in the BASE24 network. Normally, every terminal uses unique key registers so there would be one CSEC record for every default terminal ID in the PTDF records.

A PTDF file conversion program on the PS60UC05 subvolume (TSSPTDF) can be used to create or update the CSEC.

For more information about configuring the CSEC Data Source, refer to the ***BASE24 Transaction Security Services Processing Guide***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Security Device

APACS MACing is only supported when using Thales/Racal 8000 series v2.3 hardware security modules.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The minimum version of the BASE24-es Transaction Security Services (TSS) required for the implementation of this enhancement is version 06.2.

File Conversions

A PTDF file conversion is supplied with the TSS enhancement. There are no structural changes in the PTDF file, but TSS files can be created and data populated from the PTDF file. The PTDF file is updated with key locators to point BASE24-pos and TSS processes to the correct TSS records. The TSS data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

For more information on the PTDF file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSPTDFR file on the PS60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

APACS Triple DES

This section describes the implementation requirements associated with the APACS Triple DES enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the APACS MACing enhancement.

As a result of BASE24-es TSS being enhanced to support APACS MACing, the BASE24-pos APACS Device Handler can be configured to utilize TSS for all its current encryption needs. This then allows the BASE24 user to set up double length keys for both PIN and MAC processing between the APACS DH and an APACS70 compatible terminal.

No code changes have been made for this enhancement.

A Thales/Racal 8000 series v2.3 is the only hardware security module that supports APACS Triple DES MACing.

Prerequisite: The minimum version of TSS required for the implementation of this enhancement is version 06.2. If Triple DES PIN processing is to be used, Triple DES MACing must also be configured, thus requiring the changes for the APACS MACing enhancement described earlier in Section 6.

Software Impacts

No code changes have been made for this enhancement.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

<u>BASE24-pos Terminal Data File (PTDF)</u>

All key fields in the PTDF records must be updated to contain key locators. The key locators are used by TSS to read the Channel Security Data Source (CSEC) records to retrieve the actual key registers used by the APACS device handler's terminals.

<u>TSS Channel Security Data Source (CSEC)</u>

A CSEC record must exist for every set of key registers used by the APACS device handler's terminals in the BASE24 network. Normally, every terminal uses unique key registers so there would be one CSEC record for every default terminal ID in the PTDF records.

A PTDF file conversion program on the PS60UC05 subvolume (TSSPTDF) can be used to create or update the CSEC.

For more information about configuring the CSEC Data Source, refer to the ***BASE24 Transaction Security Services Processing Guide***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

<u>Security Device</u>

APACS Triple DES MACing is only supported when using a Thales/Racal 8000 series v2.3 hardware security module.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The minimum version of the BASE24-es Transaction Security Services (TSS) required for the implementation of this enhancement is version 06.2.

File Conversions

A PTDF file conversion is supplied with the TSS enhancement. There are no structural changes in the PTDF file, but TSS files can be created and data populated from the PTDF file. The PTDF file is updated with key locators to point BASE24-pos and TSS processes to the correct TSS records. The TSS data can be entered manually using the ACI Desktop Graphical User Interface instead of using the conversion programs.

For more information on the PTDF file conversion, refer to the AAREAD05 on the BA60UC05 subvolume and TSSPTDFR file on the PS60UC05 subvolume.

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Institution Level Security for CRT Auth Enhancement

This section describes the implementation requirements associated with the Institution Level Security for CRT Auth Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Institution Level Security for CRT Auth Enhancement.

The BASE24-pos CRT Authorization product has been modified to restrict access to referral transactions by retailer FIID on the Multiple Referral File (MRF) and POS Referral File (PRF) screens. The BASE24 user is allowed access to transactions for only those retailers whose FIIDs defined in the POS Retailer Definition File (PRDF) match one of the allowed FIIDs specified for the user on Security (SEC) screen 2.

The BASE24-pos CRT Authorization product has also been modified to support the configuration of CRT Authorization device handler processes in multiple logical networks when those logical networks share a common Pathway environment. Each device handler process is configured as a separate server class in Pathway by using the new server class name SERVER-CRTDxxxx, where xxxx is the identifier of the logical network where the device handler process resides. This allows transactions originating from the Online Transaction (OLT) screen and POS Referral File (PRF) screen to be routed to the CRT Authorization device handler process in the same logical network under which the BASE24 user logged in.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPRDFS

All BASE24-pos CRT Authorization device handler customers:

- Inline changes to BP60CRTA.RQMRFS, RQOLTS and RQPRFS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNCRTA.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Define one CRT Authorization device handler server class for each logical network that contains a CRT Authorization device handler process. The format of the server class name is SERVER-CRTDxxxx, where xxxx is the identifier of the logical network where the device handler process resides.

If the server class definition SERVER-CRTDH exists in the PATHCONF, delete it.

CRT Access Security Changes

For each BASE24 user that will access a retailer's transactions on the MRF or PRF screens, give them permission to access that retailer's transactions by adding the retailer's FIID to screen 2 of the user's SEC record.

Routing Changes

Not applicable

Network Configuration

Configure a CRT Authorization device handler process for each logical network that supports CRT Authorization.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Service Code Validation Enhancement

This section describes the implementation requirements associated with the Service Code Validation Enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Service Code Validation Enhancement.

The purpose of this enhancement is to provide better detection of counterfeit cards. In card-read transactions, the card is authenticated by validating the Card Verification Digits encoded in the track data (CVD1). For the algorithm used to generate the CVD1, the input data includes the Service Code, also encoded in the track data. In manually-entered transactions, the card is authenticated by validating the Card Verification Digits printed or embossed on the card itself (CVD2). For the algorithm used to generate the CVD2, the input data includes a value of “000” in place of the Service Code, which is typically not available in manually-entered transaction data.

A counterfeit card can be created by generating fraudulent track data, which includes the CVD2 printed or embossed on the card and a Service Code value of “000”. The authorizing system may successfully validate the CVD2 (using the Service Code of “000”), believing it to be the CVD1. This means that the card is assumed to be valid when in fact it is fraudulent. To detect these fraudulent cards, the BASE24-pos Router/Authorization module has been modified to reject transactions which contain a Service Code of “000” in card-read transactions.

Software Impacts

All BASE24-pos customers:
• Inline code changes to PS60RTAU.AUTHLIBS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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SPDH & RTAU Enhancements for Visa and MasterCard

This section describes the implementation requirements associated with the SPDH & RTAU Enhancements for Visa and MasterCard.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH & RTAU Enhancements for Visa and MasterCard.

The ACI Standard POS Device Handler (SPDH) and Router/Authorization module (RTAU) have been modified to support a number of business enhancements implemented by Visa and MasterCard. These enhancements include auto-substantiation transactions for healthcare and transit purchases, healthcare eligibility inquiry transactions, card level results, and recurring payment cancellations.

<u>Healthcare/Transit Auto-Substantiation Transactions</u>

The SPDH has been modified to accept FID 6 subFID l (Healthcare/Transit Data) in transaction requests from the terminal. Included in subFID l is the amount of the qualified healthcare or transit purchase. The SPDH moves the auto-substantiation data received in subFID l to the Healthcare/Transit Token (CV) and adds the token to the 0200 request PSTM.

The SPDH returns subFID l to the terminal when the 0210 response PSTM contains the Healthcare/Transit Token and when auto-substantiation data is present in the token.

<u>Healthcare Eligibility Inquiry Transactions</u>

The SPDH has been modified to accept FID 6 subFID m (Healthcare Service Data) in transaction requests from the terminal. Included in subFID m is the provider id and service type code for up to 5 healthcare services. The SPDH moves the healthcare service data received in subFID m to the Healthcare Service Token (CW) and add the token to the 0200 request PSTM.

The SPDH returns subFID m to the terminal when the 0210 response PSTM contains the Healthcare Service Token and when healthcare service data is present in the token.

Card Level Results

The SPDH has been modified to return FID 6 subFID k (Card Level Results) to the terminal when the 0210 response PSTM contains the Switch Common Data Token (BY) and when the card level product ID is present in the token.

Router/Authorization has been modified to populate the approval code field in the PSTM with Visa's card level product ID when Router/Authorization is the generator of the approval code and when the card level product ID is present in the Switch Common Data Token. Such an approval code has the format ***NNNNNPb*B**, where *NNNNN* = a 5-byte non-zero numeric approval code, *P* = the card level product ID, *b* = blank, and B indicates BASE24 generated this approval code. The card level product ID is A (Visa Traditional), B (Visa Traditional Rewards), C (Visa Signature), D (Visa Infinite) or E-Z, 0-9 (reserved for future use).

Recurring Payment Cancellations

The SPDH has been modified to return FID 6 subFID n (Error Flag) to the terminal when the 0210 response PSTM contains the BASE24-pos Release 5.0 Token (04) and when error flag data is present in the token. Among other values, the Error Flag can contain specific codes notifying the terminal of a recurring payment cancellation, those codes being:

U	Recurring payment cancellation service
V	Stop payment order
W	Revocation of authorization order
X	Revocation of all authorization orders

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.COBTKN
- Inline changes to BA60DDL.DDLPSTKN
- Inline changes to BA60SRC.PSTKNCVS and PSTKNID

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS and RTAUG

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHG and ASPDHS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNSRVM.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

The following BASE24-pos tokens have been added for this enhancement:

- Healthcare/Transit Token (CV)
- Healthcare Service Token (CW)

If these tokens are to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token records must be configured in the Token file.

If these tokens are to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

ACI Standard Device Configuration File (ACNF)

For all portions of this enhancement, FID 6 (Product subFIDs) should be set to ‘Y’ in the ACNF response message field map.

For the Healthcare/Transit Auto-Substantiation Transaction portion of this enhancement, FID B (Amount 1) should be set to ‘Y’ in the ACNF response message field map. If the Issuer approves a healthcare/transit auto-substantiation transaction for a partial amount (partial authorization), FID B contains the approved partial amount.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

If the Healthcare/Transit Token (CV) or Healthcare Service Token (CW) is to be included in the PTLF or ILF Extracts, BASE24-pos users must consider this additional data in host programs that use the log file data.

User/Operational Impacts

Disk Impacts

The PTLF and ILF may require additional disk space if the Healthcare/Transit Token (CV) and Healthcare Service Token (CW) are included in the transaction records logged to these files.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-pos users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

ACI Standard POS Message

The following sub field identifiers (subFIDs) have been added to FID 6 (Product subFIDs) of the ACI Standard POS Message.

FID 6 subFID k – Card Level Results

This new subFID contains a code indicating the participation program for the card involved in the transaction. The first byte has one of the following values:

A	Visa Traditional
B	Visa Traditional Rewards
C	Visa Signature
D	Visa Infinite
E-Z, 0-9	Reserved for future use

The second byte is a blank, A-Z or 0-9. All are reserved values.

Request: Not applicable.
Response: Optional. Fixed length of 2 bytes.

FID 6 subFID l – Healthcare/Transit Data

This new subFID contains information associated with healthcare/transit auto-substantiation transactions and healthcare eligibility inquiry transactions. The format is as follows:

Additional Amount	occurs 1 to 6 times
Account Type	PIC X(2)
Amount Type	PIC X(2)
Currency Code	PIC X(3)
Amount Sign	PIC X
Amount	PIC X(12)

For healthcare/transit auto-substantiation transactions, only entry 1 of the Additional Amount table is used.

For healthcare eligibility inquiry transactions, entries 1 through 6 of the Additional Amount table can be used.

These fields are further described below:

Position	Length	Description
01-120	20-120	Additional Amount (occurs 1 to 6 times) Additional amount information.
01-02	2	Account Type The type of account being used. A value of 00 indicates a non-specified type will be used for healthcare/transit auto-substantiation transactions and healthcare eligibility inquiry transactions.
03-04	2	Amount Type The type of payment amount. Valid values are as follows. 3S = amount co-payment 4S = amount healthcare 4T = amount transit 57 = original amount
05-07	3	Currency Code The standard ISO currency code of the amount.
08	1	Amount Sign A code indicating whether the amount is positive or negative. Valid values are as follows. C = credit, positive balance D = debit, negative balance
09-20	12	Amount The amount specified by the Amount Type field. Right-justified, zero-filled on the left.

Request: Optional. Variable length of 20-120 bytes.
Response: Optional. Variable length of 20-120 bytes.

FID 6 subFID m – Healthcare Service Data

This new subFID contains information associated with healthcare eligibility inquiry transactions. The format is as follows:

Service	occurs 1 to 5 times
Provider ID	PIC X(9)
Type Code	PIC X(2)
Payer ID	PIC X(6)
Reason Code	PIC X(2)

For healthcare/transit auto-substantiation transactions, this subFID is not used.

For healthcare eligibility inquiry transactions, entries 1 through 5 of the Service table can be used.

These fields are further described below:

Position	Length	Description
01-95	19-95	Service (occurs 1 to 5 times) Service information.
01-09	9	Provider ID The medical license number of the healthcare service provider.
10-11	2	Type Code The healthcare service type code as defined by the Health Insurance Portability and Accountability Act (HIPAA).
12-17	6	Payer ID The healthcare insurance carrier/payer ID.
18-19	2	Reason Code The eligibility approval or rejection reason code as defined by the HIPAA.

Request: Optional. Variable length of 19-95 bytes.
Response: Optional. Variable length of 19-95 bytes.

FID 6 subFID n – Error Flag

This new subFID contains a code providing additional information about the disposition of the transaction. Valid values are as follows:

A	BASE24-pos Terminal Data files adjustment limit exceeded
C	Card verification failed
E	BASE24-pos Terminal Data files return limit exceeded
I	Invalid MAC
K	KMAC synchronization error
L	Invalid PIN length
M	MAC failure
P	Invalid PIN block
R	Sanity check error—previous zone
S	Sanity check error
T	BASE24 system error (token error)
U	**Recurring payment cancellation service**
V	**Stop payment order**
W	**Revocation of authorization order**
X	**Revocation of all authorizations order**
1	New account information available for recurring payments transaction
2	Try again later, recurring payments transaction
3	Do not try again for recurring payments transaction
blank	No information available

Request: Not applicable.
Response: Optional. Fixed length of 1 byte.

Healthcare/Transit Auto-Substantiation Transaction Request

The terminal must set the following fields when formatting a healthcare/transit auto-substantiation transaction request:

FID B – Amount 1

The terminal must set FID B to the full transaction amount.

FID 1 – PS2000 Data

Market Specific Data ID	PIC X	**M** = healthcare (medical) or
		T = transit

FID 6 subFID I – Healthcare/Transit Data

Additional Amount (entry 1)

Account Type	PIC X(2)	**00** = unspecified
Amount Type	PIC X(2)	**4S** = amount healthcare or **4T** = amount transit
Currency Code	PIC X(3)	ISO currency code of the amount
Amount Sign	PIC X	**C** = credit, positive balance
Amount	PIC X(12)	The amount of the qualified healthcare or transit purchase. Right-justified, zero-filled on the left.

Although the Additional Amount table can contain from 1 to 6 entries, only entry 1 should be used when sending a healthcare/transit auto-substantiation transaction request.

Healthcare/Transit Auto-Substantiation Transaction Response

The terminal receives FID B (Amount 1) and FID 6 subFID I (Healthcare/Transit Data) in the healthcare/transit auto-substantiation transaction response.

If the Issuer approved the healthcare/transit auto-substantiation transaction for the full amount, FID B contains the original requested amount.

If the Issuer approved the healthcare/transit auto-substantiation transaction for a partial amount (partial authorization), FID B contains the approved partial amount, which is less than the original requested amount. In entry 1 of the Additional Amount table in FID 6 subFID I, the Amount Type field contains a value of **'57'** (original amount) and the Amount field contains the original requested amount.

Although the Additional Amount table in subFID I can contain from 1 to 6 entries, only entry 1 is used in a healthcare/transit auto-substantiation transaction response.

Healthcare Eligibility Inquiry Transaction Request

The terminal must set the following fields when formatting a healthcare eligibility inquiry transaction request:

Transaction Code

The terminal must set the Transaction Code in the SPDH message header to **'07'** (balance inquiry).

FID 6 subFID U – Transaction Subtype Data

Transaction Subtype	PIC X(4)	**C002** = healthcare eligibility inquiry
Acquirer Processing Code	PIC X(6)	blanks
Issuer Processing Code	PIC X(6)	blanks

FID 6 subFID m – Healthcare Service Data

Service (occurs 1 to 5 times)		
Provider ID	PIC X(9)	medical license number of provider
Type Code	PIC X(2)	HIPAA code for healthcare treatment
Payer ID	PIC X(6)	blanks
Reason Code	PIC X(2)	blanks

The terminal may request eligibility information for up to 5 healthcare services within a single request message. Each entry in subFID m's Service table represents a single healthcare service.

Healthcare Eligibility Inquiry Transaction Response

The terminal receives FID 6 subFID l (Healthcare/Transit Data) and FID 6 subFID m (Healthcare Service Data) in the healthcare eligibility inquiry transaction response.

SubFID l contains from 1 to 6 entries in the Additional Amount table. The Amount Type field in each entry contains a value of **'3S'** (amount co-payment).

SubFID m contains from 1 to 5 entries in the Service table.

Reference Documents

Banknet Release 06.2 Document, May 5, 2006

October 2005 VisaNet Business Enhancements, April 15, 2005

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SPDH New Fields to Support Network Requirements Enhancement

This section describes the implementation requirements associated with the SPDH New Fields to Support Network Requirements Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH New Fields to Support Network Requirements Enhancement.

American Express Card Acceptance and Processing Network

The ACI Standard POS Device Handler (SPDH) has been modified to support additional data in transaction requests bound for the American Express Card Acceptance and Processing Network (AMEX CAPN). This additional data consists of Card Not Present (ITD) data from merchants in the mail, telephone, and internet order industries, and Airline Passenger Data (APD) from merchants in the airline industry. FID 6 subFID o (American Express Additional Data) has been created to carry this data. The SPDH moves the data received in subFID o to the American Express Additional Data Token (CU) and adds the token to the 0200 request PSTM.

Message Reason Code

FID 6 subFID N (Reason Online Code) has been renamed to the more general “Message Reason Code”. ISO message reason codes 1004 through 1007 have been added to the list of possible values. Acquirers can use the message reason code to determine the appropriate EMV Authorization Response Code (ARC) for transactions authorized at an EMV terminal. If an approved advice message is received with a message reason code of ‘10xx’, the ARC is ‘Y1’ (offline approved). If an approved advice message is received with a message reason code of ‘15xx’, the ARC is ‘Y3’ (unable to go online, offline approved).

Data Encryption Key

FID I (Data Encryption Key) and DID j (Data Encryption Key) have been expanded from 16 bytes to also support 32- and 48-byte keys. This is to satisfy a new Interac requirement that states the account balance, if returned to the terminal, must be encrypted using at least a 32-byte data encryption key.

Software Impacts

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHG, ASPDHS, SPDHDDLS and SPDHLDRS

All BASE24-pos ACI Standard POS Device Handler (SPDH) EMV customers:

- Inline changes to PS60ISPD.SPDHEMVS

All BASE24-pos ACI Standard POS Device Handler (SPDH) Dynamic Key Management (DKM) customers:

- Inline changes to PS60KSPD.SPDHDKMS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNSPNR.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Token file (TKN)

If the American Express Additional Data Token (CU) is to be logged to the BASE24-pos Transaction Log File (PTLF) or the BASE24 Interchange Log File (ILF), the appropriate token record must be configured in the Token file.

If the American Express Additional Data Token (CU) is to be extracted with the PTLF or ILF, sent to the host, or passed through the BIC ISO Interchange Interface, the appropriate record must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

POS Terminal Data File (PTD)

For customers converting from release 5.3 to release 6.0 version 7, the PTD conversion program must be run to convert the PTD. Running the conversion program is required for any BASE24-pos customer utilizing the PTD in transaction processing. Users must create the new PTD file using the PSDDLFUP file.

For customers converting from release 6.0 version 6 to release 6.0 version 7, the PTD conversion program is not required.

The PTD conversion program on the PS60UC04 subvolume can be used to convert the PTD records from the release 5.3 format to the release 6.0 version 7 format. If modifications to the conversion program are necessary, the source code can be modified by editing the CNVPTDS file.

To compile the conversion program, enter at the TACL prompt:

$<volume>.makev500.makev501/out $s.#ptdcnv.t,name $ptdm,nowait/ cnvptdm –force –list –listreplace

Release 5.3 users should update the CNVPTDR obey file on the PS60UC04 subvolume with the appropriate information. The CNVPTDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the BA60UC04 subvolume.

Host/Co-Network Impacts

Message Reason Code

Host applications should query the RSN-ONL-CDE of the EMV Status Token (B4) to determine the appropriate EMV Authorization Response Code (ARC) for transactions authorized at an EMV terminal. This field gets its value from FID 6 subFID N (Message Reason Code) of the SPDH message. If the token's RSN-ONL-CDE is '10xx', the ARC is 'Y1' (offline approved). If the token's RSN-ONL-CDE is '15xx', the ARC is 'Y3' (unable to go online, offline approved).

User/Operational Impacts

Disk Impacts

The PTLF and ILF may require additional disk space if the American Express Additional Data Token (CU) is included in the transaction records logged to these files.

System Limitations

BASE24-pos users installing this enhancement must be aware of their internal system limitations for inter-process messaging and writing to the PTLF and ILF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24-pos users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Vendor Impacts

ACI Standard POS Message

The following field identifiers (FIDs) and sub field identifiers (subFIDs) have been modified, have had new values defined or have been added to the ACI Standard POS Message.

FID I – Data Encryption Key

This FID has been expanded to support double-length and triple-length data encryption keys.

The data encryption key is a working key generated by the security module and provided to the terminal. The data encryption key is the data encryption communications key (KME), encrypted under the data encryption terminal master key. This key is used to encrypt the Available Balance field. The number and type of keys used determines the length of this field, as follows.

- A single-length key contains 16 bytes.
- A double-length key contains 32 bytes.
- A triple-length key contains 48 bytes.

Request: Not available.
Response: Optional. Variable length of 16, 32, or 48 bytes. This field can be configured in any response and is included if more than the configured number of message encryption errors occur. This field can also be included optionally in downloads.

FID 6 subFID N – Message Reason Code (formerly Reason Online Code)

This subFID has been renamed from Reason Online Code.

The Message Reason Code specifies why a transaction is to be authorized online (rather than being completed locally), or why a transaction has been completed locally (rather than being authorized online). Values are defined in the ISO 8583 (1993) Standard.

In a force-post (message subtype 'F') or store-and-forward (message subtype 'S') message that the terminal has previously attempted to send to BASE24-pos as an online request message (message subtype 'O'), the Message Reason Code shall contain the same value as in the online request message.

Acquirers can use the Message Reason Code to determine the appropriate EMV Authorization Response Code (ARC) for transactions authorized at a terminal. If an approved advice message is received with a Message Reason Code of '10xx', then the ARC is 'Y1' (offline approved). If an approved advice message is received with a Message Reason Code of '15xx', then the ARC is 'Y3' (unable to go online, offline approved).

Request: Optional. Fixed length of 4 bytes.
Response: Not applicable.

FID 6 subFID o – American Express Additional Data

This new subFID contains additional data for American Express transactions. This subFID is variable-length and the format is as follows.

Additional Data ID	PIC X(3)
Additional Data	PIC X(297)

These fields are further described below.

Position	Length	Description
01-03	3	Additional Data ID Indicates the type of additional data that follows. Valid values are as follows. ITD = Card Not Present Data (mail, telephone and internet order) APD = Airline Passenger Data

04-300	0-297	Additional Data Contains either Card Not Present Data or Airline Passenger Data, depending on the value of Additional Data ID.
04-270	0-267	Card Not Present Data The following data fields may be used when the Additional Data ID field is set to a value of 'ITD'. Each data field is preceded by a three-byte data ID and a two-byte data length indicator. Each data ID/data length/data field group is optional and the groups may occur in any order.
	3	Data ID CE~ = Customer Email (~ = blank character)
	2	Data Length Length of the following field.
	1-60	Customer Email Customer's email address.
	3	Data ID CH~ = Customer Host Name (~ = blank character)
	2	Data Length Length of the following field.
	1-60	Customer Host Name Name of the server to which the customer is connected.
	3	Data ID HBT = HTTP Browser Type
	2	Data Length Length of the following field.
	1-60	HTTP Browser Type Customer's HTTP browser type.

3	Data ID
	STC = Ship To Country
2	Data Length
	Length of the following field.
3	Ship To Country
	Three-byte numeric ISO country code.
3	Data ID
	SM~ = Shipping Method (~ = blank character)
2	Data Length
	Length of the following field.
2	Shipping Method
	Shipping method code. Valid values are as follows. 01 = same day 02 = overnight/next day 03 = priority, 2-3 days 04 = ground, 4 or more days 05 = electronic delivery 06 – ZZ = reserved for future use
3	Data ID
	MPS = Merchant Product SKU
2	Data Length
	Length of the following field.
1-15	Merchant Product SKU
	Unique SKU (Stock Keeping Unit) inventory reference number of the product associated with this authorization request. For multiple items, enter the SKU of the most expensive item.

3	Data ID
	+IP = Customer IP (ACI-defined data ID)
2	Data Length
	Length of the following field.
15	Customer IP
	Customer's internet IP address in the format nnn.nnn.nnn.nnn
3	Data ID
	+AN = Customer ANI (ACI-defined data ID)
2	Data Length
	Length of the following two fields.
10	Customer ANI
	ANI (Automatic Number Identification) -specified phone number that the customer used to place the order with the merchant.
2	Customer II Digits
	Telephone company-provided ANI II (Information Identifier) code digits that are associated with the Customer ANI phone number and correspond to the call type (e.g., cellular, government institution).

04-300	0-297	Airline Passenger Data The following data fields may be used when the Additional Data ID field is set to a value of 'APD'. Each data field is preceded by a three-byte data ID and a two-byte data length indicator. Each data ID/data length/data field group is optional and the groups may occur in any order.
	3	Data ID +DD = Departure Date (ACI-defined data ID)
	2	Data Length Length of the following field.
	8	Departure Date Departure date in the format YYYYMMDD.
	3	Data ID APN = Airline Passenger Name
	2	Data Length Length of the following field.
	23-40	Passenger Name Passenger name in the format SURNAME~FIRSTNAME~ MIDDLEINITIAL~TITLE, where ~ = blank character. Data must be 23 bytes minimum, left-justified and blank-filled as needed. 40 bytes maximum, truncate if necessary.

3	Data ID CN~ = Cardmember Name (~ = blank character)
2	Data Length Length of the following field.
23-40	Cardmember Name Cardmember name in the format SURNAME~ FIRSTNAME~MIDDLEINITIAL~TITLE, where ~ = blank character. Data must be 23 bytes minimum, left-justified and blank-filled as needed. 40 bytes maximum, truncate if necessary.
3	Data ID +AP = Origination/Destination Airport (ACI-defined data ID)
2	Data Length Length of the following two fields.
5	Origination Airport Origination airport for the first segment of the trip. Five-character airport code allows for the anticipated expansion of the current three-character code. Left-justify and blank-fill as needed.
5	Destination Airport Destination airport for the first segment of the trip; not necessarily the final destination. Five-character airport code allows for the anticipated expansion of the current three-character code. Left-justify and blank-fill as needed.

3	Data ID RTG = Routing
2	Data Length Length of the following two fields.
2	Number of Cities Number of airports or cities on the ticket, maximum of 10.
11-59	Routing Cities Routing airport or city code for each leg on the ticket (including Origination Airport and Destination Airport), in five-byte segments each separated by a "/". Example: "ABC~~/DEF~~/GHI~~", where ~ = blank character.
3	Data ID ALC = Airline Carriers
2	Data Length Length of the following two fields.
2	Number of Airline Carriers Number of airline carrier entries in the following field, maximum of 9.
5-53	Airline Carriers Airline carrier code for each leg on the ticket (including Origination Airport and Destination Airport), in five-byte segments each separated by a "/". Each leg must have an airline carrier code entry, even if multiple (or all) legs are on the same airline. Example: "AB~~~/AB~~~", where ~ = blank character.

3	Data ID
	+FR = Fare Data (ACI-defined data ID)
2	Data Length
	Length of the following three fields.
24	Fare Basis
	Primary and secondary discount codes indicating the class of service and fare level associated with the ticket. Truncate to 24 bytes, if necessary.
3	Number of Passengers
	Number of passengers in the party.
1	E-Ticket Indicator
	Indicates if the ticket is electronic. Valid values are as follows.
	E = electronic ticket blank = non-electronic ticket
3	Data ID
	RES = Reservation Code
2	Data Length
	Length of the following field.
6-15	Reservation Code
	A precursor to a ticket number. Corresponds to an airline ticket purchase made by an airline or a global distribution system (GDS).

Request: Optional. Variable length of up to 300 bytes.
Response: Not applicable.

Download Data Elements

The following download field identifiers (DIDs) have been modified.

DID j – Data Encryption Key

DID j has been expanded to support double-length and triple-length data encryption keys.

DID j contains the data encryption key used by this terminal. The number and type of keys used determines the length of this field, as follows.

- A single-length key contains 16 bytes.
- A double-length key contains 32 bytes.
- A triple-length key contains 48 bytes.

Reference Documents

Not applicable

Visa DUKPT MACing Enhancement

This section describes the implementation requirements associated with the Visa DUKPT MACing Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Visa DUKPT MACing Enhancement.

BASE24-pos has been enhanced to address an issue in the initial TSS implementation of DUKPT MACing. The Atalla MACing commands in use do not provide sufficient flexibility, and as a result, do not return the appropriate MAC values. These commands have since been enhanced by Atalla. BASE24 and TSS were enhanced to include an extra flag in the command. This enhancement is required for all customers that wish to use DUKPT MACing.

DUKPT MACing is mandated in the Scandinavian countries. DUKPT MACing cannot be implemented by Scandinavian customers in its current form. These customers cannot migrate to TSS without this enhancement.

This enhancement provides the functionality required by the Scandinavian customers implementing DUKPT MACing. Atalla has modified their new DUKPT MACing commands to include support for the Visa DUKPT algorithm used by the older Atalla 5C command. This modification was made for BASE24 Scandinavian customers. The standard product direction is to support the ISO-9791-1 Algorithm 1 and ISO-9791-1 Algorithm 3.

Atalla has only modified command 386 (Generate DUKPT MAC) to include additional MAC Type values. Atalla command 348 (Verify DUKPT MAC) has not been enhanced to support the additional MAC Type values.

Therefore, this enhancement and the enhancement made by Atalla only addresses Scandinavian DUKPT MACing scenarios between the device and the Host (e.g., BASE24).

Thales/Racal Hardware Security Modules do not support the “Old Visa DUKPT” algorithm.

Prerequisite: The minimum version of TSS required for the implementation of this enhancement is version 06.2.

Prerequisite: The Atalla enhancement for command 386 was implemented for the 2.80 release of the Ax100-AKB HSM family.

Software Impacts

All BASE24-pos customers:

- Inline changes to the TSS Command Message DDL file (BA60DDL.DDLGTSS)
- Inline changes to the Security Utilities (BA60SRC.SECUTILS)

All BASE24-pos SPDH device handler customers:

- Inline changes to the SPDH device handler module (PS60SPDH.ASPDHS)

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNDKPT.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

TSS Channel Security Data Source (CSEC)

A new value for MAC Option has been added to configure Visa DUKPT MACing to the CSEC.

For more information about these initial setup procedures, refer to the ***BASE24 Transaction Security Services Processing Guide***.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Security Device

The minimum version for Visa DUKPT MACing support for the Atalla AKB Hardware Security Module is version 2.80.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

The minimum version of TSS required for the implementation of this enhancement is version 06.2.

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Setup for BASE24

When configuring BASE24 to use the “Old Visa DUKPT” algorithm, BASE24 must be configured to use the same fields used in the MAC verify on the request and the MAC generate on the response. Also, the fields being MAC’d cannot change during the transaction. This is how the Scandanavian customers are configured today using their current functionality.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

Optional Enhancements

This section identifies the BASE24-pos enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Contactless Chip/Magnetic Stripe Enhancement
- Correct Declined Advices to Host
- CVD/CVD2 Checking in a Single Request
- EMV CCD Issuer Support
- Hypercom (HPDH) Software MACing Support Enhancement
- Improve EMV Transaction Certificates in SPDH Enhancement
- Log Stored Value Dormancy/Escheatment Trans to PTLF

BASE24-pos users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements for future reference.

ACI Worldwide Inc

Contactless Chip/Magnetic Stripe Enhancement

This section describes the implementation requirements associated with the Contactless Chip/Magnetic Stripe Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Contactless Chip/Magnetic Stripe Enhancement.

The term "proximity technologies" include a wide variety of wireless technologies such as Radio Frequency IDentifier (RFID) transponders, Bluetooth, Infrared, Wireless Fidelity (WiFi), and contactless chip cards (based on the ISO/IEC 14443 standard). Contactless transactions for magnetic stripe and chip(EMV) card transactions take advantage of these technologies. Note: The terms Contactless Chip and Contactless EMV are used interchangeably in this document.

MasterCard is extending its product range to include "proximity technologies". These technologies enable terminals and cards to exchange data without physical contact. MasterCard PayPass™ is the first program to be implemented under the MasterCard Proximity Payments Program. As with traditional MasterCard magnetic stripe card transactions, MasterCard PayPass requires a Card Verification Code (CVC) value. The MasterCard PayPass CVC is referred to as CVC3 and is used for contactless magnetic stripe transactions. Support for dynamic CVC3 is included in this enhancement. <u>Static CVC3 is not supported at this time</u>. Contactless EMV transactions for MasterCard will use M/Chip4 data and verification methods. Support for M/Chip4 is already provided by the BASE24 EMV extension.

Visa has adopted a similar program called Visa Wave. It uses a card verification method designated as dCVV. The dCVV is also used only for contactless magnetic stripe transactions. Support for dCVV is included in this enhancement. Contactless EMV transactions for Visa will use VSDC or qVSDC data and verification methods. A new cryptogram version number (CVN 17) is defined primarily for qVSDC. Support for this CVN is added with this enhancement.

Contactless magnetic stripe transactions contain data that makes it look like a swiped magnetic stripe transaction with two exceptions. The first exception is the POS entry mode is a value of "91". The second exception is in the Issuer Discretionary Data (IDD) of the Track 1 or Track 2 data. The IDD contains the data necessary to perform Dynamic Card Verification (DCV). The format of the data depends on the card scheme being supported. DCV is performed only on POS transactions at this time.

MasterCard's PayPass implementation for contactless magnetic stripe cards has a number of options that can be modified during personalization of the ICC on the contactless card. This controls the information sent from the terminal in the Track 1 and Track 2 data fields. The three data elements that can be present in the Issuer Discretionary Data portion of the magnetic stripe data and the possible lengths are:

Unpredictable number (UN): This is a random number generated by the terminal and sent to the card to be used in calculating the dynamic CVC3 value. Zero to 8 bytes of data can be included in the track data. The length of the Unpredictable Number is always given in the last byte of the Issuer Discretionary Data. If this length is zero, it means that no Unpredictable Number was used in the calculation of the dynamic CVC3. This value is always zero if the card is using a static CVC3 value. For issuers doing dynamic CVC3, they should plan to issue cards that include an Unpredictable Number of at least 1 digit. This will facilitate determining the type of CVC3 calculation (static or dynamic) from the length of the Unpredictable Number.

Application Transaction Counter (ATC): This is a counter that is kept on the ICC chip used to generate the RFID signal. It is incremented once for each transaction. A similar field is maintained by the issuer. The ATC is stored on the card and on the BASE24 Cardholder Authorization File (CAF) as 2 binary bytes. However, in contactless magnetic stripe transactions, only a portion of the card's ATC is sent in the transaction data. The length and position of this field in the Track 1 or Track 2 data is defined in the ICC personalization data and the BASE24 Card Prefix File (CPF). It can be different for Track 1 and Track 2. Only the least significant digits of the ATC are sent in the transaction data. These digits are used to "synchronize" the value on the card with the value maintained in the database. However, the entire ATC value is used in calculating the dynamic CVC3 and must be sent to TSS for CVC3 verification.

Dynamic CVC3 value: This is a value calculated by the ICC on the card using the values in the ATC and the UN (and other data elements). The CVC3 value can be sent as the least significant 3, 4 or 5 bytes as part of the track data. The length and position of this field in the Track 1 or Track 2 data is defined in the ICC personalization data and the BASE24 Card Prefix File (CPF).

Visa's Wave contactless magnetic stripe implementation uses different information to calculate the Visa dynamic card verification value called dCVV. No Unpredictable Number is used in the Visa calculation. The dCVV is always 3 bytes in length. The ATC is always sent as 4 bytes of numeric (not binary) data. The largest value sent is 9999. For Track 1 data, the ATC may be split with 2 bytes before and 2 bytes after the 3 bytes that contain the dCVV. It is assumed that the entire ATC on the card and in the CAF is the ATC used in the dCVV calculation. When checking the ATC value in the transaction against the value stored on the CAF, it will be necessary to remove the most significant digit from the decimal representation of the CAF value (subtract 10,000 or 20,000, etc.), if the value on the CAF is over 9999.

Dynamic Card Verification (DCV) support within BASE24-pos is a separately licensable module, bound into POS Router Authorization. The core module has calls to dummy procedures to allow the DCV module to be bound-in, if it is purchased. There are also new security utilities in BA60QDCV that are bound into the POS Router Authorization module if DCV is purchased.

The POS Router Authorization process was modified to validate Dynamic Card Verification digits on contactless magnetic stripe transactions. This was done by adding calls to Dynamic Card Verification (DCV) extension procs in AUTHLIBS, AUTHS and ROUTERS. A separate DCV extension module was created on PS60QRTA. The DCV extension formats additional data received in Track 1 or Track 2, creates messages to TSS to have the Dynamic Card Verification Digits (DCVD) verified and processes the response from TSS. The extension also manages a new ATC on the Base segment of the CAF. Changes were also made in POS EMV Router Authorization to maintain the ATC on the Base segment, if desired.

Because of the new ATC counter on the Base segment of the CAF, it is necessary for ATM EMV transactions to be able to check and update the new fields. Changes were made to ATM EMV Authorization to allow this.

The new DCV procedures are in 2 separate subvolumes: BA60QDCV and PS60QRTA. Make files to compile the various pieces are located on each subvolume. The Make file on PS60RTAU was modified to bind in the new files. New CUSTMACS flags were defined to add the DCV functionality into POS Authorization.

Items to consider for contactless magnetic stripe transactions and dynamic card verification:

- Modifications are required in Transaction Security Service (TSS) to support this enhancement. The changes were made in TSS version 06.4.
- For contactless magnetic stripe transactions, only Dynamic Card Verification will be performed. This type of card verification is mutually exclusive from other types of card verification. It is the only type of card verification that can be performed on contactless magnetic stripe transactions and only contactless magnetic stripe transactions use this type of card verification.
- Only MasterCard's dynamic CVC3 and Visa's dCVV are supported in this release. Support for MasterCard's static CVC3 may be a future consideration.
- The fields on the CPF Base Segment that apply to Dynamic Card Verification will apply only to magnetic stripe transactions.
- The Application Transaction Counter (ATC) is a field in the transaction track data that is required to verify the incoming Dynamic Card Verification value. In order to verify the cryptogram, the full value used in the cryptogram must be present in the track data or in a CAF file on BASE24. If the CAF is not present on BASE24, it will not be possible to verify if the ATC is a reasonable value based on information from previous transactions. Also, it may not be possible to verify the DCVD, depending on how many ATC digits are sent in the transaction data and how many digits are used in the DCVC calculation. This is especially a problem if customers are using MasterCard PayPass and sending only one or two digits of ATC in the transaction data.
- There are no changes to the CAF Refresh or FHM formats.
- If a full file refresh of the CAF is done after the initial refresh to update the CAF format, the ATC counter will be set to zero in the Base and EMV segments. Special processing will be required to rebuild the ATC in the Base segment. The value in the EMV segment will be reset from the first EMV transaction, as it is now.
- In the SPDH message, FID 6 SFID E (POS Entry Mode) is required for all contactless transactions, both magnetic stripe and EMV. This data element will tell the device handler/router/authorization process that the transaction is a contactless magnetic stripe or EMV transaction.
- ATC checking must be turned on (set to a non-zero value on CPF Screen 3) if the ATC sent in the transaction data contains fewer digits than the value sent to the HSM for verification. The ATC checking logic also "rebuilds" the full ATC value that is needed in DCVD verification. If transactions should not be declined for ATC checking failure, that can be set in the Bad ATC Action field on the same screen.
- The concept of ATC Limit Checking has not been implemented in ATC checking for contactless magnetic stripe transactions. ATC Limit Checking will still be available for all EMV transactions.

- For contactless magnetic stripe transactions from a MasterCard card, 1-5 digits of the ATC value may be sent to be compared with the ATC kept on the CAF. Values of one digit will be compared to the value on the CAF incremented by up to 9. For example, if the CAF has 00123 and a value of 8 is sent, the ATC will be incremented to 00128 for verifying the CVC3. If a value of 2 is sent, the ATC will be incremented to 00132.
- For MasterCard and Visa transactions, the entire ATC value will be used in verifying the cryptogram, regardless of how many digits are sent in the transaction.
- For contactless magnetic stripe transactions, when the ATC value on the CAF Base segment ends with a value between 95 and 99, it will be allowed to rollover to a value that ends between 00 and 05, depending on the ATC value and number of ATC digits from the transaction. This will ensure transactions will not fail ATC checking when the ATC value in the last 2 positions change from 99 to 00. The value of 99 will be allowed to changed to 100, 999 will change to 1000 and 9999 will change to 10000. This limit checking will not be implemented for the Second Card ATC.
- If DCV verification fails, the ATC on the Base segment of the CAF will not be updated.
- A contactless transaction with a bad PIN will not decrement the ATC. With contactless transactions, the ATC will have been incremented before the terminal or host can do PIN checking. The card will no longer be in communication with the terminal to make any changes to the ATC held on the card.
- Track preference (Track 1 or Track 2) when formatting the DCV information for contactless magnetic stripe transaction will be the same as when formatting a contact (swiped) magnetic stripe transaction, using CPF.BASE.TRK-PREF.
- The member number used in contactless transactions will be the Application Pan Sequence Number (APSN). If a value is not provided in the incoming transaction information, the value will default to zeros. This value is not expected to be provided in a contactless magnetic stripe transaction.
- The value in the ivcvc3 field sent to TSS in the call to validate the CVC3 will be default values. At this time, this value will not be stored on BASE24.
- The position of the service code in track data is assumed to be in the position defined by MasterCard and Visa. This means it will always begin in the first position following the expiration date in Track 1 or Track 2. No offset will be provided on the CPF screen 3 to allow customers to specify a different location for the service code used in dynamic card verification.
- If PIN information is provided in the contactless Track 1 or Track 2, it must be in the same location as in the contact Track 1 or Track 2 data. The same POFST/PVV LOC field will be used for PIN verification for contact or contactless magnetic stripe transactions.

- If a customer issues cards using MasterCard dynamic CVC3, the length of the Unpredictable Number used in the cryptogram calculation must be at least 1. Should BASE24 support static CVC3 in the future, if the length of the Unpredictable Number is 0, it will be assumed the CVC3 calculation for that transaction used the static CVC algorithm.

Items to consider for contactless EMV transactions:

- Modifications are required in Transaction Security Service (TSS) to support this enhancement. The changes were made in TSS version 06.4.
- For contactless EMV transactions, dynamic card verification will not be performed. These transactions will go through EMV verification and iCVV.
- There will be no fallback checking associated with a contactless chip transaction that is different than the current fallback checking.
- There is a new Visa cryptogram version number, %h17. This will be for contactless qVSDC transactions. These "quick" EMV transactions will not pass all of the EMV data elements and there is no EMV script processing.
- If a full file refresh of the CAF is done after the initial refresh to update the CAF format, the ATC counter will be set to zero in the Base and EMV segments. Special processing will be required to rebuild the ATC in the Base segment. The value in the EMV segment will be reset from the first EMV transaction, as it is now.
- The concept of ATC Limit Checking has not been implemented in ATC checking for contactless magnetic stripe transactions. ATC Limit Checking will still be available for all EMV transactions.
- MasterCard M/Chip processing will be based on M/Chip 4.

Prerequisite: The minimum version of TSS required for the implementation of this enhancement is version 06.4.

Software Impacts

All BASE24 customers:

- Changes to BA60MAKE.CUSTMACS and FLGSDOC
- Inline changes to BA60SRC.BASEFM, BASEM and BASEMM
- Inline changes to BA60DDL.DDLBCNST, DDLFCAF, DDLFCPF and DDLGTSS.
- Inline changes to BA60AFT.SCRNCAF, RQCAFS, SCRNCPF, RQCPFS, and SVCPFS.
- Comment changes to BA60DDL.DDGPSTM

All BASE24-pos customers:

- Inline changes to PS60SRC.POSFM and POSMM
- Inline changes to PS60RTA.AUTHLIBS, AUTHS, ROUTERS, RTAUD, RTAUG and RTAUM
- Changes to log message manuals BA60LOGM.PSAUTH and PSROUTER

All BASE24 EMV customers:

- Inline changes to BA60IEMV.CAMUTILS and EMVG

All BASE24-atm EMV customers:

- Inline changes to AT60IATH.AUTHEMVG and AUTHEMVS

All BASE24-pos EMV customers:

- Inline changes to PS60IRTA.RTAUEMVS

All BASE24-pos SPDH EMV customers:

- Inline changes to PS60ISPD.SPDHEMVS

All BASE24 Banknet Interface customers

- Inline changes to SW60BNET.BNETG and BNETLIBS

All BASE24 Banknet EMV Interface customers

- Inline changes to SW60IBNT.BNETEMVS

All BASE24 Europay Interface customers

- Inline changes to SW60EURO.EUROLIBS

All BASE24 Europay EMV Interface customers

- Inline changes to SW60IEUR.EUROEMVS

All BASE24 Link EMV Interface customers

- Inline changes to SW60ILNK.LINKEMVS

All BASE24 MDS Interface customers

- Inline changes to SW60MDS.MDSG

All BASE24 MDSM Interface customers

- Inline changes to SW60MDSM.MDSMS

All BASE24 MDSM EMV Interface customers

- Inline changes to SW60IMDM.MDSMEMVS

All BASE24 MACI Interface customers

- Inline changes to SW60MACI.MACIG and MACIS

All BASE24 NYCI Interface customers

- Inline changes to SW60NYCI.NYCIG and NYCIS

All BASE24 STRI Interface customers

- Inline changes to SW60STRI.STRIG and STRIS

All BASE24 Visa Interface customers

- Inline changes to SW60VISA.VISAG and VISAFMTS

All BASE24 Visa EMV Interface customers

- Inline changes to SW60IVIS.VISAEMVS

All BASE24-pos customers implementing the Contactless Chip/Magnetic Stripe Enhancement:

- New subvolume, BA60QDCV, with files for the Dynamic Card Verification
- New subvolume, PS60QRTA, with files for the POS Contactless Chip/Magnetic Stripe extension
- New log message manual BA60LOGM.BADCVUTL

A complete list of the files affected by the Contactless Chip/Magnetic Stripe Enhancement can be found in the SCN file BA60UD0A.SCNCTLS for the Contactless Chip/Magnetic Stripe Enhancement.

CUSTMACS Changes

Set the following flag to true to compile the Dynamic Card Verification Utilities:

- dyn_crd_vrfy_on

Set the following flag to true to bind the Contactless Chip/Magnetic Stripe module to the POS Router/Authorization process. This will also bind in the Dynamic Card Verification Utilities.

- pos_dyn_crd_vrfy_on

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ACI SPDH Configuration File (ACNF)

The SPDH terminal configuration (ACNF) must be set to expect FID6 SFID E (POS Entry Mode) on all contactless transactions, both EMV and contactless magnetic stripe.

Card Prefix File (CPF)

The following additional new fields have been added to the CPF Screen 3 to configure dynamic card verification for contactless magnetic stripe transactions.

This field identifies the Dynamic Card Verification group to which this record belongs. If this field is not blank, the Device Handler/Router/Authorization module uses the value of this field, among others, to read the Dynamic Card Verification record in TSS. If this field contains blanks, then Dynamic Card Verification is not performed. The default value is spaces.

Screen Name:	DYN CARD VERIF KEY LOCATOR
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.DCV-KEY-LOC
Valid Values are:	Any combination of alphanumeric characters and leading and trailing spaces

A flag will be added to indicate when Dynamic Card Verification is to be attempted. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name:	DCV CHECK
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.DCV-CHK-TYP

Valid Values are:

- 0 = Dynamic Card Verification disabled. Do not check Dynamic Card Verification Digits (DCVD). Default Value.
- 1 = CVC3 checking enabled.
- 2 = CVC3 checking enabled. If the value of the ATC in the Base segment of the CAF is zero, the transaction continues.
- 5 = dCVV checking enabled.
- 6 = dCVV checking with split ATC in Track 1 enabled.

TRACK 1 SETTINGS

BASE24 products use the following five fields to locate certain pieces of information on Track 1 of cards with this prefix.

A new field will be added to define the offset of the Dynamic Card Verification digits on Track 1. If this field is equal to zero, Dynamic Card Verification is not performed for transactions with Track 1. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name:	TRK1 DCVD OFST
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK1-DCVD-OFST
Valid Values:	0-99

The length of the Dynamic Card Verification digits on Track 1 will also need to be entered. The default value is 0.

Screen Name:	DCVD LEN
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK1-DCVD-LEN
Valid Values:	0-9

The offset of the Application Transaction Counter on Track 1 will be included on CPF screen 3. If this field is equal to zero, ATC checking is not for transactions with Track 1. Additionally, the ATC cannot be part of the DCVD calculation. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name:	ATC OFST
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK1-ATC-OFST
Valid Values:	0-99

The length of the Application Transaction Counter on Track 1 will also need to be entered. The default value is 0.

Screen Name:	ATC LEN
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK1-ATC-LEN
Valid Values:	0-9
Default Value:	0

The offset of the Unpredictable Number on Track 1 will be included on CPF screen 3. This field is used by cards following the MasterCard PayPass implementation; otherwise this field should be set to zero. If this field is equal to zero, the Unpredictable Number cannot be part of the DCVD calculation. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name: UNPRED NUM OFST
Screen: CPF Screen 3
DDL Name: CPF.BASE.TRK1-UNPREDIC-NUM
Valid Values: 0-99

TRACK 2 SETTINGS

BASE24 products use the following five fields to locate certain pieces of information on Track 2 of cards with this prefix.

The offset of the Dynamic Card Verification digits on Track 2 will be included on CPF screen 3. If this field is equal to zero, Dynamic Card Verification is not performed for transactions with Track 2. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name: TRK2 DCVD OFST
Screen: CPF Screen 3
DDL Name: CPF.BASE.TRK2-DCVD-OFST
Valid Values: 0-99

The length of the Dynamic Card Verification digits on Track 2 will also need to be entered. The default value is 0.

Screen Name: DCVD LEN
Screen: CPF Screen 3
DDL Name: CPF.BASE.TRK2-DCVD-LEN
Valid Values: 0-9

The offset of the Application Transaction Counter on Track 2 will be included on CPF screen 3. If this field is equal to zero, ATC checking is not performed for transactions with Track 2. Additionally, the ATC cannot be part of the DCVD calculation. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name: ATC OFST
Screen: CPF Screen 3
DDL Name: CPF.BASE.TRK2-ATC-OFST
Valid Values: 0-99

The length of the Application Transaction Counter on Track 2 will also need to be entered. The default value is 0.

Screen Name:	ATC LEN
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK2-ATC-LEN
Valid Values:	0-9

The offset of the Unpredictable Number on Track 2 will be included on CPF screen 3. This field is used by cards following the MasterCard PayPass implementation; otherwise this field should be set to zero. If this field is equal to zero, the Unpredictable Number cannot be part of the DCVD calculation. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name:	UNPRED NUM OFST
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.TRK2-UNPREDIC-NUM
Valid Values:	0-99

A new flag will be added to indicate whether the Application Transaction Counter (ATC) check is to be performed. Setting this field to a non-zero value causes the ATC fields in the Base segment of the CAF to be updated. This field must be set to a non-zero value if the ATC value sent in the transaction data contains fewer digits than the value sent to the HSM for verification. If transactions should not be declined for ATC checking failure, that can be set in the BAD ATC ACTION field on this screen. The default value is 0.

Screen Name:	ATC CHECK
Screen:	CPF Screen 3
DDL Name:	CPF.BASE.ATC-CHK

Valid Values:

- 0 = Do not perform ATC check.
- 1 = Perform ATC check on EMV transactions.
- 2 = Perform ATC check on contactless magnetic stripe transactions.
- 3 = Perform ATC check on EMV and contactless magnetic stripe transactions.

A new flag will be added to indicate whether a pre-screen Dynamic Card Verification check is performed before sending a transaction to the host. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 0.

Screen Name: CHECK IF HOST ONLINE DCV
Screen: CPF Screen 3
DDL Name: CPF.BASE.PRE-SCRN-DCVD
Valid Values:

0 = No, do not verify the Dynamic Card Verification Digits (DCVD) before sending the transaction request to the host if the host is online.
1 = Yes, verify the Dynamic Card Verification Digits (DCVD) before sending the transaction request to the host. If the DCVD is invalid, the Authorization processes perform the action defined in the BAD DCV ACTION field.

A new flag will be added to indicate whether a pre-screen Application Transaction Counter (ATC) check is performed before sending a transaction to the host. The default value is 0.

Screen Name: ATC
Screen: CPF Screen 3
DDL Name: CPF.BASE.PRE-SCRN-ATC
Valid Values:

0 = No, do not verify the Application Transaction Counter (ATC) before sending the transaction request to the host if the host is online.
1 = Yes, verify the Application Transaction Counter (ATC) before sending the transaction request to the host. If the ATC is invalid, the Authorization processes perform the action defined in the BAD ATC ACTION field.

A new flag will be added to indicate the action taken when an ATC check fails. The default value is 1.

Screen Name: BAD ATC ACTION
Screen: CPF Screen 3
DDL Name: CPF.BASE.ATC-BAD-DISP
Valid Values:

0 = Note the situation and continue.
1 = Deny and return the card.
2 = Deny and retain the card.
3 = Refer the transaction (BASE24-pos only).

A new flag will be added to indicate the action taken by the Device Handler/ Router/Authorization module when a cardholder uses a card that contains invalid Dynamic Card Verification data. This field is used only if the DYN CARD VERIF KEY LOCATOR field is not equal to blanks. The default value is 1.

Screen Name: **BAD DCV ACTION**
Screen: CPF Screen 3
DDL Name: CPF.BASE. DCV-BAD-DISP
Valid Values:

0 = Note the situation and continue.
1 = Deny and return the card.
2 = Deny and retain the card.
3 = Refer the transaction (BASE24-pos only).

The following new values have been added to the fields have been added to the CPF Screen 11 to configure ATC checking.

A new flag will be added to indicate whether the Application Transaction Counter (ATC) check is to be performed for EMV transactions. The default value is 0.

Screen Name: ATC CHECK TYPE
Screen: CPF Screen 11
DDL Name: CPF.EMV.ATC-CHK
Valid Values:

0 = Do not perform ATC check. (was “N” in previous release)
1 = Perform ATC check on EMV transactions. (was “Y” in previous release)
9 = Refer to ATC-CHK field in Base segment, Screen 3.

A full description of these fields can be found in the ***BASE24 Files Maintenance Manual.***

ATC Checking Configuration

The following table summarizes the possible combinations of values for the ATC CHECK fields on CPF screens 3 and 11 and how the ATC counter is checked or updated in the CAF base or EMV segment for each possible combination.

ATC CHECK on CPF Screen 3	ATC CHECK TYPE on CPF Screen 11	CAF Segment ATC checked/updated in Contactless Magnetic Stripe transactions	CAF Segment ATC checked/updated in EMV transactions
0	0	None	None
0	1	None	EMV
0	9	None	None
1	0	None	None
1	1	None	EMV
1	9	None	BASE
2	0	BASE	None
2	1	BASE	EMV
2	9	BASE	None
3	0	BASE	None
3	1	BASE	EMV
3	9	BASE	BASE

POS Terminal Data File (PTD)

The terminal capabilities for a contactless terminal must be updated on Screen 19 of the POS Terminal Data file (PTD). This will allow the correct information to be sent to Visa or MasterCard.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Issuers requiring support for Dynamic Card Verification or Visa Cryptogram version number 17 must implement version 06.4 of TSS at a minimum.

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Cardholder Authorization File (CAF)

For customers converting from release 5.3 to release 6.0 version 7, the CAF conversion program must be run to convert the CAF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

No separate conversion program was created to move the CAF from version 6 to version 7. Customers must do a full file refresh using the new CAF layout. This will initialize the new ATC fields to zero.

Host/Co-Network Impacts

If a host is sending transactions to BASE24 that require dynamic card verification, the host must put the appropriate point of service entry mode (Data Element 22) in the incoming message. This field must be set to a “07” for a contactless EMV transaction and a “91” for a contactless magnetic stripe transaction.

If a host will be validating Dynamic Card Verification Digits, changes must be made at the host to retrieve the data elements required and to perform the validation.

The Host might receive values of “07” and “91” in Data Element 22 (Point of Service Entry Mode). Changes might be required to handle these new values in POS Entry mode field.

User/Operational Impacts

In order to facilitate static CVC3 verification at a future time, it is important that issuers doing dynamic CVC3 always use an Unpredictable Number with a length of at least 1. This will allow the Unpredictable Number length of 0 in the last position of the Issuer Discretionary Data to indicate that static CVC3 is being used in a particular transaction.

Visa says the ATC sent in the transaction data must be 4 digits in length. MasterCard says the ATC sent in the transaction data can be as short as 1 digit. When fewer ATC digits are sent in a transaction, especially if only 1 is sent, it is more likely the full ATC value in the database will get out of sync with the full ATC kept on the card. Once they are out of sync, it is unlikely they will ever match up and the Dynamic Card Verification Digits sent will never be verified. ACI recommends sending the maximum number of ATC digits possible in the track data. The new ATC checking and dynamic card verification logic will attempt to determine the correct ATC value, even if only one digit is sent. However, the issuer must recognize there is greater chance that the value will be wrong when fewer digits are included in the transaction data.

<u>Performance Impacts</u>

Dynamic card verification takes the place of standard card verification in the transaction path. The performance impact of doing DCV should be the same as doing standard card verification.

Vendor Impacts.

This enhancement will require device firmware changes to be able to capture and send the data required for this enhancement.

The terminal must send FID 6 SFID E (POS Entry Mode) in the SPDH message for all contactless transactions, both magnetic stripe and EMV. This data element tells the device handler/router/authorization process if the transaction is a contactless magnetic stripe or EMV transaction.

Reference Documents

- MasterCard PayPass Mag Stripe – Technical Specifications, Version 3.1, November 2003
- MasterCard PayPass Product Guide, February 2004
- MasterCard PayPass Mag Stripe, Acquirer Implementation Guide, PayPass 17, v.2, 2005
- MasterCard PayPass Mag Stripe, Issuer Implementation Guide, PayPass 16,v.2, 2006
- MasterCard PayPass M/Chip, Acquirer Implementation Guide, PayPass 23, v. 1-A4, June 2006
- MasterCard PayPass M/Chip, Issuer Implementation Guide, PayPass 24, v.1-A4, June 2006
- Visa dCVV and MasterCard CVC3 TSS Support, Version 2, February 16, 2006
- Visa Contactless Payment Specification, December 2004, Version 1.4.2.
- Visa Contactless Payment Specification, Additions and Clarifications, February 2006, Version 1.1.
- Visa Contactless Payment Specification, March 2006, Version 2.0.1

Correct Declined Advices to Host

This section describes the implementation requirements associated with the Correct Declined Advices to Host enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Forward Declined Advices to Host enhancement.

The BASE24-pos Router/Authorization process has been modified to allow declined advices (e.g. transactions declined by an interchange or interchange interface) to be sent to a HISO process, if required. Previously, these types of transactions were logged to the POS Transaction Log File (PTLF) only, and a Host would only see them if it processed PTLF extracts.

This enhancement is an extension to the "Forward Declined Advices to Host" enhancement, implemented in Release 6.0 Version 5. Please refer to the ***BASE24 Release 6.0 Version 5 Implementation Guide*** for full details of this enhancement.

The following considerations apply to this enhancement:

- The ability to forward and log declined advices will be available for all authorization levels.
- Declined advices for which a CPF record cannot be successfully retrieved and processed will not be forwarded as declined advices to the Host.
- All advices rejected before the IDF record is successfully retrieved and processed cannot be forwarded as declined advices to the Host, since no routing information is available without an IDF record.
- Declined advices for which an IPCF record cannot be successfully retrieved and processed will not be forwarded as declined advices to the Host.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60RTAU.ROUTERS, AUTHS, AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file on BA60UD0A.SCNDADV.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

ISO Host Interface Process

If the BASE24 user wishes to forward declined advices, then the Host system must be capable of accepting and processing non-approved advices.

User/Operational Impacts

Performance Impacts

Performance may be impacted in systems that forward all declined advices to a back-end Host. The impact to performance of the system will depend on the number of extra transactions being forwarded to a Host.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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CVD/CVD2 Checking in a Single Request

This section describes the implementation requirements associated with the CVD/CVD2 Checking in a Single Request enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the CVD/CVD2 Checking in a Single Request enhancement.

BASE24 has been enhanced to support the validation of CVD1 and CVD2 in the same transaction, for card types in addition to AMEX. Logic has been added to ensure that the card verification results are recorded in the correct field for card-read and manually entered transactions. Additional processing has been implemented to allow issuers to configure the validation of CVD2 separately from CVD1.

A manual card verification check type field on the Card Prefix File (CPF) screen 2 specifies the circumstances (if any) under which manual card verification should be performed. This field replaces the value in the card verification check type field for manually entered transactions, thereby allowing CVD1 and CVD2 validation to be configured separately.

The logic in the POS Router/Authorization modules has been enhanced to utilize this new field to determine whether and how to perform card verification on transactions. The following new functions are supported:

- It is possible for issuers to perform CVD2 validation (when CVD2 is present) on card-read transactions (as well as CVD1 validation), using a single message exchange with TSS.

- When a security device error is encountered during CVD1 validation, the transaction is no longer declined unconditionally; instead, the authorization response is based on the setting of the appropriate BAD CV ACTION field on CPF Screen 2.

- Whenever CVD2 is present in an authorization request, the CRD-VRFY-FLG2 field in the POS Data1 Token is set to indicate whether CVD2 validation was performed, and, if so, the result.

- It is possible for issuers to configure BASE24-pos to perform CVD2 validation on manually-entered transactions but not CVD1 validation on card-read transactions, or vice versa.

The processing added in Release 6.0 version 5 to support AMEX CSC validation is not affected by this enhancement. This means that the manually entered AMEX CSC will always be validated when present in a card-read transaction.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.SCRNCPF, RQCPFS, SVCPFS
- Inline changes to BA60SRC.CVUTILS and SECUTILS

All BASE24-pos customers:

- Inline changes to PS60RTAU.AUTHLIBS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNCVDS.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Card Prefix File (CPF)

The following new field makes it possible to configure CVD2 validation separately from CVD1 validation:

Screen Name:	MANUAL CV CHECK TYPE
Screen:	CPF Screen 2
DDL Name:	MANUAL-CV-CHK-TYP
Valid Values are:	0 – Do not check CVD2 (this is the default value) 1 – Check CVD2 when present in manually entered transactions only 2 – Check CVD2 when present in all transactions

A full description of this field can be found in the ***BASE24 Files Maintenance Manual.***

CVD1 and CVD2 validation in the same transaction can be performed only if the same card verification keys are used for both CVD1 and CVD2, because only one KEYA Group can be included in the message to TSS.

CVD1 and CVD2 validation in the same transaction can be performed only if YYMM is configured as the preferred CVD2 expiration date format, because this is the required format of the card expiration date in the message to TSS.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Issuers requiring support for CVD1 and CVD2 validation in the same transaction must implement version 06.4 of TSS at a minimum.

TSS will always attempt to validate both CVD1 and CVD2 if both are included in the TSS message.

File Conversions

Card Prefix File (CPF)

The CPF conversion program must be run to convert the CPF. Details of this conversion program can be found in the **General Changes** earlier in Section 4.

Host/Co-Network Impacts

Not Applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

EMV CCD Issuer Support

This section describes the implementation requirements associated with the EMV CCD Issuer Support enhancement.

The implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the EMV CCD Issuer Support enhancement.

For additional detailed information on this enhancement see Section 4 (Base).

Software Impacts

For full details on the software impacts for this enhancement see Software Impacts in Section 4 (Base) of this document.

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNECCD.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

Seg-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Operational Impacts:

The Card Verification Results field is a different format on a CCD transaction. It is a 5 byte field (all bits in Byte 5 are reserved for future use). The decisional information contained in these bytes is as follows:

Action Index	Byte	Bit	Force Online	Auth Action	Meaning
“n”	1	4	“Y”	auth_act_cont_d	CDA performed
“n”	1	3	“Y”	auth_act_cont_d	Offline DDA performed
“n”	1	2	“Y”	auth_act_cont_d	Issuer authentication not performed
“n”	1	1	“Y”	auth_act_cont_d	Issuer authentication failed
“n”	2	4	“Y”	auth_act_cont_d	Offline PIN verification performed
“n”	2	3	“Y”	auth_act_cont_d	Offline PIN verification performed and PIN not successfully verified
“n”	2	2	“Y”	auth_act_cont_d	PIN try limit exceeded
“n”	2	1	“Y”	auth_act_cont_d	Last online transaction not completed
“n”	3	8	“Y”	auth_act_cont_d	Lower offline transaction count limit exceeded
“n”	3	7	“Y”	auth_act_cont_d	Upper offline transaction count limit exceeded
“n”	3	6	“Y”	auth_act_cont_d	Lower cumulative offline amount limit exceeded
“n”	3	5	“Y”	auth_act_cont_d	Upper cumulative offline amount limit exceeded
“n”	4	4	“Y”	auth_act_cont_d	Issuer script processing failed
“n”	4	3	“Y”	auth_act_cont_d	Offline data authentication failed on previous transaction
“n”	4	2	“Y”	auth_act_cont_d	Go online on next transaction was set
“n”	4	1	“Y”	auth_act_cont_d	Unable to go online

Users may set the “Force Online” and “Auth Action” for any bit setting, according to business requirements.

The default base set for CCD cards is:

"4", 1, 1, “Y”, auth_act_ref_d

This is interpreted as Issuer Authentication failed on the last transaction, and will result in the transaction being referred.

Vendor Impacts

Not applicable

Reference Documents

- EMVCo – EMV 2004 version 4.1, May 2004

Hypercom (HPDH) Software MACing Support Enhancement

This section describes the implementation requirements associated with the Hypercom (HPDH) Software MACing Support Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Hypercom (HPDH) Software MACing Support Enhancement.

The BASE24-pos Terminal Data files (PTD) server has been modified to allow Hypercom terminals to be configured for hardware (hardware security module) PIN encryption using Transaction Security Services (TSS) and software (DES) MACing. Hypercom terminals use a proprietary MACing algorithm that is supported by the BASE24-pos Hypercom Device Handler (HPDH) in software but not hardware.

Specifically, the PTD server has been changed so if the terminal type on PTD screen 1 is set to D0 (Hypercom TDS) and TSS is enabled then the MAC type field on PTD screen 7 can be set to '01' (software message authentication) and the PIN encryption type can be set to '03' (security device, PIN/PAD PIN block) or '04' (security device, ANSI standard PIN/PAN PIN block),. This combination of MAC type and PIN encryption type remains prohibited for all other terminal types.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60AFT.SVPTDTS

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

BASE24-pos Terminal Data files (PTD) Screen

Users should be aware that if the terminal type on PTD screen 1 is set to D0 (Hypercom TDS) and TSS is enabled then the MAC type field on PTD screen 7 can now be set to ’01’ (software message authentication) and the PIN encryption type can be set to ‘03’ (security device, PIN/PAD PIN block) or ‘04’ (security device, ANSI standard PIN/PAN PIN block). Previously the MAC type and PIN encryption type could not be set to software and hardware, respectively, under these circumstances.

This combination of MAC type and PIN encryption type remains prohibited for all other terminal types.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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Improve EMV Transaction Certificates in SPDH Enhancement

This section describes the implementation requirements associated with the Improve EMV Transaction Certificates in SPDH Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Improve EMV Transaction Certificates in SPDH Enhancement.

The ACI Standard POS Device Handler (SPDH) has been modified to improve the handling of EMV transaction certificates. A previous enhancement enabled EMV transaction certificates to be uploaded to the SPDH in the form of EMV Log-Only transactions (Message Subtype = E) and logged to the BASE24-pos Transaction Log File (PTLF) in records with message type 9920. The Improve EMV Transaction Certificates in SPDH Enhancement enables the terminal to cancel (reverse) those transactions.

A new SPDH transaction, the EMV Log-Only Cancellation (Message Subtype = V), has been created. If the terminal times out before the SPDH responds to an EMV Log-Only transaction request, the terminal can cancel the transaction by sending an EMV Log-Only Cancellation. The EMV Log-Only Cancellation is logged to the PTLF using the new message type 9921. From the terminal's perspective, an EMV Log-Only Cancellation is like a reversal. The terminal should process an EMV Log-Only Cancellation similarly to a Timeout Reversal. An EMV Log-Only Cancellation must immediately follow the original transaction. It must take precedence in the queue over online transaction requests and SAF requests. And as with Timeout Reversals, the Transmission Number of the EMV Log-Only Cancellation request must match that of the original EMV Log-Only transaction.

The BASE24-pos PTLF Extract module has been enhanced to extract the new 9921 EMV Log-Only Cancellation transaction records from the PTLF.

Software Impacts

All BASE24-pos customers:

- Inline changes to PS60EXT.PTLFXS

All BASE24-pos ACI Standard POS Device Handler (SPDH) customers:

- Inline changes to PS60SPDH.ASPDHG and ASPDHS
- Updated event (log) message manual BA60LOGM.PSSPDH

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNIEMV.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-pos users that utilize the PTLF Extract process and want to accept and process the new 9921 EMV Log-Only Cancellation PTLF records.

User/Operational Impacts

Not applicable

Vendor Impacts

EMV Log-Only Cancellation Request

If the terminal times out before the SPDH responds to an EMV Log-Only transaction request, the terminal should cancel the transaction by sending an EMV Log-Only Cancellation (Message Subtype = V). The terminal must do the following:

- Set the Message Subtype in the SPDH message header to “V”.
- Set the Transmission Number in the SPDH message header to match that of the original EMV Log-Only transaction that is being cancelled.
- Add FID 6 subFID O (EMV Request Data) to the EMV Log-Only Cancellation request. The ARQC field within subFID O should contain the same EMV transaction certificate that was sent in the original transaction.
- Like a reversal, give the EMV Log-Only Cancellation transaction priority over other transactions in the queue, including online transaction requests and SAF requests.
- Send the EMV Log-Only Cancellation request immediately after the original EMV Log-Only transaction.

Reference Documents

Not applicable

Log Stored Value Dormancy/Escheatment Trans to PTLF

This section describes the implementation requirements associated with the Log Stored Value Dormancy/Escheatment Trans to PTLF Enhancement.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Log Stored Value Dormancy/Escheatment Trans to POS Transaction Log File (PTLF) Enhancement.

The Dormancy/Escheatment module has been modified to log Dormancy and Escheatment transactions to the PTLF. A new optional LCONF parameter POS-SV-PTLF-LOGGING specifies whether Dormancy and Escheatment transactions should be logged to the PTLF or not.

Dormancy transactions will be logged as a Purchase transaction with the Subtype Token set to “C003”. Escheatment transactions will be logged as a Balance Inquiry transaction with the Subtype Token set to “C004”. The PTLF record type is set to “22” for both types of transactions.

If logging is turned on and the account status is modified during dormancy processing then the current and the previous account status will be written to the Stored Value Token (U2). The Stored Value Token has been redefined to hold the data. The previous account status will be written to Stored Value token Previous Account Status field. The current account status will be written to Stored Value token Current Status field.

Software Impacts

All BASE24-pos customers:

- Inline changes to BA60DDL.DDLPSTKN

All BASE24-pos Dormancy/Escheatment (DORMESC) customers:

- Inline changes to PS60SV.DORMESCG, DORMESCM and DORMESCS

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0A.SCNSVDE.

CUSTMACS Changes

Not applicable

LCONF Assign Changes

This section describes the assign record that must be added to the LCONF to support this enhancement. The LCONFPS file located on the BA60MISC subvolume has been modified to include these assigns.

POS-SV-PTLF- EXTMEM-SWAPVOL

The fully qualified name of the BASE24-pos Stored Value PTLF extended memory swap volume. This assign specifies where to temporarily store data when building extended memory. If this assign is not present or contains an invalid value, the NSK operating system assigns a swap volume location.

LCONF Param Changes

This section describes the param records that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFPS file located on BA60MISC subvolume.

POS-SV-PTLF-LOGGING

This param specifies whether Dormancy/Escheatment transactions should be logged to the PTLF or not. If this param is not present in the LCONF or contains a value other than “Y”, PTLF logging is disabled (default). A value of “Y” indicates that PTLF logging is enabled. The default value is “N”.

Y = Log Dormancy/Escheatment transactions to the PTLF.

N = Do not Log Dormancy/Escheatment transactions to the PTLF.

POS-SV-RETAILER-ID

This param specifies the POS Retailer ID that is to be included in the PTLF record logged by the Dormancy/Escheatment process. Specifically, the value in this param will be logged in the PTLF Retailer ID field (ptlf.head.retl.ky.rdfkey.id). This param is used during Dormancy/Escheatment POS Transaction Log File (PTLF) logging only. The default is “DORMANCY-RETAILER”.

POS-SV-TERMINAL-ID

This param specifies the POS Terminal ID that is to be included in the PTLF record logged by the Dormancy/Escheatment process. Specifically, the value in this param will be logged in the PTLF Terminal ID field (ptlf.head.retl.term_id). This param is used during Dormancy/Escheatment POS Transaction Log File (PTLF) logging only. The default is “DORMANCY-TERM”.

Database Configuration

Token file (TKN)

The BASE24-pos Stored Value Token (U2) has been modified for this enhancement:

- The new field PREV-ACCT-STAT is a redefine of the BAL-AS-CASH field.
- The new field CUR-ACCT-STAT is a redefine of the SV-RVSL-FLG field.

If the Stored Value Token is to be logged to the BASE24-pos Transaction Log File (PTLF), the appropriate token records must be configured in the Token file.

If this Stored Value Token is to be extracted with the PTLF, the appropriate records must be configured in the Token file.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

The host may wish to interrogate the Subtype token when processing dormancy and escheatment PTLF transactions.

The host may wish to interrogate the Stored Value token (U2) when processing dormancy transaction in order to find possible account status changes within the dormancy PTLF records.

User/Operational Impacts

System Limitations

BASE24 users installing this enhancement must be aware of their internal system limitations for messaging to the PTLF. There are system limitations regarding the amount of data that can be passed through the BASE24 system and logged to a log file. BASE24 users should be aware of these limitations and be cognizant that support of new data in online messages and log files in the form of tokens can cause them to exceed these limitations. Appendix A of the ***BASE24 Tokens Manual*** provides information on assessing the impacts of tokens on log file disk space requirements.

Disk Impacts

The introduction of an additional record logged to the PTLF will result in the use of additional disk space. The additional disk space required for the PTLF is dependent upon the volume of dormancy and escheatment transactions processed. This impact is customer-specific and must be evaluated by each BASE24 customer.

Vendor Impacts

Not applicable

Reference Documents

Not applicable

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7: BASE24-teller

Section 7 discusses the changes and enhancements made to BASE24-teller in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24-teller users must consider when upgrading from release 6.0 version 6.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-teller users, along with the associated conversion and implementation requirements that all BASE24-teller users must address.
- Optional enhancements describe the changes that affect only those BASE24-teller users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-teller users must address to implement the enhancements.

Section 7 must be used in conjunction with Section 4 to identify the changes that BASE24-teller users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-teller users must address in order to upgrade to release 6.0 version 7.

There are no general enhancements inherent to BASE24-teller.

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Optional Enhancements

This section identifies the BASE24-teller enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-teller.

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8: BASE24-remote banking

Section 8 discusses the changes and enhancements made to BASE24-remote banking in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24-remote banking users must consider when upgrading from release 6.0 version 6. This section covers changes made to the following products within the BASE24 family:

- BASE24-telebanking
- BASE24-billpay
- BASE24 Kernel/IEP

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24-remote banking users, along with the associated conversion and implementation requirements that all BASE24-remote banking users must address.
- Optional enhancements describe the changes that affect only those BASE24-remote banking users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24-remote banking users must address to implement the enhancements.

Section 8 must be used in conjunction with Section 4 to identify the changes that BASE24-remote banking users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24-remote banking users must address in order to upgrade to release 6.0 version 7.

There are no general enhancements inherent to BASE24-remote banking.

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Optional Enhancements

This section identifies the BASE24-remote banking enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no new optional enhancements available for BASE24-remote banking.

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9: BASE24 Interchange Interfaces

Section 9 discusses the changes and enhancements made to BASE24 Interchange Interfaces in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing BASE24 Interchange Interface users must consider when upgrading from release 6.0 version 6.

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 Interchange Interface users, along with the associated conversion and implementation requirements that all BASE24 Interchange Interface users must address.
- Optional enhancements describe the changes that affect only those BASE24 Interchange Interface users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 Interchange Interface users must address to implement the enhancements.

Section 9 must be used in conjunction with Section 4 to identify the changes that BASE24 Interchange Interface users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all existing BASE24 Interchange Interface users must address in order to upgrade to release 6.0 version 7.

Changes and requirements are identified using a standard format defined in the introduction to Section 4.

General Changes

SPDH & RTAU Enhancements for Visa and MasterCard

These areas are summarized on the following pages.

Section 2 contains a functional overview of the general enhancements described here.

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General Changes

The LINK EMV Interchange was modified due to changes done as part of the Contactless Chip/Magnetic Stripe enhancement. The interface referenced the define pt_**svc**_entry_mde_chip_d in BA60IEMV.EMVG which has been changed. Therefore, the Interface was modified to use the new Base define for Point of Service Entry Mode. References to pt_**svc**_entry_mde_chip_d have been changed to pt_**srv**_entry_mde_chip_d

Software Impacts

All BASE24 LINK EMV Interchange customers:

- Inline changes to SW60ILNK.LINKEMVS.

CUSTMACS Changes

Not Applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not Applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

SPDH & RTAU Enhancements for Visa and MasterCard

This section describes the implementation requirements associated with the enhancement to provide support for the SPDH & RTAU Enhancements for Visa and MasterCard.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the SPDH & RTAU Enhancements for Visa and MasterCard.

All customers who use the healthcare portion of the SPDH & RTAU Enhancements for Visa and MasterCard must implement this change to the switch interfaces.

The Healthcare Token CP will no longer be used by BASE24 and will be replaced by the following two tokens:

- Healthcare Service Token CW
- Healthcare Transit Token CV

This enhancement impacts the following interfaces:

- BankNet
- Plus ISO/Interlink
- VISA DPS
- VisaNet

Refer to the interface's respective manual and updated guide for further details on the implementation of this enhancement.

Software Impacts

All BASE24 customers:

- Inline changes to BA60AFT.COBTKN
- Inline changes to BA60DDL.DDLPSTKN
- Inline changes to BA60SRC.PSTKNCVS and PSTKNID

All BankNet Switch Interface customers

Inline changes to SW60BNET.BNETG and SW60BNET.BNETLIBS
Documentation changes to SW60BNET.BNETMAN

All PLUS ISO/Interlink Switch Interface customers

Inline changes to SW60PISO.PISOG and SW60PISI.PISIS

All VISA DPS Switch Interface customers

Inline changes to SW60VDPS.VDPSG and SW60VDPS.VDPSS
Documentation changes to SW60VDPS.VDPSMAN, SW60VDPS.VDPSUPDT, and SW60VDPS.VDPSLOGM

All VisaNet Switch Interface customers

Inline changes to SW60VISA.VISAFMTS and SW60VISA.VISAG
Documentation changes to SW60VISA.VISAMNWD, SW60VISA.VISAUPDT, and SW60VISA.VISALOGM

All BASE24-pos SPDH customers
For complete details, refer to the SPDH & RTAU Enhancements for Visa and MasterCard enhancement within Section 6 of this document.

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable.

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Optional Enhancements

This section identifies the BASE24 Interchange Interface enhancements that can be enabled or disabled with release 6.0 version 7, along with the changes, conversion requirements, and implementation requirements associated with each. These enhancements include the following:

- Contactless Chip/Magnetic Stripe Support

BASE24 Interchange Interface users can choose whether or not to implement each of the above enhancements. Users who plan to implement a particular enhancement must address the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements. Users who do not plan to implement a particular enhancement can simply review the corresponding release 6.0 version 7 changes, conversion requirements, and implementation requirements for future reference.

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Contactless Chip/Magnetic Stripe Support

This section describes the implementation requirements associated with the enhancement to provide support for the Contactless Chip/Magnetic Stripe Support.

These implementation requirements are identified using a standard format defined in the introduction to Section 4.

Section 2 contains a functional overview of the Contactless Chip/Magnetic Stripe Support.

All customers who wish to acquire or authorize transactions initiated using the contactless method will need to implement this enhancement.

This enhancement impacts the following interfaces:

- BankNet
- EuroPay
- MAC MASM
- MDS Maestro
- NYCI ISO
- STAR ISO
- VisaNet

Refer to the interface's respective manual and updated guide for further details on the implementation of this enhancement.

Software Impacts

All BASE24-pos customers
For complete details, refer to the Contactless Chip/Magnetic Stripe Support enhancement within Section 6 of this document.

All BankNet Switch Interface customers

Inline changes to SW60BNET.BNETG and SW60BNET.BNETLIBS
Documentation changes to

All BankNet EMV Interface customers:

Inline changes to SW60IBNT.BNETEMVS

All EuroPay Switch Interface customers

Inline changes to SW60EURO.EUROLIBS

All EuroPay EMV Interface customers:

Inline changes to SW60IEUR.EUROEMVS

All MDS Maestro Interface customers:
Inline changes to SW60MDS.MDSG and SW60MDSM.MDSMS

All MDS Maestro EMV Interface customers:
Inline changes to SW60IMDM.MDSMEMVS

All MAC MASM Switch Interface customers
Inline changes to SW60MACI.MACIG and SW60MACI.MACIS
Documentation changes to

All NYCI ISO Switch Interface customers
Inline changes to SW60NYCI.NYCIG and SW60NYCI.NYCIS
Documentation changes to

All STAR ISO Switch Interface customers
Inline changes to SW60STRI.STRIG and SW60STRI.STRIS
Documentation changes to

All VisaNet Switch Interface customers
Inline changes to SW60VISA.VISAG and SW60VISA.VISAFMTS
Documentation changes to

All VisaNet EMV Interface customers:
Inline changes to SW60IVIS.VISAEMVS

CUSTMACS Changes

Not applicable.

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Not applicable

Vendor Impacts

Not applicable

Reference Documents

Not applicable

10: BASE24 Value-Added Products

Section 10 discusses the changes and enhancements made to the BASE24 value-added products in release 6.0 version 7, along with the resulting conversion and implementation requirements that existing users of these products must consider when upgrading from release 6.0 version 6.

The value-added products discussed in this section include the following:

BASE24-card
BASE24-customer service
BASE24-InfoBase
BASE24-inventory Voucher Manager

This section is divided into two parts: General Enhancements and Optional Enhancements.

- General enhancements describe the changes that affect all BASE24 value-added product users, along with the associated conversion and implementation requirements that all BASE24 value-added product users must address.
- Optional enhancements describe the changes that affect only those BASE24 value-added product users that choose to implement the enhancements, along with the associated conversion and implementation requirements that BASE24 value-added product users must address to implement the enhancements.

Section 10 must be used in conjunction with Section 4 to identify the changes that BASE24 value-added product users need to consider when upgrading to release 6.0 version 7.

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General Enhancements

This section describes the changes, conversion requirements, and implementation requirements that all BASE24 Value-added product users must address in order to upgrade to release 6.0 version 7.

There are no general enhancements affecting the following BASE24 value-added product users:

BASE24-card
BASE24-customer service
BASE24-InfoBase
BASE24-inventory Voucher Manager

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Optional Enhancements

This section identifies the BASE24 enhancements that can be enabled or disabled with release 6.0 version 7 that have an impact on the following BASE24 value-added products, along with the changes, conversion requirements, and implementation requirements associated with each.

There are no optional enhancements affecting the following BASE24 value-added product users:

BASE24-card
BASE24-customer service
BASE24-InfoBase
BASE24-inventory Voucher Manager

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A: Logical Network Configuration File Changes

Appendix A contains the Logical Network Configuration File (LCONF) changes.

This information includes assigns and params that have been added, deleted or modified with release 6.0 version 7 along with their associated page numbers. Refer to the specified page numbers for explanations of how these assigns and params can or must be set.

Refer to the ***BASE24 Logical Network Configuration File (LCONF) Manual*** for instructions on adding them to the LCONF.

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ASSIGNS

The following assigns were added or modified for release 6.0 version 7. Refer to the specified page numbers for explanations of how these assigns can or must be set.

Locked ATD Records Causing Cutover Failure

ATM-SETL-ATD-LOCKED-RECS-SKIP..5-59, 5-60

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PARAMS

The following params were added or modified for release 6.0 version 7. Refer to the specified page numbers for explanations of how these params can or must be set.

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B: Host/CO-Network Impacts

Appendix B contains a list of release 6.0 version 7 enhancements that have host/co-network impacts, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

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Enhancements

The following enhancements have host/co-network impacts. Refer to the specified page numbers for more information about these enhancements.

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C: CUSTMACS File Changes

Appendix C contains a list of release 6.0 version 7 enhancements that have CUSTMACS changes, along with their associated page numbers. Refer to the specified page numbers for more information about these enhancements.

BASE24 users should review these changes to determine how the CUSTMACS file should be set for their BASE24 environment before running MAKE to compile their system. More information on all MAKE flags used within BASE24 can be found in the FLGSDOC file located on the BA60MAKE subvolume delivered with BASE24 Release 6.0 Version 7 code.

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Enhancements

The following enhancements have CUSTMACS impacts:

<u>Contactless Chip/Magnetic Stripe Support</u>

<u>Triton Mobile Top-Up</u>

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D: Re-Sequenced Files and New Files

Appendix D consists of the following information:

- No files were resequenced as part of release 6.0 version 7.
- The table later in this section illustrating the new files added to BASE24 for the release 6.0 version 7 enhancements.

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New Files Added to BASE24 with Version 7

The following table lists the files, by subvolume, added to Release 6.0 by the version 7 enhancements:

Subvolume	Filename	Language	Enhancement Name
AT60DDLD	TDF08	Documentation	Tidel General Enhancements Triton Token Retrieval Option
	TDF0A	Documentation	Tidel General Enhancements Triton Token Retrieval Option
AT60IN50	RQWEMAIM	MAKE	Shared NDC+ EMV Upgrade
	RQWEMAIS	COBOL	Shared NDC+ EMV Upgrade
	SCNWEMAI	Screen File	Shared NDC+ EMV Upgrade
AT60TTRT	ATTTRTFM	MAKE	Triton Mobile Top-Up
	ATTTRTMM	MAKE	Triton Mobile Top-Up
	TRITMTUG	TAL	Triton Mobile Top-Up
	TRITMTUM	MAKE	Triton Mobile Top-Up
	TRITMTUS	TAL	Triton Mobile Top-Up
	TRITNAMS	TAL	Triton Mobile Top-Up
AT60UC0A	AAREAD0A	Documentation	Tidel General Enhancements Triton Token Retrieval Option
	ATUC0AFM	MAKE	Tidel General Enhancements Triton Token Retrieval Option
	ATUC0AM	MAKE	Tidel General Enhancements Triton Token Retrieval Option
	ATUC0AMM	MAKE	Tidel General Enhancements Triton Token Retrieval Option
	CNVTDFM	MAKE	Tidel General Enhancements Triton Token Retrieval Option
	CNVTDFR	Run File	Tidel General Enhancements Triton Token Retrieval Option
	CNVTDFS	TAL	Tidel General Enhancements Triton Token Retrieval Option
BA60DDLD	CAF08	Documentation	Contactless Chip/Magnetic Stripe Support
	CAF0A	Documentation	Contactless Chip/Magnetic Stripe Support
	CPF0A	Documentation	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support
BA60LOGM	BADCVUTL	Documentation	Contactless Chip/Magnetic Stripe Support
BA60QDCV	BAQDCVFM	MAKE	Contactless Chip/Magnetic Stripe Support
	BAQDCVM	MAKE	Contactless Chip/Magnetic Stripe Support
	BAQDCVMM	MAKE	Contactless Chip/Magnetic Stripe Support
	DCVG	TAL	Contactless Chip/Magnetic Stripe Support
	DCVUTILG	TAL	Contactless Chip/Magnetic Stripe Support
	DCVUTILM	MAKE	Contactless Chip/Magnetic Stripe Support
	DCVUTILS	TAL	Contactless Chip/Magnetic Stripe Support

Subvolume	Filename	Language	Enhancement Name
BA60UC04	CNVHCFM	MAKE	HISO Miscellaneous Changes
	CNVHCFR	Run File	HISO Miscellaneous Changes
	CNVHCFS	TAL	HISO Miscellaneous Changes
BA60UC0A	AAREAD0A	Documentation	Add Additional Values to ATD/PTD AMEX SE Number Additional Validation Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support HISO Full Message Encryption HISO Miscellaneous Changes Tidel General Enhancements Triton Token Retrieval Option
	BAUC0AFM	MAKE	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support HISO Full Message Encryption HISO Miscellaneous Changes
	BAUC0AM	MAKE	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support HISO Full Message Encryption HISO Miscellaneous Changes
	BAUC0AMM	MAKE	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support HISO Full Message Encryption HISO Miscellaneous Changes
	CNVCPFM	MAKE	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support
	CNVCPFR	Run File	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support
	CNVCPFS	TAL	Contactless Chip/Magnetic Stripe Support EMV CCD Issuer Support
	CNVHCFM	MAKE	HISO Miscellaneous Changes
	CNVHCFR	Run File	HISO Miscellaneous Changes
	CNVHCFS	TAL	HISO Miscellaneous Changes
	CNVKEYFM	MAKE	HISO Full Message Encryption
	CNVKEYFR	Run File	HISO Full Message Encryption
	CNVKEYFS	MAKE	HISO Full Message Encryption
PS60DDLD	PRDF07	Documentation	AMEX SE Number Additional Validation
	PRDF0A	Documentation	AMEX SE Number Additional Validation
	PTD07	Documentation	Add Additional Values to ATD/PTD
	PTD0A	Documentation	Add Additional Values to ATD/PTD
PS60QRTA	PSQRTAFM	MAKE	Contactless Chip/Magnetic Stripe Support
	PSQRTAMM	MAKE	Contactless Chip/Magnetic Stripe Support
	RTAUDCVG	TAL	Contactless Chip/Magnetic Stripe Support
	RTAUDCVM	MAKE	Contactless Chip/Magnetic Stripe Support
	RTAUDCVS	TAL	Contactless Chip/Magnetic Stripe Support

Subvolume	Filename	Language	Enhancement Name
PS60UC0A	AAREAD0A	Documentation	Add Additional Values to ATD/PTD AMEX SE Number Additional Validation
	CNVPRDFM	MAKE	AMEX SE Number Additional Validation
	CNVPRDFR	Run File	AMEX SE Number Additional Validation
	CNVPRDFS	TAL	AMEX SE Number Additional Validation
	CNVPTDM	MAKE	Add Additional Values to ATD/PTD
	CNVPTDR	Run File	Add Additional Values to ATD/PTD
	CNVPTDS	TAL	Add Additional Values to ATD/PTD
	PSUC0AFM	MAKE	Add Additional Values to ATD/PTD AMEX SE Number Additional Validation
	PSUC0AM	MAKE	Add Additional Values to ATD/PTD AMEX SE Number Additional Validation
	PSUC0AMM	MAKE	Add Additional Values to ATD/PTD AMEX SE Number Additional Validation

The following table lists the same files but by version 7 enhancement name:

Enhancement Name	Subvolume	Filename	Language
Add Additional Values to ATD/PTD	BA60UC0A	AAREAD0A	Documentation
	PS60DDLD	PTD07	Documentation
		PTD0A	Documentation
	PS60UC0A	AAREAD0A	Documentation
		CNVPTDM	MAKE
		CNVPTDR	Run File
		CNVPTDS	TAL
		PSUC0AFM	MAKE
		PSUC0AM	MAKE
		PSUC0AMM	MAKE
AMEX SE Number Additional Validation	BA60UC0A	AAREAD0A	Documentation
	PS60DDLD	PRDF07	Documentation
		PRDF0A	Documentation
	PS60UC0A	AAREAD0A	Documentation
		CNVPRDFM	MAKE
		CNVPRDFR	Run File
		CNVPRDFS	TAL
		PSUC0AFM	MAKE
		PSUC0AM	MAKE
		PSUC0AMM	MAKE

Enhancement Name	Subvolume	Filename	Language
Contactless Chip/Magnetic Stripe Support	BA60DDLD	CAF08	Documentation
		CAF0A	Documentation
		CPF0A	Documentation
	BA60LOGM	BADCVUTL	Documentation
	BA60QDCV	BAQDCVFM	MAKE
		BAQDCVM	MAKE
		BAQDCVMM	MAKE
		DCVG	TAL
		DCVUTILG	TAL
		DCVUTILM	MAKE
		DCVUTILS	TAL
	BA60UC0A	AAREAD0A	Documentation
		BAUC0AFM	MAKE
		BAUC0AM	MAKE
		BAUC0AMM	MAKE
		CNVCPFM	MAKE
		CNVCPFR	Run File
		CNVCPFS	TAL
	PS60QRTA	PSQRTAFM	MAKE
		PSQRTAMM	MAKE
		RTAUDCVG	TAL
		RTAUDCVM	MAKE
		RTAUDCVS	TAL
EMV CCD Issuer Support	BA60DDLD	CPF0A	Documentation
	BA60UC0A	AAREAD0A	Documentation
		BAUC0AFM	MAKE
		BAUC0AM	MAKE
		BAUC0AMM	MAKE
		CNVCPFM	MAKE
		CNVCPFR	Run File
		CNVCPFS	TAL
HISO Full Message Encryption	BA60UC0A	AAREAD0A	Documentation
		BAUC0AFM	MAKE
		BAUC0AM	MAKE
		BAUC0AMM	MAKE
		CNVKEYFM	MAKE
		CNVKEYFR	Run File
		CNVKEYFS	MAKE

Enhancement Name	Subvolume	Filename	Language
HISO Miscellaneous Changes	BA60UC04	CNVHCFM	MAKE
		CNVHCFR	Run File
		CNVHCFS	TAL
	BA60UC0A	AAREAD0A	Documentation
		BAUC0AFM	MAKE
		BAUC0AM	MAKE
		BAUC0AMM	MAKE
		CNVHCFM	MAKE
		CNVHCFR	Run File
		CNVHCFS	TAL
Shared NDC+ EMV Upgrade	AT60IN50	RQWEMAIM	MAKE
		RQWEMAIS	COBOL
		SCNWEMAI	Screen File
Tidel General Enhancements	AT60DDLD	TDF08	Documentation
		TDF0A	Documentation
	AT60UC0A	AAREAD0A	Documentation
		ATUC0AFM	MAKE
		ATUC0AM	MAKE
		ATUC0AMM	MAKE
		CNVTDFM	MAKE
		CNVTDFR	Run File
		CNVTDFS	TAL
	BA60UC0A	AAREAD0A	Documentation
Triton Mobile Top-Up	AT60TTRT	ATTTRTFM	MAKE
		ATTTRTMM	MAKE
		TRITMTUG	TAL
		TRITMTUM	MAKE
		TRITMTUS	TAL
		TRITNAMS	TAL
Triton Token Retrieval Option	AT60DDLD	TDF08	Documentation
		TDF0A	Documentation
	AT60UC0A	AAREAD0A	Documentation
		ATUC0AFM	MAKE
		ATUC0AM	MAKE
		ATUC0AMM	MAKE
		CNVTDFM	MAKE
		CNVTDFR	Run File
		CNVTDFS	TAL
	BA60UC0A	AAREAD0A	Documentation

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E: Moved, Renamed, Deleted Files

Appendix E contains a list of files that were Moved, Renamed, or Deleted in BASE24 Release 6.0 Version 7.

No files were moved, renamed or deleted in release 6.0 version 7.

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F: Regional Specific Information

Appendix F contains the information that is required to implement the changes for release 6.0 version 7 that are specific to each region.

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